# LEED v4.1 BUILDING DESIGN AND CONSTRUCTION

July 10, 2020

U.S. Green Building Council

## INTEGRATIVE PROCESS (IP) ..... 8

**Prerequisite: Integrative Project Planning and Design ..... 8**
Healthcare ..... 8

**Credit: Integrative Process ..... 9**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 9

## LOCATION AND TRANSPORATION (LT) ..... 11

**LT Credit: LEED for Neighborhood Development Location ..... 11**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 11

**LT Credit: Sensitive Land Protection ..... 13**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 13

**LT Credit: High-Priority Site ..... 14**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 14

**LT Credit: Surrounding Density and Diverse Uses ..... 16**
NC, CS, Schools, Retail, Hospitality ..... 16
Data Centers, Warehouses & Distribution Centers ..... 17
Healthcare ..... 18

**LT Credit: Access to Quality Transit ..... 19**
NC, CS, Data Centers, Warehouses & Distribution Centers, Hospitality, Retail ..... 19
Schools ..... 20
Healthcare ..... 21

**LT Credit: Bicycle Facilities ..... 22**
New Construction, Core and Shell, Data Centers, Warehouses & Distribution Centers, Hospitality, Retail, Healthcare ..... 22
New Construction, Schools, Data Centers, Warehouses & Distribution Centers, Hospitality, Retail, Healthcare ..... 23
Core and Shell ..... 23
Schools ..... 24
Retail ..... 24
Healthcare ..... 25

**LT Credit: Reduced Parking Footprint ..... 26**
NC, CS, Retail, Schools, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 26

**LT Credit: Electric Vehicles ..... 27**
NC, CS, Data Centers, Hospitality, Retail, Healthcare ..... 27
Schools ..... 29
Warehouses & Distribution Centers ..... 29

## SUSTAINABLE SITES (SS) ..... 30

**SS Prerequisite: Construction Activity Pollution Prevention ..... 30**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 30

**SS Prerequisite: Environmental Site Assessment ..... 31**
Schools, Healthcare ..... 31

**SS Credit: Site Assessment ..... 32**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 32

**SS Credit: Protect or Restore Habitat ..... 33**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 34

**SS Credit: Open Space ..... 35**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 36

**SS Credit: Rainwater Management ..... 37**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 37

**SS Credit: Heat Island Reduction ..... 38**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 39

**SS Credit: Light Pollution Reduction ..... 41**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 41

**SS Credit: Site Master Plan ..... 45**
Schools ..... 45

**SS Credit: Tenant Design and Construction Guidelines ..... 46**
CS ..... 46

**SS Credit: Places of Respite ..... 47**
Healthcare ..... 47

**SS Credit: Direct Exterior Access ..... 48**
Healthcare ..... 48

**SS Credit: Joint Use of Facilities ..... 49**
Schools ..... 49

## WATER EFFICIENCY (WE) ..... 51

**WE Prerequisite: Outdoor Water Use Reduction ..... 51**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 51

**WE Prerequisite: Indoor Water Use Reduction ..... 52**
NC, CS, Schools, NC-Retail, Data Centers, Warehouses & Distribution Centers, NC-Hospitality, Healthcare ..... 52

**WE Prerequisite: Building-Level Water Metering ..... 54**

NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 55

**WE Credit: Outdoor Water Use Reduction ..... 56**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 56

**WE Credit: Indoor Water Use Reduction ..... 58**
NC, CS, Schools, NC-Retail, Data Centers, Warehouses & Distribution Centers, NC-Hospitality, Healthcare ..... 58

**WE Credit: Cooling Tower and Process Water Use ..... 61**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare, ..... 61

**WE Credit: Water Metering ..... 64**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 64

**ENERGY AND ATMOSPHERE ..... 66**

**EA Prerequisite: Fundamental Commissioning and Verification ..... 66**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 66

**EA Prerequisite: Minimum Energy Performance ..... 68**
NC, CS, Schools, Retail, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 68

**EA Prerequisite: Building-Level Energy Metering ..... 70**
NC, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare 70
CS ..... 70

**EA Prerequisite: Fundamental Refrigerant Management ..... 71**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 71

**EA Credit: Enhanced Commissioning ..... 72**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 72

**EA Credit: Optimize Energy Performance ..... 75**
NC, CS, Schools, Retail, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 75
Data Centers only ..... 77

**EA Credit: Advanced Energy Metering ..... 81**
NC, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare 81
CS ..... 81

**EA Credit: Grid Harmonization ..... 83**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 83

**EA Credit: Renewable Energy ..... 85**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 85

**EA Credit: Enhanced Refrigerant Management ..... 88**

NC, CS, Schools, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ....88
Retail NC ....89

## MATERIALS AND RESOURCES (MR) ....92

**MR Prerequisite: Storage and Collection of Recyclables ....92**
NC, CS, Schools, Data Centers, Warehouses & Distribution Centers, Hospitality NC, Healthcare92
Retail NC ....92

**MR Prerequisite: Construction and Demolition Waste Management Planning ....93**
NC, CS, Schools, Retail NC, Data Centers, Warehouses & Distribution Centers, Hospitality NC, Healthcare ....93

**MR Prerequisite: PBT Source Reduction—Mercury ....94**
Healthcare ....94

**MR Credit: Building Life-Cycle Impact Reduction....96**
NC, CS, Schools, Retail NC, Data Centers, Warehouses & Distribution Centers, Hospitality NC, Healthcare ....96

**MR Credit: Building Product Disclosure and Optimization—Environmental Product Declarations ....99**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ....99

**MR Credit: Building Product Disclosure and Optimization – Sourcing of Raw Materials ....102**
NC, CS, Schools, Retail NC, Data Centers, Warehouses & Distribution Centers, Hospitality NC, Healthcare ....102

**MR Credit: Building Product Disclosure and Optimization – Material Ingredients ....103**
NC, CS, Schools, Retail NC, Data Centers, Warehouses & Distribution Centers, Hospitality NC, Healthcare ....103

**MR Credit: PBT Source Reduction—Mercury....107**
Healthcare ....107

**MR Credit: PBT Source Reduction—Lead, Cadmium, and Copper....109**
Healthcare....109

**MR Credit: Furniture and Medical Furnishings ....110**
Healthcare ....110

**MR Credit: Design for Flexibility ....113**
Healthcare ....113

**MR Credit: Construction and Demolition Waste Management ....114**
NC, CS, Schools, Retail NC, Data Centers, Warehouses & Distribution Centers, Hospitality NC, Healthcare ....114

## INDOOR ENVIRONMENTAL QUALITY (EQ) .... 116

**EQ Prerequisite: Minimum Indoor Air Quality Performance ....116**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality ....116
Healthcare ....118

**EQ Prerequisite: Environmental Tobacco Smoke Control....119**

NC, CS, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 119
Schools ..... 120

**EQ Prerequisite: Minimum Acoustic Performance ..... 121**
Schools ..... 121

**EQ Credit: Enhanced Indoor Air Quality Strategies ..... 123**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 123

**EQ Credit: Low-Emitting Materials ..... 126**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 126

**EQ Credit: Construction Indoor Air Quality Management Plan ..... 131**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality ..... 131
Healthcare ..... 131

**EQ Credit: Indoor Air Quality Assessment ..... 133**
NC, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare 133

**EQ Credit: Thermal Comfort ..... 136**
NC, Schools, Retail, Data Centers, Hospitality, Healthcare ..... 136
Warehouses & Distribution Centers ..... 136
NC, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality ..... 137
Healthcare ..... 137

**EQ Credit: Interior Lighting ..... 138**
NC, Schools, Data Centers, Warehouses & Distribution Centers, Hospitality ..... 138
Retail NC ..... 139
Healthcare ..... 139

**EQ Credit: Daylight ..... 140**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 140

**EQ Credit: Quality Views ..... 144**
NC, CS, Schools, Retail, Data Centers, Hospitality ..... 144
Warehouses & Distribution Centers ..... 144
Healthcare ..... 144

**EQ Credit: Acoustic Performance ..... 146**
NC, Data Centers, Warehouses & Distribution Centers, Hospitality ..... 146
Schools ..... 148
Healthcare ..... 148

## INNOVATION (IN) ..... 150

**IN Credit: Innovation ..... 150**
NC, CS, Schools, Retail, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 150

**IN Credit: LEED Accredited Professional ..... 152**
NC, CS, Schools, Retail NC, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 152

**REGIONAL PRIORITY (RP) ..... 153**

NC, CS, Schools, Retail NC, Data Centers, Warehouses & Distribution Centers, Hospitality, Healthcare ..... 153

**APPENDICES ..... 154**

**Appendix 1. Use Types and Categories ..... 154**

**Appendix 2. Default Occupancy Counts ..... 155**

**Appendix 3. Retail Process Load Baselines ..... 156**

# INTEGRATIVE PROCESS (IP)

## PREREQUISITE: INTEGRATIVE PROJECT PLANNING AND DESIGN
**Required**

This prerequisite applies to:
- Healthcare

**Intent**

Maximize opportunities for integrated, cost-effective adoption of green design and construction strategies, emphasizing human health as a fundamental evaluative criterion for building design, construction and operational strategies. Utilize innovative approaches and techniques for green design and construction.

**Requirements**

**HEALTHCARE**

Use cross-discipline design and decision making, beginning in the programming and pre-design phase. At a minimum, ensure the following process:

**Owner's Project Requirements Document.** Prepare an Owner's Project Requirements (OPR) document. Develop a health mission statement and incorporate it in the OPR. The health mission statement must address "triple bottom line" values—economic, environmental and social. Include goals and strategies to safeguard the health of building occupants, the local community and the global environment, while creating a high-performance healing environment for the building's patients, caregivers and staff.

**Preliminary Rating Goals.** As early as practical and preferably before schematic design, conduct a preliminary LEED meeting with a minimum of four key project team members and the owner or owner's representative. As part of the meeting, create a LEED® action plan that, at a minimum:
- Determines the LEED certification level to pursue (Certified, Silver, Gold, or Platinum);
- Selects the LEED credits to meet the targeted certification level; and
- Identifies the responsible parties to ensure the LEED requirements for each prerequisite and selected credit are met.

**Integrated Project Team.** Assemble an integrated project team and include as many of the following professionals as feasible (minimum of four), in addition to the owner or owner's representative.
- Owner's capital budget manager
- Architect or building designer
- Mechanical engineer
- Structural engineer
- Energy modeler
- Equipment planner
- Acoustical consultant
- Telecommunications designer
- Controls designer
- Food Service Consultant
- Infection Control Staff
- Building science or performance testing agents
- Green building or sustainable design consultant
- Facility green teams
- Physician and nursing teams
- Facility managers
- Environmental services staff
- Functional and space programmers
- Commissioning agent
- Community representatives
- Civil engineer
- Landscape architect
- Ecologist
- Land planner
- Construction manager or general contractor
- Life cycle cost analyst; construction cost estimator
- Lighting Designer
- Other disciplines appropriate to the specific project type

**Design Charrette.** As early as practical and preferably before schematic design, conduct a minimum four-hour, integrated design charrette with the project team as defined above. The goal is to optimize the integration of green strategies across all aspects of building design, construction and operations, drawing on the expertise of all participants.

# CREDIT: INTEGRATIVE PROCESS

BD+C
**1 point**
This credit applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

## Intent

To support high-performance, cost-effective project outcomes through an early analysis of the interrelationships among systems.

## Requirements

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Beginning in pre-design and continuing throughout the design phases, identify and use opportunities to achieve synergies across disciplines and building systems. Use the analyses described below to inform the owner's project requirements (OPR), basis of design (BOD), design documents, and construction documents.

**Discovery:**

Energy-Related Systems

Establish an energy performance target no later than the schematic design phase. The target must be established using one of the following metrics:

- kBtu per square foot-year (kWh per square meter-year) of site energy use
- KBtu per square foot-year (kWh per square meter-year) of source energy use
- pounds per square foot-year (Kg per square meter-year) of greenhouse gas emissions
- energy cost per square foot-year (cost per square meter-year)

Perform a preliminary "simple box" energy modeling analysis before the completion of schematic design that explores how to reduce energy loads in the building and accomplish related sustainability goals by questioning default assumptions. Assess strategies associated with each of the following, as applicable:

- *Site conditions.* Assess shading, exterior lighting, hardscape, landscaping, and adjacent site conditions.
- *Massing and orientation.* Assess how massing and orientation affect HVAC sizing, energy consumption, lighting, and renewable energy opportunities.
- *Basic envelope attributes.* Assess insulation values, window-to-wall ratios, glazing characteristics, shading, and window operability.

- *Lighting levels.* Assess interior surface reflectance values and lighting levels in occupied spaces.
- *Thermal comfort ranges.* Assess thermal comfort range options.
- *Plug and process load needs.* Assess reducing plug and process loads through programmatic solutions (e.g., equipment and purchasing policies, layout options).
- *Programmatic and operational parameters.* Assess multifunctioning spaces, operating schedules, space allotment per person, teleworking, reduction of building area, and anticipated operations and maintenance.

AND

Water-Related Systems

Perform a preliminary water budget analysis before the completion of schematic design that explores how to reduce potable water loads in the building, reduce the burden on municipal supply or wastewater treatment systems, and accomplish related sustainability goals. Assess and estimate the project's potential nonpotable water supply sources and water demand volumes, including the following, as applicable:

- *Indoor water demand.* Assess flow and flush fixture design case demand volumes, calculated in accordance with WE Prerequisite Indoor Water Use Reduction.
- *Outdoor water demand.* Assess landscape irrigation design case demand volume calculated in accordance with WE Credit Outdoor Water-Use Reduction.
- *Process water demand.* Assess kitchen, laundry, cooling tower, and other equipment demand volumes, as applicable.
- *Supply sources.* Assess all potential nonpotable water supply source volumes, such as on-site rainwater and graywater, municipally supplied nonpotable water, and HVAC equipment condensate. Analyze how nonpotable water supply sources can contribute to the water demand components listed above.

**Implementation:**

**Develop a Project Team Letter.** Provide a dated letter on the letterhead of the Integrative Process Facilitator that summarizes the team's integrative process approach and describes the difference that this integrative approach made in terms of improving project team interaction and project performance.

- Describe the approach developed by the project team for engaging a clearly defined and manageable integrative design process beginning in pre-design and continuing throughout the design phases.
- The letter must include a separate summary for each issue area analyzed by the project team, describing how the analysis informed the design and building form decisions in the project's OPR and BOD and the eventual design of the project. Describe the most important goals for each issue area and provide clear guidance on how to evaluate the project's impact on the selected goals.

The creation of this letter should be a team effort facilitated by the Integrative Process Facilitator. The letter must be signed by all principal project team members and made available to key stakeholders including, but not limited to the owner(s), facility manager(s), tenant(s), and community members. Describe how the letter was distributed to these stakeholders and/or made publicly available.

# LOCATION AND TRANSPORATION (LT)

## LT CREDIT: LEED FOR NEIGHBORHOOD DEVELOPMENT LOCATION

BD+C
**5–20 points**
This credit applies to

- New Construction (8–16 points)
- Core & Shell (8–20 points)
- Schools (8–15 points)
- Retail (8–16 points)
- Data Centers (8–16 points)
- Warehouses & Distribution Centers (8–16 points)
- Hospitality (8–16 points)
- Healthcare (5–9 points)

### Intent

To avoid development on inappropriate sites. To reduce vehicle distance traveled. To enhance livability and improve human health by encouraging daily physical activity.

### Requirements

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Locate the project within the boundary of a development certified under LEED for Neighborhood Development (Stage 2 or Stage 3 under the Pilot or v2009 rating systems, Certified Plan or Certified Built Project under the LEED v4 rating system).

Projects attempting this credit are not eligible to earn points under other Location and Transportation credits.

**Table 1. Points for LEED ND location.**

| Certification level | Points BD+C | Points BD+C (Core and Shell) | Points BD+C (Schools) | Points BD+C (Healthcare) |
|---|---|---|---|---|
| Certified | 8 | 8 | 8 | 5 |
| Silver | 10 | 12 | 10 | 6 |
| Gold | 12 | 16 | 12 | 7 |
| Platinum | 16 | 20 | 15 | 9 |

# LT CREDIT: SENSITIVE LAND PROTECTION

BD+C
**1-2 points**
This credit applies to

- New Construction (1 point)
- Core & Shell (2 points)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1 point)

**Intent**

To avoid the development of environmentally sensitive lands and reduce the environmental impact from the location of a building on a site.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

**Option 1. Previously Developed Land**
Locate the development footprint on land that has been *previously developed*.

OR

**Option 2. Avoidance of Sensitive Land**
Locate the development footprint on land that does not meet the following criteria for sensitive land:

- *Prime farmland.* Prime farmland, unique farmland, or farmland of statewide or local importance as defined by the U.S. Code of Federal Regulations, Title 7, Volume 6, Parts 400 to 699, Section 657.5 (or local equivalent for projects outside the U.S.) and identified in a state Natural Resources Conservation Service soil survey (or local equivalent for projects outside the U.S.).
- *Floodplains*. A flood hazard area shown on a legally adopted flood hazard map or otherwise legally designated by the local jurisdiction or the state. For projects in places without legally adopted flood hazard maps or legal designations, locate on a site that is entirely outside any floodplain subject to a 1% or greater chance of flooding in any given year.
- *Habitat*. Land identified as habitat for the following:
    - species listed as threatened or endangered under the U.S. Endangered Species Act or the state's endangered species act, or
    - species or ecological communities classified by NatureServe as GH (possibly extinct), G1 (critically imperiled), or G2 (imperiled), or
    - species listed as threatened or endangered specifies under local equivalent standards (for projects outside the U.S.) that are not covered by NatureServe data.
- *Water bodies.* Areas on or within 100 feet (30 meters) of a *water body*, except for minor improvements.
- *Wetlands*. Areas on or within 50 feet (15 meters) of a *wetland*, except for minor improvements.

# LT CREDIT: HIGH-PRIORITY SITE AND EQUITABLE DEVELOPMENT

BD+C
**2-3 points**
This credit applies to

- New Construction (1-2 points)
- Core & Shell (2-3 points)
- Schools (1-2 points)
- Retail (1-2 points)
- Data Centers (1-2 points)
- Warehouses & Distribution Centers (1-2 points)
- Hospitality (1-2 points)
- Healthcare (1-2 points)

## Intent

To build the economic and social vitality of communities, encourage project location in areas with development constraints and promote the ecological and community health of the surrounding area.

## Requirements

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

**BD+C**
**Option 1. High Priority (1 point BD+C except Core and Shell, 2 points Core and Shell)**

**Path 1. Economically Disadvantaged Community Location (1 point BD+C except Core and Shell, 2 points Core and Shell)**
Locate within one of the following areas

- Census tract* in which average household income is at or below 80% AMI
- Census tract in which at least 20% of population is at or below poverty rate of state, provincial, or other regional jurisdiction
- Census tract in which unemployment is at least 150% of the state, provincial, or other regional jurisdiction.

*or local equivalent government-defined municipal tract for projects outside the U.S.

OR

**Path 2. Brownfield Remediation (1 point BD+C except Core and Shell, 2 points Core and Shell)**

Locate on a *brownfield* where soil or groundwater contamination has been identified, and where the local, state, or national authority (whichever has jurisdiction) requires its remediation. In cases of voluntary remediation by the project team, provide confirmation by the local, state, or national authority (whichever has jurisdiction) to verify that the site is a brownfield. Perform remediation to the satisfaction of the relevant authority.

AND/OR

**Option 2. Equitable Development (1 point BD+C except Core and Shell, 2 points Core and Shell)**

**Path 1. Equity & Community Benefits (1 point BD+C except Core and Shell, 2 points Core and Shell)**

Develop and implement and equity plan.

OR

**Path 2. Affordable Housing in Residential or Mixed-Use Projects (1 point)**

Include a proportion of new rental and/or for-sale dwelling units priced for households earning less than the area median income (AMI). Rental units must be maintained at affordable levels for a minimum of 15 years. Existing dwelling units are exempt from requirement calculations. Meet or exceed the minimum thresholds in Table 1. Projects must meet or exceed the minimum percentage of units mandated through inclusionary zoning by their local jurisdictions.

Table 1. Minimum affordable units

| | |
|---|---|
| Rental Dwelling Units | 10 percent of total rental units (or at least one unit) priced up to 60% AMI |
| For-Sale Dwelling Units | 5 percent of total for-sale units (or at least one unit) priced up to 80% AMI |

# LT CREDIT: SURROUNDING DENSITY AND DIVERSE USES

BD+C
**1–6 points**
This credit applies to

- New Construction (1–5 points)
- Core & Shell (1–6 points)
- Schools (1–5 points)
- Retail (1–5 points)
- Data Centers (1–5 points)
- Warehouses & Distribution Centers (1–5 points)
- Hospitality (1–5 points)
- Healthcare (1 point)

## Intent

To conserve land and protect farmland and wildlife habitat by encouraging development in areas with existing infrastructure. To promote walkability, and transportation efficiency and reduce vehicle distance traveled. To improve public health by encouraging daily physical activity.

## Requirements

**NC, CS, SCHOOLS, RETAIL, HOSPITALITY**

**Option 1. Surrounding Density (2–3 points BD+C except Core and Shell, 2-4 points Core and Shell)**

Locate on a site whose surrounding existing density within a ¼-mile (400-meter) offset of the project boundary meets the values in Table 1. Use either the "separate residential and nonresidential densities" or the "combined density" values.

**Table 1a. Points for average density within 1/4 mile of project (IP units)**

| Combined density | Separate residential and nonresidential densities | | Points BD+C (except Core and Shell) | Points BD+C (Core and Shell) |
|---|---|---|---|---|
| Square feet per acre of buildable land | Residential density (DU/acre) | Nonresidential density (FAR) | | |
| 22,000 | 7 | 0.5 | 2 | 2 |
| 35,000 | 12 | 0.8 | 3 | 4 |

**Table 1b. Points for average density within 400 meters of project (SI units)**

| Combined density | Separate residential and nonresidential densities | | Points BD+C (except Core and Shell) | Points BD+C (Core and Shell) |
|---|---|---|---|---|
| Square meters per hectare of buildable land | Residential density (DU/hectare) | Nonresidential density (FAR) | | |

| | | | | |
|---|---|---|---|---|
| 5,050 | 17.5 | 0.5 | 2 | 2 |
| 8,035 | 30 | 0.8 | 3 | 4 |

DU = dwelling unit; FAR = floor-area ratio.

Schools only
Physical education spaces that are part of the project site, such as playing fields and associated buildings used during sporting events only (e.g., concession stands) and playgrounds with play equipment, are excluded from the development density calculations.

AND/OR

**Option 2. Diverse Uses (1–2 points)**

Construct or renovate a building or a space within a building such that the building's main entrance is within a ½-mile (800-meter) walking distance from the following number of uses (see Appendix 1), as listed below.

**Table 1. Points for proximity to uses**

| Uses | Points |
|---|---|
| 4–7 | 1 |
| ≥ 8 | 2 |

The following restrictions apply.

- A use counts as only one type (e.g., a retail store may be counted only once even if it sells products in several categories).
- No more than two uses in each use type may be counted (e.g. if five restaurants are within walking distance, only two may be counted).
- The counted uses must represent at least three of the five categories, exclusive of the building's primary use.

**DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS**

**Option 1. Development and Adjacency (2–3 points)**

Construct or renovate the project on a previously developed site that was used for industrial or commercial purposes. (2 points).

OR

Construct or renovate the project on a site that is both a previously developed and an adjacent site. The adjacent sites must be currently used for industrial or commercial purposes (3 points).

AND/OR

**Option 2. Transportation Resources (1–2 points)**

Construct or renovate the project on a site that has two or three (1 point) or four (2 points) of the following

transportation resources:

- The site is within a 10-mile (16 kilometer) driving distance of a main logistics hub, defined as an airport, seaport, *intermodal facility*, or *freight village* with intermodal transportation.
- The site is within a 1-mile (1600-meter) driving distance of an on-off ramp to a *highway.*
- The site is within a 1-mile (1600-meter) driving distance of an access point to an active freight rail line.
- The site is served by an active freight rail spur.

In all cases, a planned transportation resource must be sited, funded, and under construction by the date of the certificate of occupancy and complete within 24 months of that date.

**HEALTHCARE**

**Option 1. Surrounding Density (1 point)**

Locate on a site whose surrounding existing density within a ¼-mile (400-meter) offset of the project boundary is:

1. At least 7 dwelling units per acre (17.5 DU per hectare) with a 0.5 floor-area ratio. The counted density must be *existing* density, not zoned density, or
2. At least 22,000 square feet per acre (5 050 square meters per hectare) of buildable land.

For previously developed existing rural healthcare campus sites, achieve a minimum development density of 30,000 square feet per acre (6890 square meters per hectare).

OR

**Option 2. Diverse Uses (1 point)**

Construct or renovate a building on a site such that the building's main entrance is within a ½-mile (800-meter) walking distance of the main entrance of at least seven operational and publicly accessible uses (listed in Appendix 1).

The following restrictions apply.

- A use may be counted as only one type (e.g., a retail store may be counted only once even if it sells products in several categories).
- No more than two uses in each use type may be counted (e.g., if five restaurants are within walking distance, only two may be counted).
- The counted uses must represent at least three of the five categories, exclusive of the building's primary use.

# LT CREDIT: ACCESS TO QUALITY TRANSIT

BD+C
**1–6 points**
This credit applies to

- New Construction (1–5 points)
- Core & Shell (1–6 points)
- Schools (1–4 points)
- Data Centers (1–5 points)
- Warehouses & Distribution Centers (1–5 points)
- Hospitality (1–5 points)
- Retail (1–5 points)
- Healthcare (1–2 points)

## Intent

To encourage development in locations shown to have multimodal transportation choices or otherwise reduced motor vehicle use, thereby reducing greenhouse gas emissions, air pollution, and other environmental and public health harms associated with motor vehicle use.

## Requirements

**NC, CS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, RETAIL**

Locate any *functional entry* of the project within a ¼-mile (400-meter) *walking distance* of existing or planned bus, *streetcar*, or *informal transit stops,* or within a ½-mile (800-meter) walking distance of existing or planned *bus rapid transit* stops, passenger rail stations (i.e. light, heavy, or commuter rail) or commuter ferry terminals. The transit service at those stops and stations in aggregate must meet the minimums listed in Table 1. Planned stops and stations may count if they are sited, funded, and under construction by the date of the certificate of occupancy and are complete within 24 months of that date.

Both weekday and weekend trip minimums must be met.

- For each qualifying transit route, only trips in one direction are counted towards the threshold.
- For weekend trips, only trips on the day with the higher number of trips are counted towards the threshold.
- If a qualifying transit route has multiple stops within the required walking distance, only trips from one stop are counted towards the threshold.
- Privately-run shuttles are only acceptable if the service is also made available to the public.

**Table 1. Minimum daily transit service**

| Weekday trips | Weekend trips | Points BD+C (except Core and Shell) | Points BD+C (Core and shell |
|---|---|---|---|
| 72 | 30 | 1 | 1 |
| 100 | 70 | 2 | 2 |
| 144 | 108 | 3 | 3 |
| 250 | 160 | 4 | 4 |
| 360 | 216 | 5 | 6 |

If *existing* transit service is temporarily rerouted outside the required distances for less than two years, the project may meet the requirements, provided the local transit agency has committed to restoring the routes with service at or above the prior level.

**SCHOOLS**

**Option 1. Transit-Served Location (1–4 points)**

Locate any *functional entry* of the project within a ¼-mile (400-meter) *walking distance* of existing or planned bus, *streetcar*, or *informal transit* stops, or within a ½-mile (800-meter) walking distance of existing or planned *bus rapid transit* stops, passenger rail stations, or commuter ferry terminals. The transit service at those stops and stations must meet the minimums listed in Tables 1 and 2. Planned stops and stations may count if they are sited, funded, and under construction by the date of the certificate of occupancy and are complete within 24 months of that date.

- For each qualifying transit route, only trips in one direction are counted towards the threshold.
- If a qualifying transit route has multiple stops within the required walking distance, only trips from one stop are counted towards the threshold.

**Table 1. Minimum daily transit service**

| Weekday trips | Points |
|---|---|
| 72 | 1 |
| 144 | 2 |
| 250 | 3 |
| 360 | 4 |

Projects served by two or more transit routes such that no one route provides more than 60% of the prescribed levels may earn one additional point, up to the maximum number of points.

If existing transit service is temporarily rerouted outside the required distances for less than two years, the project may meet the requirements, provided the local transit agency has committed to restoring the routes with service at or above the prior level.

OR

**Option 2. Pedestrian Access (1–4 points)**
Show that the project has an *attendance boundary* such that the specified percentages of dwelling units are within no more than a 3/4-mile (1200-meter) walking distance (for grades 8 and below, or ages 14 and below), and 1 1/2-mile (2400-meter) walking distance (for grades 9 and above or ages 15 and above) of a functional entry of a school building. Points are awarded according to Table 3.

**Table 2. Points for dwelling units within walking distance**

| Percentage of dwelling units in attendance boundary | Points |
|---|---|
| 50% | 1 |
| 60% | 2 |
| 70% or more | 4 |

In addition, locate the project on a site that allows pedestrian access to the site from all residential areas in the attendance boundary.

**HEALTHCARE**

Locate any *functional entry* of the project within a ¼-mile (400-meter) *walking distance* of existing or planned bus, *streetcar*, or *informal transit* stops, or within a ½-mile (800-meter) walking distance of existing or planned *bus rapid transit* stops, passenger rail stations or commuter ferry terminals. The transit service at those stops and stations in aggregate must meet the minimums listed in Tables 1 and 2. Planned stops and stations may count if they are sited, funded, and under construction by the date of the certificate of occupancy and are complete within 24 months of that date.

Both weekday and weekend trip minimums must be met.

- 
- For each qualifying transit route, only trips in one direction are counted towards the threshold.
- If a qualifying transit route has multiple stops within the required walking distance, only trips from one stop are counted towards the threshold.

**Table 1. Minimum daily transit service**

| Weekday trips | Weekend trips | Points |
|---|---|---|
| 72 | 30 | 1 |
| 144 | 108 | 2 |

Projects served by two or more transit routes such that no one route provides more than 60% of the prescribed levels may earn one additional point, up to the maximum number of points.

If *existing* transit service is temporarily rerouted outside the required distances for less than two years, the project may meet the requirements, provided the local transit agency has committed to restoring the routes with service at or above the prior level.

## LT CREDIT: BICYCLE FACILITIES

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Retail (1 point)
- Healthcare (1 point)

### Intent

To promote bicycling and transportation efficiency and reduce vehicle distance traveled. To improve public health by encouraging utilitarian and recreational physical activity.

### Requirements

**NEW CONSTRUCTION, CORE AND SHELL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, RETAIL, HEALTHCARE**

**Bicycle Network**

Design or locate the *project* such that a *functional entry* or bicycle storage is within a 200-yard (180-meter) *walking distance* or *bicycling distance* from a *bicycle network* that connects to at least one of the following:

- at least 10 diverse uses (see Appendix 1);
- a school or *employment center,* if the project total floor area is 50% or more residential; or
- *a bus rapid transit* stop, passenger rail station, or ferry terminal.

All destinations must be within a 3-mile (4800-meter) bicycling distance of the project boundary.

Planned bicycle trails or lanes may be counted if they are fully funded by the date of the certificate of occupancy and are scheduled for completion within one year of that date.

**Bicycle Storage and Shower Rooms**

**Case 1. Commercial or Institutional Projects**
Provide *short-term bicycle storage* for at least 2.5% of all peak visitors, but no fewer than four storage spaces per building.

Provide *long-term bicycle storage* for at least 5% of all regular building occupants, but no fewer than four storage spaces per building in addition to the short-term bicycle storage spaces.

Provide at least one on-site shower with changing facility for the first 100 regular building occupants and one additional shower for every 150 regular building occupants thereafter.

**Case 2. Residential Projects**
Provide *short-term bicycle storage* for at least 2.5% of all peak visitors but no fewer than four storage spaces per building.

Provide *long-term bicycle storage* for at least 15% of all regular building occupants, but no less than one storage space per three residential units.

**Case 3. Mixed-Use Projects**
Meet the Case 1 and Case 2 storage requirements for the nonresidential and residential portions of the project, respectively.

**Large-Occupancy Projects Only:**

The following guidance should be applied when determining the number of showers needed for projects with a large number of occupants:

**NEW CONSTRUCTION, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, RETAIL, HEALTHCARE**

Provide at least one on-site shower with changing facility for the first 100 regular building occupants and one additional shower for every 150 regular building occupants thereafter, up to 999 regular building occupants.

- one additional shower for every 500 regular building occupants, for the additional 1,000 – 4,999 regular building occupants
- one additional shower for every 1,000 regular building occupants, for the additional 5,000 + regular building occupants

**CORE AND SHELL**

Provide at least one on-site shower with changing facility for the first 100 regular building occupants and one additional shower for every 150 regular building occupants thereafter, up to 999 regular building occupants.

- one additional shower for every 750 regular building occupants, for the additional 1,000 – 4,999 regular building occupants
- one additional shower for every 1,500 regular building occupants, for the additional 5,000 + regular building occupants

**For All Projects**
*Short-term bicycle storage* must be within 200 feet (60meters) walking distance of any main entrance. *Long-term bicycle storage* must be within 300 feet (90 meters) walking distance of any *functional entry*. Vertical distances travelled by elevator are exempt from counting towards the walking distance requirements.

Bicycle storage capacity may not be double-counted: storage that is fully allocated to the occupants of non-project facilities cannot also serve project occupants. Indoor storage is acceptable as long as it meets the walking distance requirements. On-site bicycle sharing stations within the project boundary may count for 50% of the long-term and short-term bicycle storage space. Zero lot line projects may count publicly available bicycle parking towards their short-term storage requirements if it meets the maximum allowable walking distance.

Provide at least one on-site shower with changing facility for the first 100 regular building occupants and one additional shower for every 150 regular building occupants thereafter. Exclude patients and K-12 students from the regular building occupant count.

**SCHOOLS**

**Bicycle Network**
Design or locate the *project* such that a *functional entry* and/or bicycle storage is within a 200-yard (180-meter) *walking distance* or *bicycling distance* of a *bicycle network* that connects to either of the following:

- a *bus rapid transit* stop or passenger rail station or ferry terminal; or
- 50% of dwelling units within the school's attendance boundary.

The stops/stations or dwelling units must be within no more than a 1 1/2-mile (2400-meter) biking distance (for grades 8 and below, or ages 14 and below), and 3-mile (4800-meter) biking distance (for grades 9 and above or ages 15 and above).
Provide dedicated bicycle lanes that extend at least to the end of the school property with no barriers (e.g., fences) on school property.

**Bicycle Storage and Shower Rooms**

Meet storage and shower requirements for all projects and provide *long-term bicycle storage* for at least 5% of all regular building occupants (excluding students grade 3 and younger), but no fewer than four storage spaces per building.

**RETAIL**

**Bicycle Network**
Meet Bicycle Network requirements for all projects.

**Bicycle Storage and Shower Rooms**

Meet storage distance and shower requirements for all projects and provide at least two *short-term bicycle storage* spaces for every 5,000 square feet (465 square meters), but no fewer than two storage spaces per building.

Provide *long-term bicycle storage* for at least 5% of regular building occupants, but no fewer than two storage spaces per building in addition to the short-term bicycle storage spaces.

Provide a bicycle maintenance program for employees or bicycle route assistance for employees and customers. Route assistance must be provided in a manner easily accessible to both employees and customers.

## HEALTHCARE

**Bicycle Network**
Meet Bicycle Network requirements for all projects.

**Bicycle Storage and Shower Rooms**
Meet storage and shower requirements for all projects and provide *short-term bicycle storage* for at least 2.5% of all peak visitors, but no fewer than four storage spaces per building.

Provide *long-term bicycle storage* for at least 5% of regular building occupants (excluding patients), but no fewer than four storage spaces per building in addition to the short-term bicycle storage spaces.

# LT CREDIT: REDUCED PARKING FOOTPRINT

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Retail (1 point)
- Healthcare (1 point)

## Intent

To minimize the environmental harms associated with parking facilities, including automobile dependence, land consumption, and rainwater runoff.

## Requirements

**NC, CS, RETAIL, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

**Option 1. No Off-Street Parking (1 point)**

Do not provide off-street parking.

OR

**Option 2. Reduce Parking (1 point)**

Do not exceed the minimum local code requirements for parking capacity.

Provide parking capacity that is a 30% reduction below the base ratios recommended by the Parking Consultants Council, as shown in the Institute of Transportation Engineers' Transportation Planning Handbook, 4th edition, Table 11-12.

OR

**Option 3. Carshare (1 point)**

Provide dedicated parking for carshare vehicles. Provide at least one vehicle parking space for every 100 occupants, rounded up. If the project has fewer than 100 occupants, provide one carshare vehicle parking space.

Existing carshare vehicles located in nearby on- or off-street parking areas do not contribute to credit achievement.

OR

**Option 4. Unbundling Parking (1 point)**

Sell parking separately from all property sales or leases.

Implement a daily parking fee at a cost equal to or greater than the daily cost of municipal public transit.

**For All Projects**
The credit calculations must include all existing and new off-street parking spaces that are leased or owned by the project, including parking that is outside the project boundary but is used by the project. On-street parking in public rights-of-way is excluded from these calculations.

## LT CREDIT: ELECTRIC VEHICLES

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Data Centers (1 point)
- Hospitality (1 point)
- Retail (1 point)
- Healthcare (1 point)
- Schools (1 point)
- Warehouses & Distribution Centers (1 point)

### Intent

To reduce pollution by promoting alternatives to conventionally fueled automobiles.

### Requirements

**NC, CS, DATA CENTERS, HOSPITALITY, RETAIL, HEALTHCARE, SCHOOLS**

Provide charging infrastructure for electric vehicles for on-site parking.

**Option 1. Electric Vehicle Charging (1 point)**
Install *electrical vehicle supply equipment (EVSE)* in 2% of all parking spaces used by the project or at least two spaces, whichever is greater. Clearly identify and reserve these spaces for the sole use by plug-in electric vehicles.

The EVSE must:

- Provide a Level 2 charging capacity (208 – 240 volts) or greater.
- Comply with the relevant regional or local standard for electrical connectors, such as SAE Surface Vehicle Recommended Practice J1772, SAE Electric Vehicle Conductive Charge Coupler or IEC 62196 of the International Electrotechnical Commission for projects outside the U.S.
- Be capable of responding to time-of-use market signals (e.g. price). Projects pursuing EA credit Grid Harmonization should incorporate EVSE into any demand response program or load flexibility and management strategies.

OR

**Option 2. Electric Vehicle Charging Infrastructure (1 point)**
Make 6% of parking spaces or at least 6 spaces, whichever is greater, *EV Ready*.

To be *EV Ready,* meet all of the following:

- Install listed raceway capable of accommodating a 208/240-volt dedicated branch circuit.
- The raceway shall not be less than trade size 1 (nominal 1-inch inside diameter).
- The raceway shall originate at the main service or subpanel and shall terminate into a listed cabinet, box or enclosure in close proximity to the proposed location of the EV space.
- The service panel and/or subpanel shall provide capacity to install a 40-ampere minimum dedicated branch circuit and space(s) reserved to permit installation of a branch circuit overcurrent protective device.

Multiple Panel Spaces required:

- When multiple charging spaces are required, raceway(s) is/are required to be installed at the time of construction.
- The raceway(s) shall originate at a service panel or subpanel(s) serving the area, and shall terminate in close proximity to the proposed location of the charging equipment into listed cabinet(s), box(es), enclosure(s) or equivalent.
- Construction documents shall indicate raceway termination point and proposed location of future EV spaces and EV chargers. Construction documents shall also provide information on amperage of future EVSE, raceway method(s), wiring schematics and electrical load calculations to verify electrical panel service capacity and electrical system, including any on-site distribution transformer(s), have sufficient capacity to simultaneously charge all EVs at all required EV spaces at full rated amperage of the EVSE
- Plan design shall be based upon a 40-ampere minimum branch circuit.
- Electrical calculations shall substantiate the design of the electrical system, to include the rating of equipment and any on-site distribution transformers and have sufficient capacity to simultaneously charge all required EVs at its full rated amperage.
- The service panel or subpanel(s) shall have sufficient capacity to accommodate the required number of dedicated branch circuit(s) for the future installation of the EVSE.

**SCHOOLS**

Meet Option 1 or Option 2 above.

OR

**Option 3: Electric buses or school-owned vehicles (1 point)**

Develop and implement a plan for acquiring at least 1 electric bus and/or for every other bus serving the school to meet the following emissions standards within seven years of the building certificate of occupancy:

- nitrogen oxide (NOx) emissions of 0.50 grams or less per brake horsepower-hour; and
- particulate matter emissions of 0.01 grams or less per brake horsepower-hour.

Emission standards must be met for each bus and not by an average of the entire fleet serving the school.

Develop and implement a plan for 50% of all other (non-bus) vehicles owned or leased to serve the school to be electric vehicles.

**WAREHOUSES & DISTRIBUTION CENTERS**

**Option 1. Electric Vehicle Charging (1 point)**

Provide an on-site fleet with at least one *yard tractor* that is powered by electricity and provide on-site charging for the vehicles.

OR

**Option 2. Reduced Truck Idling (1 point)**

Provide an electrical connection for at least 50% of all dock door locations to limit truck idling at the dock.

# SUSTAINABLE SITES (SS)

## SS PREREQUISITE: CONSTRUCTION ACTIVITY POLLUTION PREVENTION Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**
To reduce pollution from construction activities by controlling soil erosion, waterway sedimentation, and airborne dust.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Create and implement an erosion and sedimentation control plan for all construction activities associated with the project. The plan must conform to the erosion and sedimentation requirements of the 2017 U.S. Environmental Protection Agency (EPA) Construction General Permit (CGP) or local equivalent, whichever is more stringent. Projects must apply the CGP regardless of size. The plan must describe the measures implemented.

## SS PREREQUISITE: ENVIRONMENTAL SITE ASSESSMENT
## Required

BD+C
This prerequisite applies to

- Schools
- Healthcare

**Intent**
To protect the health of vulnerable populations by ensuring that the site is assessed for environmental contamination and that any environmental contamination has been remediated.

**Requirements**

**SCHOOLS, HEALTHCARE**

Conduct a Phase I Environmental Site Assessment as described in ASTM E1527-13 (or a local equivalent) to determine whether environmental contamination exists at the site. If contamination is suspected, conduct a Phase II Environmental Site Assessment as described in ASTM E1903–11 (or a local equivalent).

If a site is contaminated, remediate the site to meet local, state, or national environmental protection agency region residential (unrestricted) standards, whichever are most stringent.

## SS CREDIT: SITE ASSESSMENT

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1 point)

**Intent**
To assess site conditions before design to evaluate sustainable options and inform related decisions about site design.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Complete and document a site survey or assessment[1] that includes the following information:

*Topography.*

- Contour mapping
- Unique topographic features
- Slope stability risks

*Hydrology.*

- Special Flood Hazard Areas (SPFHA) as determined by FEMA's Flood Insurance Rate Map (FIRM) (or local equivalent for projects outside the U.S.)
- Delineated natural water bodies wetlands, lakes, streams, and shorelines (refer to U.S. EPA's Clean Water Act or local equivalent for projects outside the U.S.)
- Rainwater collection and reuse opportunities
- Impervious and pervious surfaces within the site boundary

*Climate.*

- Solar exposure and shading opportunities
- Heat island effect potential
- Seasonal sun angles
- Prevailing winds
- Average monthly precipitation and temperature ranges

*Vegetation.*

- Primary vegetation types
- Greenfield area
- Significant tree mapping
- Federal or state threatened or endangered species lists; for projects outside the U.S., International Union for Conservation of Nature (IUCN) Red List of Threatened Species

[1] Components adapted from the Sustainable Sites Initiative: Guidelines and Performance Benchmarks 2009, Prerequisite 2.1: Site Assessment.

- Invasive plant species listed by regional, state, or federal entities
- EPA Level III ecoregion description (or local equivalent)

*Soils.*

- Natural Resources Conservation Service soils delineation (or local equivalent for projects outside the U.S.)
- U.S. Natural Resources Conservation Service (or local equivalent for projects outside the United States) prime farmland, unique farmland, farmland of statewide importance, or farmland of local importance
- Healthy soils
- Previous development
- Disturbed soils

*Human use.*

- Views
- Adjacent transportation infrastructure, bicycle network, and bicycle storage
- Adjacent diverse uses
- Construction materials with existing recycle or reuse potential

*Human health effects.*

- Proximity of vulnerable populations
- Adjacent physical activity opportunities
- Proximity to major sources of air and water pollution

The survey or assessment should demonstrate the relationships between the site features and topics listed above and how these features influenced the project design; give the reasons for not addressing any of those topics.

## SS CREDIT: PROTECT OR RESTORE HABITAT

BD+C
**1–2 points**
This credit applies to

- New Construction (1–2 points)
- Core & Shell (1–2 points)
- Schools (1–2 points)
- Retail (1–2 points)
- Data Centers (1–2 points)
- Warehouses & Distribution Centers (1–2 points)
- Hospitality (1–2 points)
- Healthcare (1 point)

**Intent**
To conserve existing natural areas and restore damaged areas to provide habitat and promote biodiversity.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Preserve and protect from all development and construction activity 40% of the *greenfield* area on the site (if such areas exist).

AND

**Option 1. On-Site Restoration (2 points except Healthcare, 1 point Healthcare)**

Using native or adapted vegetation, restore 25% (including the building footprint) of all portions of the site identified as previously disturbed. Vegetated roof surfaces may be included if the plants are native or adapted and provide habitat.

**Soils**

Restore all soils on site that have been disturbed or replace all soils removed by current construction activities that will later serve as the final vegetated area.

- Restore or replace soils to a minimum depth of 12 inches (30.48 centimeters) or depth of root ball for larger plant materials.
- Provide a soils test of imported soils that includes recommended amendments. Incorporate test recommended amendments prior to planting.
- Imported soils must be reused for functions comparable to their original function.
- Imported soils may not include the following:
  - soils defined regionally by the Natural Resources Conservation Service web soil survey (or local equivalent for projects outside the U.S.) as *prime farmland*, unique farmland, or farmland of statewide or local importance; or
  - soils from other *greenfield* sites
  - sphagnum peat moss

**Vegetation**

Use only plant species that are appropriate for the project's EPA Level III ecoregion and that are suitable for site conditions, climate, and design intent. Both native and adapted vegetation may qualify. Native grasses may be used in conjunction with a variety (two or more) of native or adapted vegetation species.

- Use only plant species not currently listed as invasive on any federal or qualifying regional lists.
- Protect the root zone of trees found on site. Planting within the one foot (0.30 meter) radius per inch (2.54 centimeters) Diameter Breast Height (DBH) should be avoided.
- Conserve endangered species.

Schools only:
Dedicated athletic fields that are solely for athletic uses are exempted from the soil restoration criteria. These areas may not count toward the minimum required area.

OR
**Option 2. Financial Support (1 point)**
Provide financial support equivalent to at least $0.20 per square foot (US$2 per square meter) for the total site area (including the building footprint).

Financial support must be provided to a conservation land trust or accredited conservation organization within the same EPA Level III ecoregion or the project's state (or within 100 miles of the project [160 kilometers] for projects outside the U.S.). For U.S. projects, the land trust must be accredited by the Land Trust Alliance.

For projects outside of the U.S., the conservation land trust must either be a project supported by The Nature Conservancy or World Land Trust.

## SS CREDIT: OPEN SPACE

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1 point)

**Intent**
To create exterior open space that encourages interaction with the environment, social interaction, passive recreation, and physical activities.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Provide outdoor space greater than or equal to 30% of the total site area (including building footprint).

25% of the minimum 30% total outdoor space requirement must be planted with two or more types of vegetation or have *overhead vegetated canopy*.

The outdoor space must be physically accessible and be one or more of the following:

- a pedestrian-oriented paving or landscape area that accommodate outdoor social activities
- a recreation-oriented paving or landscape area that encourage physical activity;
- a landscape area with a two or more of vegetation types that provide opportunities for year-round visual interest;
- a garden space dedicated to community gardens or urban food production;
- preserved or created habitat that meets the criteria of SS Credit Protect or Restore Habitat and also includes elements of human interaction.

Extensive or intensive vegetated roofs that are physically accessible can be used toward the minimum 25% vegetation requirement, and qualifying roof-based physically accessible paving areas can be used toward credit compliance.

Wetlands or naturally designed ponds may count as open space if the side slope gradients average 1:4 (vertical:horizontal) or less and are vegetated.

## SS CREDIT: RAINWATER MANAGEMENT

BD+C
**1–3 points**
This credit applies to

- New Construction (1–3 points)
- Core & Shell (1–3 points)
- Schools (1–3 points)
- Retail (1–3 points)
- Data Centers (1–3 points)
- Warehouses & Distribution Centers (1–3 points)
- Hospitality (1–3 points)
- Healthcare (1-2 points)

**Intent**
To reduce runoff volume and improve water quality by replicating the natural hydrology and water balance of the site, based on historical conditions and undeveloped ecosystems in the region.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Treat run-off from pollutant-generating impervious surfaces (i.e. vehicle pavement, service courts, trash enclosures) using *low-impact development (LID)* practice.
**Percentile of Rainfall Events (1-3 points except Healthcare, 1-2 points Healthcare )**

In a manner best replicating *natural site hydrology* processes, retain (i.e. infiltrate, evapotranspirate, or collect and reuse) *on site* the runoff from the developed site for, at minimum, the 80th percentile of regional or local rainfall events using *low-impact development (LID*) and *green infrastructure (GI) practices*. GI and LID strategies can be either structural or non-structural. Points are awarded according to Table 1.

For all projects, the use of coal tar sealants shall be prohibited in any application exposed to stormwater, wash waters, condensates, irrigation water, snowmelt, or icemelt.

Examples of acceptable techniques include the following:

- planting rain gardens with *native* or *adapted* plant material (e.g. trees shrubs);
- installing a vegetated roof;
- using permeable paving, consisting of porous above-ground materials (e.g., open pavers, engineered products), a base layer designed to drain water away from the building, and (often) a 6-inch-deep (150 millimeters) subbase; and
- installing permanent infiltration or collection features (e.g., vegetated swale, rain garden, rainwater cistern) that can retain 100% of the runoff from at minimum, the 80th percentile of regional or local rainfall events.

A combination of LID approaches are recommended (but not required) as they are holistic measures which maximize benefits. In contrast to LID, conventional stormwater techniques include grey infrastructure, such as detention or retention ponds, pipes, and vaults. Conventional grey infrastructure devices may be accepted only if integrated within a holistic LID system (i.e. a combination of LID techniques).

Use daily rainfall data and the methodology in the U.S. Environmental Protection Agency (EPA) Technical Guidance on Implementing the Stormwater Runoff Requirements for Federal Projects under Section 438 of the Energy Independence and Security Act to determine the percentile amount to be retained.

**Table 1. Points for percentile of rainfall retained**

| Percentile of Rainfall Retained | Points | Points Healthcare |
|---|---|---|
| 80th Percentile | 1 | 1 |
| 85th Percentile | 2 | 2 |
| 90th Percentile | 3 | - |

**Zero Lot Line projects only (1-3 points except Healthcare, 1-2 points Healthcare)**
The following requirement applies to zero lot line projects in urban areas with a minimum density of 1.5 FAR. Treat run-off from pollutant-generating impervious surfaces (i.e. vehicle pavement, service courts, trash enclosures) using low-impact development (LID) practice/green infrastructure (GI) (or a traditional stormwater treatment device if LID/GI is not feasible for lack of space). Any above-ground setback area must be designed and used as a pedestrian-oriented space (e.g. restaurant seating, outdoor displays, private vendors, or related public purpose).

In a manner best replicating natural site hydrology processes, retain on site the runoff from the developed site for, at minimum, the 70th percentile of regional or local rainfall events, using LID/GI. Points are awarded according to Table 2.

**Table 2. Points for percentile of rainfall retained on Zero Lot Line projects**

| Percentile of Rainfall Retained | Points | Points Healthcare |
|---|---|---|
| 70th Percentile | 1 | 1 |
| 75th Percentile | 2 | 2 |
| 80th Percentile | 3 | - |

If the Zero Lot Line project meets the credit requirements and achieves at least the minimum percentile threshold of rainfall retained, additional volume can be retained offsite so long as the LID/GI system is designed to accommodate use by the project.

## SS CREDIT: HEAT ISLAND REDUCTION

BD+C
**1–2 points**
This credit applies to

- New Construction (1-2 points)
- Core & Shell (1-2 points)
- Schools (1-2 points)
- Retail (1-2 points)
- Data Centers (1-2 points)
- Warehouses & Distribution Centers (1-2 points)
- Hospitality (1-2 points)

- Healthcare (1 point)

**Intent**
To minimize effects on microclimates and human and wildlife habitats by reducing heat islands.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Choose one of the following options:

**Option 1. Nonroof and Roof (2 points except Healthcare, 1 point Healthcare)**

Meet the following criterion:

$$\frac{\text{Area of Nonroof Measures}}{0.5} + \frac{\text{Area of High-Reflectance Roof}}{0.75} + \frac{\text{Area of Vegetated Roof}}{0.75} \geq \text{Total Site Paving Area} + \text{Total Roof Area}$$

Alternatively, an SRI and SR weighted average approach may be used to calculate compliance.

Use any combination of the following strategies.

***Nonroof Measures***
- Use the existing plant material or install plants that provide shade over paving areas (including playgrounds) on the site within 10 years of planting. Install vegetated planters. Plants must be in place at the time of occupancy permit and cannot include artificial turf.
- Provide shade with structures covered by energy generation systems, such as solar thermal collectors, photovoltaics, and wind turbines.
- Provide shade with architectural devices or structures. If the device or structure is a roof, it shall have an aged *solar reflectance (SR)* value of at least 0.28 as measured in accordance with ANSI/CRRC S100. If the device or structure is not a roof, or if aged solar reflectance information is not available, it shall have at installation an initial SR of at least 0.33 as measured in accordance with ANSI/CRRC S100.
- Provide shade with vegetated structures.
- Use paving materials with an initial *solar reflectance (SR)* value of at least 0.33.
- Use an *open-grid pavement system* (at least 50% unbound).

***High-Reflectance Roof***
Use roofing materials that have an aged SRI equal to or greater than the values in Table 1. If aged SRI is not available, the roofing material shall have an initial SRI equal to or greater than the values in Table 1.

**Table 1. Minimum solar reflectance index value, by roof slope**

| | **Slope** | **Initial SRI** | **Aged SRI** |
|---|---|---|---|
| Low-sloped roof | ≤ 2:12 | 82 | 64 |
| Steep-sloped roof | > 2:12 | 39 | 32 |

Roof area that consists of functional, usable spaces (such as helipads, recreation courts, and similar amenity areas) may meet the requirements of nonroof measures. Applicable roof area excludes roof area covered by mechanical equipment, solar energy panels, skylights, and any other appurtenances.

***Vegetated Roof***
Install a vegetated roof using native or adapted plant species.

OR

**Option 2. Parking under Cover (1 point)**
Place a minimum of 75% of *parking spaces under cover*. Any roof used to shade or cover parking must (1) have a three-year aged SRI of at least 32 (if three-year aged value information is not available, use materials with an initial SRI of at least 39 at installation), (2) be a vegetated roof, or (3) be covered by energy generation systems, such as solar thermal collectors, photovoltaics, and wind turbines.

The credit calculations must include all existing and new off-street parking spaces that are leased or owned by the project, including parking that is outside the project boundary but is used by the project. On-street parking in public rights-of-way is excluded from these calculations.

# SS CREDIT: LIGHT POLLUTION REDUCTION

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1 point)

**Intent**
To increase night sky access, improve nighttime visibility, and reduce the consequences of development for wildlife and people.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Meet uplight and light trespass requirements, using either the backlight-uplight-glare (BUG) method (Option 1) or the calculation method (Option 2). Projects may use different options for uplight and light trespass.

Meet these requirements for all exterior luminaires located inside the project boundary (except those listed under "Exemptions"), based on the following:

- the photometric characteristics of each luminaire when mounted in the same orientation and tilt as specified in the project design; and
- the lighting zone of the project property (at the time construction begins). Classify the project under one lighting zone using the lighting zones definitions provided in the Illuminating Engineering Society and International Dark Sky Association (IES/IDA) Model Lighting Ordinance (MLO) User Guide.

Additionally, meet the internally illuminated signage requirement.

**Uplight**

**Option 1. BUG Rating Method**

Do not exceed the following luminaire uplight ratings, based on the specific light source installed in the luminaire, as defined in IES TM-15-11, Addendum A.

**Table 1. Maximum uplight ratings for luminaires**

| MLO lighting zone | Luminaire uplight rating |
|---|---|
| LZ0 | U0 |
| LZ1 | U1 |
| LZ2 | U2 |

| | |
|---|---|
| LZ3 | U3 |
| LZ4 | U4 |

OR

**Option 2. Calculation Method**

Do not exceed the following percentages of total lumens emitted above horizontal.

**Table 2. Maximum percentage of total lumens emitted above horizontal, by lighting zone**

| MLO lighting zone | Maximum allowed percentage of total luminaire lumens emitted above horizontal |
|---|---|
| LZ0 | 0% |
| LZ1 | 0% |
| LZ2 | 1.5% |
| LZ3 | 3% |
| LZ4 | 6% |

AND

**Light Trespass**

**Option 1. BUG Rating Method**

Do not exceed the following luminaire backlight and glare ratings (based on the specific light source installed in the luminaire), as defined in IES TM-15-11, Addendum A, based on the mounting location and distance from the lighting boundary.

**Table 3. Maximum backlight and glare ratings**

| | MLO lighting zone | | | | |
|---|---|---|---|---|---|
| Luminaire mounting | LZ0 | LZ1 | LZ2 | LZ3 | LZ4 |
| | Allowed backlight ratings | | | | |
| > 2 mounting heights from lighting boundary | B1 | B3 | B4 | B5 | B5 |
| 1 to 2 mounting heights from lighting boundary and properly oriented | B1 | B2 | B3 | B4 | B4 |
| 0.5 to 1 mounting height to lighting boundary and properly oriented | B0 | B1 | B2 | B3 | B3 |
| < 0.5 mounting height to lighting boundary and properly oriented | B0 | B0 | B0 | B1 | B2 |
| | Allowed glare ratings | | | | |
| Building-mounted > 2 mounting heights from any lighting boundary | G0 | G1 | G2 | G3 | G4 |

| | | | | | |
|---|---|---|---|---|---|
| Building-mounted 1–2 mounting heights from any lighting boundary | G0 | G0 | G1 | G1 | G2 |
| Building-mounted 0.5 to 1 mounting heights from any lighting boundary | G0 | G0 | G0 | G1 | G1 |
| Building-mounted < 0.5 mounting heights from any lighting boundary | G0 | G0 | G0 | G0 | G1 |
| All other luminaires | G0 | G1 | G2 | G3 | G4 |

The lighting boundary is located at the property lines of the property, or properties, that the LEED project occupies. The lighting boundary can be modified under the following conditions:

- When the property line abuts a public area that includes, but is not limited to, a walkway, bikeway, plaza, or parking lot, the lighting boundary may be moved to 5 feet (1.5 meters) beyond the property line.
- When the property line abuts a public street, alley, or transit corridor, the lighting boundary may be moved to the center line of that street, alley, or corridor.
- When there are additional properties owned by the same entity that are contiguous to the property, or properties, that the LEED project is within and have the same or higher MLO lighting zone designation as the LEED project, the lighting boundary may be expanded to include those properties.

Orient all luminaires less than two mounting heights from the lighting boundary such that the backlight points toward the nearest lighting boundary line. Building-mounted luminaires with the backlight oriented toward the building are exempt from the backlight rating requirement.

OR

**Option 2. Calculation Method**

Do not exceed the following vertical illuminances at the lighting boundary (use the definition of lighting boundary in Option 1). Calculation points may be no more than 5 feet (1.5 meters) apart. Vertical illuminances must be calculated on vertical planes running parallel to the lighting boundary, with the normal to each plane oriented toward the property and perpendicular to the lighting boundary, extending from grade level to 33 feet (10 meters) above the height of the highest luminaire.

**Table 4. Maximum vertical illuminance at lighting boundary, by lighting zone**

| MLO lighting zone | Vertical illuminance |
|---|---|
| LZ0 | 0.05 fc (0.5 lux) |
| LZ1 | 0.05 fc (0.5 lux) |
| LZ2 | 0.10 fc (1 lux) |
| LZ3 | 0.20 fc (2 lux) |
| LZ4 | 0.60 fc (6 lux) |

FC = footcandle.

AND

**Internally Illuminated Exterior Signage**
Do not exceed a luminance of 200 cd/$m^2$ (nits) during nighttime hours and 2000 cd/$m^2$ (nits) during daytime hours.

**Exemptions from Uplight and Light Trespass Requirements**

The following exterior lighting is exempt from the requirements, provided it is controlled separately from the nonexempt lighting:

- specialized signal, directional, and marker lighting for transportation;
- lighting that is used solely for façade and landscape lighting in MLO lighting zones 3 and 4, and is automatically turned off from midnight until 6 a.m.;
- lighting for theatrical purposes for stage, film, and video performances;
- government-mandated roadway lighting;
- hospital emergency departments, including associated helipads;
- lighting for the national flag in MLO lighting zones 2, 3, or 4; and
- internally illuminated signage.

## SS CREDIT: SITE MASTER PLAN

BD+C
**1 point**
This credit applies to

- Schools

**Intent**
To ensure that the sustainable site benefits achieved by the project continue, regardless of future changes in programs or demographics.

**Requirements**

**SCHOOLS**

The project must achieve at least four of the following six credits, using the associated calculation methods. The achieved credits must then be recalculated using the data from the master plan.

- LT Credit: High Priority Site
- SS Credit: Site Development—Protect or Restore Habitat
- SS Credit: Open Space
- SS Credit: Rainwater Management
- SS Credit: Heat Island Reduction
- SS Credit: Light Pollution Reduction

A *site master plan* for the school must be developed in collaboration with school authorities. Previous sustainable site design measures should be considered in all master-planning efforts so that existing infrastructure is retained whenever possible. The master plan must therefore include current construction activity plus future construction (within the building's lifespan) that affects the site. The master plan development footprint must also include parking, paving, and utilities.

Projects where no future development is planned are not eligible for this credit.

## SS CREDIT: TENANT DESIGN AND CONSTRUCTION GUIDELINES

BD+C
**1 point**
This credit applies to

- Core & Shell

**Intent**
To educate tenants in implementing sustainable design and construction features in their tenant improvement build-outs.

**Requirements**

**CS**

Publish for tenants an illustrated document with the following content, as applicable:

- a description of the sustainable design and construction features incorporated in the core and shell project and the project's sustainability goals and objectives, including those for tenant spaces;
- recommendations, including examples, for sustainable strategies, products, materials, and services; and
- information that enables a tenant to coordinate space design and construction with the building systems when pursuing the following LEED v4.1 for Interior Design and Construction prerequisites and credits:
    - WE Prerequisite: Indoor Water Use Reduction
    - WE Credit: Indoor Water Use Reduction
    - EA Prerequisite: Minimum Energy Performance
    - EA Prerequisite: Fundamental Refrigerant Management
    - EA Credit: Optimize Energy Performance
    - EA Credits: Advanced Energy Metering
    - EA Credit: Renewable Energy
    - EA Credit: Enhanced Refrigerant Management
    - MR Prerequisite: Storage and Collection of Recyclables
    - EQ Prerequisite: Minimum Indoor Air Quality Performance
    - EQ Prerequisite: Environmental Tobacco Smoke Control
    - EQ Credit: Enhanced Indoor Air Quality Strategies
    - EQ Credit: Low-Emitting Materials
    - EQ Credit: Construction Indoor Air Quality Management Plan
    - EQ Credit: Indoor Air Quality Assessment
    - EQ Credit: Thermal Comfort
    - EQ Credit: Interior Lighting
    - EQ Credit: Daylight
    - EQ Credit: Quality Views
    - EQ Credit: Acoustic Performance

Provide the guidelines to all tenants before signing the lease.

## SS CREDIT: PLACES OF RESPITE

BD+C
**1 point**
This credit applies to

- Healthcare

**Intent**
To provide patients, staff, and visitors with the health benefits of the natural environment by creating outdoor places of respite on the healthcare campus.

**Requirements**

**HEALTHCARE**

Provide places of respite that are accessible to patients and visitors, equal to 5% of the *net usable program area* of the building.

Provide additional dedicated places of respite for staff, equal to 2% of the net usable program area of the building.

Places of respite must be outdoors, or be located in interior atria, greenhouses, solaria, or conditioned spaces; such interior spaces may be used to meet up to 30% of the required area if 90% of each qualifying space's gross floor area achieves a direct line of sight to unobstructed views of nature.

All areas must meet the following requirements.

- The area is accessible from within the building or located within 200 feet (60 meters) of a building entrance or access point.
- The area is located where no medical intervention or direct medical care is delivered.
- Options for shade or indirect sun are provided, with at least one seating space per 200 square feet (18.5 square meters) of each respite area, with one wheelchair space per five seating spaces.
- Horticulture therapy and other specific clinical or special-use gardens unavailable to all building occupants may account for no more than 50% of the required area.
- Universal-access natural trails that are available to visitors, staff, or patients may account for no more than 30% of the required area, provided the trailhead is within 200 feet (60 meters) of a building entrance.

Additionally, outdoor areas must meet the following requirements.

- A minimum of 25% of the total outdoor area must be planted with two or more adapted or native vegetation types, or have *overhead vegetated canopy*. Monocultures, such as conventional grass lawns or turfgrass, do not count towards this requirement.
- The area is open to fresh air, the sky, and the natural elements.
- Signage must meet the 2010 FGI Guidelines for Design and Construction of Health Care Facilities (Section 1.2-6.3 and Appendix A1.2-6.3:Wayfinding).
- Places of respite may not be within 25 feet (7.6 meters) of a smoking area (see EQ Prerequisite Environmental Tobacco Smoke Control).

Existing places of respite on the hospital campus may qualify if they otherwise meet the credit requirements.

# SS CREDIT: DIRECT EXTERIOR ACCESS

BD+C
**1 point**
This credit applies to

- Healthcare

**Intent**
To provide patients and staff with the health benefits associated with direct access to the natural environment.

**Requirements**

**HEALTHCARE**

Provide direct access to an exterior courtyard, terrace, garden, or balcony. The space must be at least 5 square feet (0.5 square meters) per patient for 75% of all inpatients and 75% of qualifying outpatients whose clinical length of stay (LOS) exceeds four hours.

Patients whose length of stay exceeds four hours, and whose treatment makes them unable to move, such as emergency, stage 1 surgical recovery, and critical care patients, may be excluded.

Places of respite outside the building envelope that meet the requirements of SS Credit Places of Respite that are immediately adjacent to clinical areas or with direct access from inpatient units may be included.

Qualifying spaces must be designated as nonsmoking  The spaces must also meet the requirements for outdoor air  contaminant concentrations enumerated in EQ Credit Enhanced Indoor Air Quality Strategies, Option 2 and be located more than 100 feet (30 meters) from building exhaust air locations, loading docks, and roadways with idling vehicles.

## SS CREDIT: JOINT USE OF FACILITIES

BD+C
**1 point**
This credit applies to
- Schools

**Intent**
To integrate the school with the community by sharing the building and its playing fields for nonschool events and functions.

**Requirements**

**SCHOOLS**

**Option 1. Make Building Space Open to the General Public (1 point)**
In collaboration with the school authorities, ensure that at least three of the following types of spaces in the school are accessible to and available for shared use by the general public:
- auditorium;
- gymnasium;
- cafeteria;
- one or more classrooms;
- playing fields and stadiums; and
- joint parking.

Provide access to toilets in joint-use areas after normal school hours.

OR

**Option 2. Contract with Specific Organizations to Share Building Space (1 point)**
In collaboration with the school authorities, contract with community or other organizations to provide at least two types of dedicated-use spaces in the building, such as the following:
- commercial office;
- health clinic;
- community service centers (provided by state or local offices);
- police office;
- library or media center;
- parking lot; and
- one or more commercial businesses.

Provide access to toilets in joint-use areas after normal school hours.

OR

**Option 3. Use Shared Space Owned by Other Organizations (1 point)**
In collaboration with the school authorities, ensure that at least two of the following six types of spaces that are owned by other organizations or agencies are accessible to students:
- auditorium;
- gymnasium;
- cafeteria;
- one or more classrooms;
- swimming pool; and
- playing fields and stadiums.

Provide direct pedestrian access to these spaces from the school. In addition, provide signed joint-use agreements with the other organizations or agencies that stipulate how these spaces will be shared.

# WATER EFFICIENCY (WE)

## WE PREREQUISITE: OUTDOOR WATER USE REDUCTION Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**

To reduce outdoor water consumption.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Reduce outdoor water use through one of the following options. Nonvegetated surfaces, such as permeable or impermeable pavement, should be excluded from the landscape area calculations. Athletic fields and playgrounds (if vegetated) and food gardens may be included or excluded at the project team's discretion.

**Option 1. No Irrigation Required**

Show that the landscape does not require a permanent irrigation system beyond a maximum two-year establishment period.

OR

**Option 2. Reduced Irrigation**

Reduce the project's landscape water requirement by at least 30% from the calculated baseline for the site's peak watering month. Reductions must be achieved through plant species selection and irrigation system efficiency, as calculated by the Environmental Protection Agency (EPA) WaterSense Water Budget Tool.

## WE PREREQUISITE: INDOOR WATER USE REDUCTION
## Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**

To reduce indoor water consumption.

**Requirements**

**NC, CS, SCHOOLS, NC-RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, NC-HOSPITALITY, HEALTHCARE**

**Building Water Use**

For the fixtures and fittings listed in Table 1, as applicable to the project scope, reduce aggregate water consumption by 20% from the baseline. Base calculations on the volumes and flow rates shown in Table 1.

All newly installed toilets, urinals, private lavatory faucets, and showerheads that are eligible for labeling must be WaterSense labeled (or a local equivalent for projects outside the U.S.).

**Table 1. Baseline water consumption of fixtures and fittings**

| *Fixture or fitting* | *Baseline (IP units)* | *Baseline (SI units)* |
|---|---|---|
| Toilet (water closet)* | 1.6 gpf | 6 lpf |
| Urinal* | 1.0 gpf | 3.8 lpf |
| Public lavatory (restroom) faucet | 0.5 gpm at 60 psi** all others except private applications | 1.9 lpm at 415 kPa, all others except private applications |
| Private lavatory faucets | 2.2 gpm at 60 psi | 8.3 lpm at 415 kPa |
| Kitchen faucet (excluding faucets used exclusively for filling operations) | 2.2 gpm at 60 psi | 8.3 lpm at 415 kPa |
| Showerhead* | 2.5 gpm at 80 psi per shower stall | 9.5 lpm at 550 kPa per shower stall |

* WaterSense label available for this product type
gpf = gallons per flush
gpm = gallons per minute
psi = pounds per square inch
lpf = liters per flush
lpm = liters per minute
kPa = kilopascals

Projects located where standard supply pressure is different than the LEED baseline supply pressure may calculate the water consumption of flow fixtures and fittings at the local standard supply pressure; the supply pressure must be consistent in the baseline and proposed case.

**Appliance and Process Water Use**

Install appliances, equipment, and processes within the project scope that meet the requirements listed in the tables below.

Existing appliances intended for reuse in the project are not required to meet the requirements in Table 2.

**Table 2. Standards for appliances**

| *Appliance* | *Requirement* |
|---|---|
| Residential clothes washers | ENERGY STAR or performance equivalent* |
| Commercial clothes washers | ENERGY STAR or performance equivalent |
| Residential dishwashers (standard and compact) | ENERGY STAR or performance equivalent* |
| Prerinse spray valves | ≤ 1.3 gpm (4.9 lpm) |
| Ice machine | ENERGY STAR or performance equivalent and use either air-cooled or closed-loop cooling, such as chilled or condenser water system |

gpm = gallons per minute

lpm = liters per minute

*Projects in Europe may install residential appliances meeting the EU A+++ label.

**Table 3. Standards for processes**

| *Process* | *Requirement* |
|---|---|
| Heat rejection and cooling | No once-through cooling with potable water for any equipment or appliances that reject heat |
| Cooling towers and evaporative condensers | Equip with<br>• makeup water meters<br>• conductivity controllers and overflow alarms<br>• efficient drift eliminators that reduce drift to maximum of 0.002% of recirculated water volume for counterflow towers and 0.005% of recirculated water flow for cross-flow towers |

Healthcare, Retail, Schools, and Hospitality Only

In addition, water-consuming appliances, equipment, and processes must meet the requirements listed in Tables 4 and 5.

**Table 4. Standards for appliances**

| *Kitchen equipment* | | *Requirement (IP units)* | *Requirement (SI units)* |
|---|---|---|---|
| Dishwasher | Undercounter | ≤ 1.6 gal/rack | ≤ 6.0 liters/rack |
| | Stationary, single tank, door | ≤ 1.4 gal/rack | ≤ 5.3 liters/rack |
| | Single tank, conveyor | ≤ 1.0 gal/rack | ≤ 3.8 liters/rack |
| | Multiple tank, conveyor | ≤ 0.9 gal/rack | ≤ 3.4 liters/rack |
| | Flight machine | ≤ 180 gal/hour | ≤ 680 liters/hour |
| Food steamer | Batch | ≤ 6 gal/hour/pan | ≤ 23 liters/hour/pan |
| | Cook-to-order | ≤ 10 gal/hour/pan | ≤ 38 liters/hour/pan |
| Combination oven, | Countertop or stand | ≤ 3.5 gal/hour/pan | ≤ 13 liters/hour/pan |
| | Roll-in | ≤ 3.5 gal/hour/pan | ≤ 13 liters/hour/pan |

**Table 5. Process requirements**

| | |
|---|---|
| Discharge water temperature tempering | Where local requirements limit discharge temperature of fluids into drainage system, use tempering device that runs water only when equipment discharges hot water<br>OR<br>Provide thermal recovery heat exchanger that cools drained discharge water below code-required maximum discharge temperatures while simultaneously preheating inlet makeup water<br>OR<br>If fluid is steam condensate, return it to boiler |
| Venturi-type flow-through vacuum generators or aspirators | Use no device that generates vacuum by means of water flow through device into drain |

## WE PREREQUISITE: BUILDING-LEVEL WATER METERING

## Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

### Intent

To support water management and identify opportunities for additional water savings by tracking water consumption.

### Requirements

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Install permanent water meters that measure the total potable water use for the building and associated grounds. Meter data must be compiled into monthly and annual summaries; meter readings can be manual or automated.

Commit to sharing with USGBC the resulting whole-project water usage data for a five-year period beginning on the date the project accepts LEED certification or typical occupancy, whichever comes first.

This commitment must carry forward for five years or until the building changes ownership or lessee.

# WE CREDIT: OUTDOOR WATER USE REDUCTION

BD+C
**1–3 points**
This credit applies to

- New Construction (1–2 points)
- Core & Shell (1–3 points)
- Schools (1–2 points)
- Retail (1–2 points)
- Data Centers (1–2 points)
- Warehouses & Distribution Centers (1–2 points)
- Hospitality (1–2 points)
- Healthcare (1 point)

## Intent

To reduce outdoor water consumption.

## Requirements

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Reduce outdoor water use through one of the following options. Nonvegetated surfaces, such as permeable or impermeable pavement, should be excluded from landscape area calculations. Athletic fields and playgrounds (if vegetated) and food gardens may be included or excluded at the project team's discretion.

**Option 1. No Irrigation Required (2 points except Healthcare and CS, 1 point Healthcare, 3 points CS)**

Show that the landscape does not require a permanent irrigation system beyond a maximum two-year establishment period.

OR

**Option 2. Reduced Irrigation (1-2 points except Healthcare and CS, 1 point Healthcare, 3 points CS)**

Reduce the project's landscape water requirement (LWR) by at least 50% from the calculated baseline for the site's peak watering month. Reductions must first be achieved through plant species selection and irrigation system efficiency as calculated in the Environmental Protection Agency (EPA) WaterSense Water Budget Tool.

Additional reductions beyond 30% may be achieved using any combination of efficiency, alternative water sources, and smart scheduling technologies.

**Table 1. Points for reducing irrigation water**

| *Percentage reduction from baseline* | *Points (except Healthcare)* | *Points (Healthcare)* | *Points (CS)* |
|---|---|---|---|
| 50% | 1 | 1 | 1 |

| | | | |
|---|---|---|---|
| 75% | -- | -- | 2 |
| 100% | 2 | — | 3 |

# WE CREDIT: INDOOR WATER USE REDUCTION

BD+C
**1–7 points**
This credit applies to

- New Construction (1–6 points)
- Core & Shell (1–4 points)
- Schools (1–7 points)
- Retail (1–7 points)
- Data Centers (1–6 points)
- Warehouses & Distribution Centers (1–6 points)
- Hospitality (1–6 points)
- Healthcare (1–7 points)

**Intent**

To reduce indoor water consumption.

**Requirements**

**NC, CS, SCHOOLS, NC-RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, NC-HOSPITALITY, HEALTHCARE**

Further reduce fixture and fitting water use from the calculated baseline in WE Prerequisite Indoor Water Use Reduction. Additional potable water savings can be earned above the prerequisite level using alternative water sources. Include fixtures and fittings necessary to meet the needs of the occupants. Some of these fittings and fixtures may be outside the tenant space (for Commercial Interiors) or project boundary (for New Construction). Points are awarded according to Table 1.

**Table 1. Points for reducing water use**

| Percentage Reduction | Points (BD+C) | Points (CS) | Points (Schools, Retail, Hospitality, Healthcare) |
|---|---|---|---|
| 25% | 1 | 1 | 1 |
| 30% | 2 | 2 | 2 |
| 35% | 3 | 3 | 3 |
| 40% | 4 | 4 | 4 |
| 45% | 5 | -- | 5 |
| 50% | 6 | -- | -- |

Schools, Retail, Hospitality, and Healthcare only

Meet the percentage reduction requirements above.

AND

**Appliance and Process Water.** Install equipment within the project scope that meets the minimum requirements in Table 2, 3, 4, or 5. One point is awarded for meeting all applicable requirements in any one table. All applicable equipment listed in each table must meet the standard.

Schools, Retail, and Healthcare projects can earn a second point for meeting the requirements of two tables.

**Table 2. Compliant commercial washing machines**

To use Table 2, the project must process at least 120,000 lbs (57 606 kg) of laundry per year.

| *Washing machine* | *Requirement (IP units)* | *Requirement (SI units)* |
|---|---|---|
| On-premise, minimum capacity 2,400 lbs (1 088 kg) per 8-hour shift | Maximum 1.8 gals per pound * | Maximum 7 liters per 0.45 kilograms * |

* Based on equal quantities of heavy, medium, and light soil laundry.

**Table 3. Standards for commercial kitchen equipment**

To use Table 3, the project must serve at least 100 meals per day of operation. All process and appliance equipment listed in the category of kitchen equipment and present on the project must comply with the standards.

| *Kitchen equipment* | | *Requirement (IP units)* | *Requirement (SI units)* |
|---|---|---|---|
| Dishwasher | Undercounter | ENERGY STAR | ENERGY STAR or performance equivalent |
| | Stationary, single tank, door | ENERGY STAR | ENERGY STAR or performance equivalent |
| | Single tank, conveyor | ENERGY STAR | ENERGY STAR or performance equivalent |
| | Multiple tank, conveyor | ENERGY STAR | ENERGY STAR or performance equivalent |
| | Flight machine | ENERGY STAR | ENERGY STAR or performance equivalent |
| Food steamer | Batch (no drain connection) | ≤ 2 gal/hour/pan including condensate cooling water | ≤ 7.5 liters/hour/pan including condensate cooling water |
| | Cook-to-order (with drain connection) | ≤ 5 gal/hour/pan including condensate cooling water | ≤ 19 liters/hour/pan including condensate cooling water |
| Combination oven, | Countertop or stand | ≤ 1.5 gal/hour/pan including condensate cooling water | ≤ 5.7 liters/hour/pan including condensate cooling water |

| | | | |
|---|---|---|---|
| | Roll-in | ≤ 1.5 gal/hour/pan including condensate cooling water | ≤ 5.7 liters/hour/pan including condensate cooling water |
| Food waste disposer | Disposer | 3-8 gpm, full load condition, 10 minute automatic shutoff; or 1 gpm, no-load condition | 11–30 lpm, full load condition, 10-min automatic shutoff; or 3.8 lpm, no-load condition |
| | Scrap collector | Maximum 2 gpm makeup water | Maximum 7.6 lpm makeup water |
| | Pulper | Maximum 2 gpm makeup water | Maximum 7.6 lpm makeup water |
| | Strainer basket | No additional water usage | No additional water usage |

gpm = gallons per minute
gph = gallons per hour
lpm = liters per minute
lph = liters per hour

**Table 4. Compliant laboratory and medical equipment**

To use Table 4, the project must be a medical or laboratory facility.

| *Lab equipment* | *Requirement (IP units)* | *Requirement (SI units)* |
|---|---|---|
| Reverse-osmosis water purifier | 75% recovery | 75% recovery |
| Steam sterilizer | For 60-inch sterilizer, 6.3 gal/U.S. tray<br>For 48-inch sterilizer, 7.5 gal/U.S. tray | For 1520-mm sterilizer, 28.5 liters/DIN tray<br>For 1220-mm sterilizer, 28.35 liters/DIN tray |
| Sterile process washer | 0.35 gal/U.S. tray | 1.3 liters/DIN tray |
| X-ray processor, 150 mm or more in any dimension | Film processor water recycling unit | |
| Digital imager, all sizes | No water use | |

**Table 5. Compliant municipal steam systems**

To use Table 5, the project must be connected to a municipal or district steam system that does not allow the return of steam condensate.

| *Steam system* | *Standard* |
|---|---|
| Steam condensate disposal | Cool municipally supplied steam condensate (no return) to drainage system with heat recovery system or reclaimed water |
| OR | |
| Reclaim and use steam condensate | 100% recovery and reuse |

# WE CREDIT: COOLING TOWER AND PROCESS WATER USE

BD+C
**1–3 points**
This credit applies to

- New Construction (1–2 points)
- Core & Shell (1–3 points)
- Schools (1–2 points)
- Retail (1–2 points)
- Data Centers (1–2 points)
- Warehouses & Distribution Centers (1–2 points)
- Hospitality (1–2 points)
- Healthcare (1–2 points)

**Intent**

To conserve water used for mechanical processes and cooling tower makeup while controlling microbes, corrosion, and scale in the condenser water system.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE,**

**Option 1. Cooling Tower Water Use (1-2 points except CS, 1-3 points CS)**

For cooling towers and evaporative condensers, conduct a one-time potable water analysis, measuring at least the five control parameters listed in Table 1.

**Table 1. Maximum concentrations for parameters in condenser water**

| *Parameter* | *Maximum level* |
|---|---|
| Ca (as $CaCO_3$) | 1000 ppm |
| Total alkalinity | 1000 ppm |
| $SiO_2$ | 100 ppm |
| $Cl^-$ | 250 ppm |
| Conductivity | 2000 µS/cm |

ppm = parts per million
µS/cm = micro siemens per centimeter

Calculate the maximum number of cooling tower cycles by dividing the maximum allowed concentration level of each parameter by the actual concentration level of each parameter found in the potable makeup water analysis. Limit cooling tower cycles to avoid exceeding maximum values for any of these parameters.

**Table 2. Points for cooling tower cycles**

| *Cooling tower cycles* | *Points (all except CS)* | *Points (CS)* |
|---|---|---|
| Maximum number of cycles achieved without exceeding any filtration levels or affecting operation of condenser water system | 1 | 1 |

| | | |
|---|---|---|
| Meet the maximum calculated number of cycles to earn 1 point, and increase the number of cycles by a minimum of 25% by increasing the level of treatment and/or maintenance in condenser or make-up water systems<br><br>OR<br><br>Meet the maximum calculated number of cycles to earn 1 point and use a minimum 20% recycled nonpotable water | 2 | 2 |
| Meet the maximum calculated number of cycles to earn 1 point, and increase the number of cycles by a minimum of 30% by increasing the level of treatment and/or maintenance in condenser or make-up water systems<br><br>OR<br><br>Meet the maximum calculated number of cycles to earn 1 point and use a minimum 30% recycled nonpotable water | -- | 3 |

Minimum percentage recycled nonpotable water used in cooling tower makeup should be based on water use during the month with the highest demand for make-up water.

Projects whose cooling is provided by district cooling systems are eligible to achieve Option 1 if the district cooling system complies with the above requirements.

OR

**Option 2. No Cooling Tower (2 points)**

For projects without cooling towers or evaporative condensers, projects may earn full credit if all conditions are met:

- the baseline system designated for the building using ASHRAE 90.1-2016 Appendix G Table G3.1.1 includes a cooling tower (systems 7 & 8)
- the project design case does not include a cooling tower
- the design case mechanical system does not use the latent heat of the evaporative cooling of water.
- the project does not receive any cooling from a District cooling system

All other system types are ineligible for credit.

OR

**Option 3. Process Water Use (1-2 points except CS, 1-3 points CS)**

Demonstrate that the project is using minimum 20% recycled alternative water to meet process water demand for 1 point, or using minimum 30% recycled alternative water to meet process water demand for 2 points. Ensure that recycled alternative water is of sufficient quality for its intended end use.

Minimum percentage of recycled alternative water used should be based on water use during the month with the highest water demand.

Process water uses eligible for achievement of Option 3 must represent at least 10% of total building water use. Eligible subsystems may include:

- Boilers
- Humidification systems
- Other subsystems using process water

Projects whose cooling is provided by district cooling systems are eligible to achieve Option 3 if the district cooling system complies with minimum thresholds for recycled alternative water use.

CS projects:

Demonstrate that the project is using minimum 20% recycled alternative water to meet process water demand for 1 point, using minimum 30% recycled alternative water to meet process water demand for 2 points, or using minimum 40% recycled alternative water to meet process water demand for 3 points. Ensure that recycled alternative water is of sufficient quality for its intended end use.

# WE CREDIT: WATER METERING

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1 point)

**Intent**

To support water management and identify opportunities for additional water savings by tracking water consumption.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Install permanent water meters for two or more of the following water subsystems, as applicable to the project:

- *Irrigation.* Meter water systems serving at least 80% of the irrigated landscaped area. Calculate the percentage of irrigated landscape area served as the total metered irrigated landscape area divided by the total irrigated landscape area. Landscape areas fully covered with xeriscaping or native vegetation that requires no routine irrigation may be excluded from the calculation.

- *Indoor plumbing fixtures and fittings.* Meter water systems serving at least 80% of the indoor fixtures and fitting described in WE Prerequisite Indoor Water Use Reduction, either directly or by deducting all other measured water use from the measured total water consumption of the building and grounds.

- *Domestic hot water.* Meter water use of at least 80% of the installed domestic hot water heating capacity (including both tanks and on-demand heaters).

- *Boiler with aggregate projected annual water use of 100,000 gallons (378 500 liters) or more, or boiler of more than 500,000 BtuH (150 kW).* A single makeup meter may record flows for multiple boilers.

- *Reclaimed water.* Meter reclaimed water, regardless of rate. A reclaimed water system with a makeup water connection must also be metered so that the true reclaimed water component can be determined.

- *Other process water.* Meter at least 80% of expected daily water consumption for process end uses, such as humidification systems, dishwashers, clothes washers, pools, and other subsystems using process water.

Healthcare Projects only
In addition to the requirements above, install water meters in any five of the following:

- purified water systems (reverse-osmosis, de-ionized);
- filter backwash water;
- water use in dietary department;
- water use in laundry;
- water use in laboratory;
- water use in central sterile and processing department;
- water use in physiotherapy and hydrotherapy and treatment areas;
- water use in surgical suite;
- closed-looped hydronic system makeup water; and
- cold-water makeup for domestic hot water systems.

# ENERGY AND ATMOSPHERE

## EA PREREQUISITE: FUNDAMENTAL COMMISSIONING AND VERIFICATION Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**

To support the design, construction, and eventual operation of a project that meets the owner's project requirements for energy, water, indoor environmental quality, and durability.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

**Commissioning Process Scope**
Complete the following commissioning (Cx) process activities for mechanical, electrical, plumbing, and renewable energy systems and assemblies, in accordance with ASHRAE Guideline 0-2013 and ASHRAE Guideline 1.1–2007 for HVAC&R Systems, as they relate to energy, water, indoor environmental quality, and durability.

- Develop the OPR.
- Develop a BOD.

The commissioning authority (CxA) must do the following:

- Review the OPR, BOD, and project design.
- Develop and implement a Cx plan.
- Confirm incorporation of Cx requirements into the construction documents.
- Develop construction checklists.
- Develop a system test procedure.
- Verify system test execution.
- Maintain an issues and benefits log throughout the Cx process.
- Prepare a final Cx process report.
- Document all findings and recommendations and report directly to the owner throughout the process.

Requirements for exterior enclosures are limited to inclusion in the owner's project requirements (OPR) and basis of design (BOD), as well as the review of the OPR, BOD and project design. ASTM E2947-16: Standard Guide for Building Enclosure Commissioning provides additional guidance.

The review of the exterior enclosure design may be performed by a qualified independent member of the

design or construction team (or an employee of that firm) who is not directly responsible for design of the building enclosure for the project.

**Commissioning Authority Qualifications**
By the end of the design development phase, engage a commissioning authority with the following qualifications.

- The CxA must have documented commissioning process experience on at least two building projects with a similar scope of work. The experience must extend from early design phase through at least 10 months of occupancy;
- The CxA may be a qualified employee of the owner, an independent consultant, or an employee of the design or construction firm who is not part of the project's design or construction team, or a disinterested subcontractor of the design or construction team.
    - For projects smaller than 20,000 square feet (1 860 square meters), the CxA may be a qualified member of the design or construction team. In all cases, the CxA must report his or her findings directly to the owner.

Project teams that intend to pursue EA Credit Enhanced Commissioning should note a difference in the CxA qualifications: for the credit, the CxA may not be an employee of the design or construction firm nor a subcontractor to the construction firm.

**Current Facilities Requirements and Operations and Maintenance Plan**
Prepare and maintain a current facilities requirements and operations and maintenance plan that contains the information necessary to operate the building efficiently. The plan must include the following:

- a sequence of operations for the building;
- the building occupancy schedule;
- equipment run-time schedules;
- setpoints for all HVAC equipment;
- set lighting levels throughout the building;
- minimum outside air requirements;
- any changes in schedules or setpoints for different seasons, days of the week, and times of day;
- a systems narrative describing the mechanical and electrical systems and equipment;
- a preventive maintenance plan for building equipment described in the systems narrative; and
- a commissioning program that includes periodic commissioning requirements, ongoing commissioning tasks, and continuous tasks for critical facilities.

<u>Data Centers only</u>
For small projects with computer room peak cooling loads less than 2,000,000 Btu/h (600 kW) or a total computer room peak cooling load less than 600,000 Btu/h (175 kW), the CxA may be a qualified employee of the design or construction team.

# EA PREREQUISITE: MINIMUM ENERGY PERFORMANCE
# Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**

To reduce the environmental and economic harms of excessive energy use by achieving a minimum level of energy efficiency for the building and its systems.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Comply with ANSI/ASHRAE/IESNA Standard 90.1–2016, with errata or a USGBC-approved equivalent standard.

ASHRAE 90.1-2016 Compliance pathways in Section 4.2.1.1 include compliance with all mandatory provisions, and compliance with one of the following:

- Prescriptive provisions of Sections 5 through 10
- Section 11 *Energy Cost Budget Method*
- Normative Appendix G *Performance Rating Method*. When using Appendix G, the Performance Cost Index (PCI) shall be less than or equal to the Performance Cost Index Target ($PCI_t$) in accordance with the methodology provided in Section 4.2.1.1. Document the PCI, $PCI_t$, and percentage improvement using metrics of cost or greenhouse gas (GHG) emissions.

**For projects using Normative Appendix G *Performance Rating Method:***

- Greenhouse gas emissions: The total greenhouse gas emissions, in terms of carbon dioxide equivalents, shall be calculated for the baseline building performance rating and for the proposed building performance rating, and the percentage improvement shall be determined using carbon dioxide equivalent emissions.

  US and Canada:
  - use U.S. Environmental Protection Agency's (EPA) regional grid mix coefficients to calculate GHG emissions by energy source; or
  - use hourly emissions profiles from U.S. Environmental Protection Agency's (EPA) AVoided Emissions and geneRation Tool (AVERT)

  International:
  - use national grid mix coefficients from the International Energy Agency CO2 Emissions from Fuel Combustion 2017 report to calculate GHG emissions by energy source
  - ISO 52000-1:2017: Greenhouse gas emission factors for each building energy source shall be determined consistently with ISO Standard 52000-1:2017 and published for the country or region where the project is located

- Exception to Mandatory Measures requirements: For ASHRAE 90.1-2016 mandatory controls provisions that are quantified in the Appendix G *Performance Rating Method*, (e.g. lighting occupancy sensor controls, lighting daylighting controls, automated receptacle controls, etc.), projects may model the *Proposed Building Performance* control parameters identically to the *Baseline Building Performance* control parameters in lieu of compliance with the mandatory provisions.

- Exceptional Calculations modeled in accordance with Section G2.5 may be modeled to document minimum prerequisite compliance.

- Only on-site or on-campus renewable energy that meets ASHRAE Standard 90.1-2016 Section G 2.4.1 requirements for on-site renewable energy may be used to meet minimum ASHRAE Standard 90.1-2016 performance requirements.

## EA PREREQUISITE: BUILDING-LEVEL ENERGY METERING
## Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**

To support energy management and identify opportunities for additional energy savings by tracking building-level energy use.

**Requirements**

**NC, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Install new or use existing building-level energy meters, or submeters that can be aggregated to provide building-level data representing total building energy consumption (electricity, natural gas, chilled water, steam, fuel oil, propane, biomass, etc). Utility-owned meters capable of aggregating building-level resource use are acceptable.

Commit to sharing with USGBC the resulting energy consumption data and electrical demand data (if metered) for a five-year period beginning on the date the project accepts LEED certification. At a minimum, energy consumption must be tracked at one-month intervals.

This commitment must carry forward for five years or until the building changes ownership or lessee.

**CS**

Install new or use existing base building-level energy meters, or submeters that can be aggregated to provide base building-level data representing total building energy consumption (electricity, natural gas, chilled water, steam, fuel oil, propane, etc.). Utility-owned meters capable of aggregating base building-level resource use are acceptable.

Commit to sharing with USGBC the resulting energy consumption data and electrical demand data (if metered) for a five-year period beginning on the date the project accepts LEED certification or typical occupancy, whichever comes first. At a minimum, energy consumption must be tracked at one-month intervals.

This commitment must carry forward for five years or until the building changes ownership or lessee.

# EA PREREQUISITE: FUNDAMENTAL REFRIGERANT MANAGEMENT Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**

To reduce stratospheric ozone depletion.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Do not use chlorofluorocarbon (CFC)-based refrigerants in new heating, ventilating, air-conditioning, and refrigeration (HVAC&R) systems. When reusing existing HVAC&R equipment, complete a comprehensive CFC phase-out conversion before project completion. Phase-out plans extending beyond the project completion date will be considered on their merits.

Existing small HVAC&R units (defined as containing less than 0.5 pound [225 grams] of refrigerant) and other equipment, such as standard refrigerators, small water coolers, and any other equipment that contains less than 0.5 pound (225 grams) of refrigerant, are exempt.

# EA CREDIT: ENHANCED COMMISSIONING

BD+C
**2-6 points**
This credit applies to

- New Construction (2-6 points)
- Core & Shell (2-6 points)
- Schools (2-6 points)
- Retail (2-6 points)
- Data Centers (2-6 points)
- Warehouses & Distribution Centers (2-6 points)
- Hospitality (2-6 points)
- Healthcare (2-6 points)

**Intent**

To further support the design, construction, and eventual operation of a project that meets the owner's project requirements for energy, water, indoor environmental quality, and durability.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Implement, or have in place a contract to implement, the following commissioning process activities in addition to those required under EA Prerequisite Fundamental Commissioning and Verification.

**Commissioning Authority Qualifications:**

- The CxA must have documented commissioning process experience on at least two building projects with a similar scope of work. The experience must extend from early design phase through at least 10 months of occupancy;
- The CxA may be a qualified employee of the owner, an independent consultant, or a disinterested subcontractor of the design team.

**Option 1. Enhanced Systems Commissioning (3-4 points)**

**Path 1: Enhanced Commissioning (3 points)**

Complete the following commissioning process (CxP) activities for mechanical, electrical, plumbing, and renewable energy systems and assemblies in accordance with ASHRAE Guideline 0–2013 and ASHRAE Guideline 1.1–2007 for HVAC&R systems, as they relate to energy, water, indoor environmental quality, and durability.

The commissioning authority must do the following:

- Review contractor submittals.
- Verify inclusion of systems manual requirements in construction documents.
- Verify inclusion of operator and occupant training requirements in construction documents.
- Verify systems manual updates and delivery.
- Verify operator and occupant training delivery and effectiveness.
- Verify seasonal testing.
- Review building operations 10 months after substantial completion.
- Develop an on-going commissioning plan.

Include all enhanced commissioning tasks in the OPR and BOD.

OR

**Path 2: Enhanced and Monitoring-Based Commissioning (4 points)**

Achieve Path 1.

AND

Develop monitoring-based procedures and identify points to be measured and evaluated to assess performance of energy- and water-consuming systems.

Include the procedures and measurement points in the commissioning plan. Address the following:

- roles and responsibilities;
- measurement requirements (meters, points, metering systems, data access);
- the points to be tracked, with frequency and duration for trend monitoring;
- the limits of acceptable values for tracked points and metered values (where appropriate, predictive algorithms may be used to compare ideal values with actual values);
- the elements used to evaluate performance, including conflict between systems, out-of-sequence operation of systems components, and energy and water usage profiles;
- an action plan for identifying and correcting operational errors and deficiencies;
- training to prevent errors;
- planning for repairs needed to maintain performance; and
- the frequency of analyses in the first year of occupancy (at least quarterly).

Update the systems manual with any modifications or new settings, and give the reason for any modifications from the original design.

AND/OR

**Option 2. Building Enclosure Commissioning (2 points)**

Fulfill the requirements in EA Prerequisite Fundamental Commissioning and Verification as they apply to the building's enclosure in addition to mechanical and electrical systems and assemblies.

Complete the following commissioning process (CxP) activities for the building's thermal envelope in accordance with ASHRAE Guideline 0–2013 and ASTM E2947-16: Standard Guide for Building Enclosure Commissioning, as they relate to energy, air and water tightness, indoor environmental quality, and durability.

The qualified independent member of the design or construction team responsible for building enclosure commissioning must complete the following:

- Review contractor submittals.
- Verify inclusion of systems manual requirements in construction documents for enclosure systems.
- For specialty enclosure systems with controls and automation:
    - Verify inclusion of operator and occupant training requirements in construction documents.
    - Verify systems manual updates and delivery.
    - Verify operator and occupant training delivery and effectiveness.
    - Verify seasonal testing.
    - Review building operations 10 months after substantial completion.

- Develop an on-going enclosure commissioning plan for maintenance, renewal and revitalization cycles.

Data Centers only

Projects that select Option 1 must complete the following commissioning process.

For small projects with peak cooling loads less than 2,000,000 Btu/h (600 kW), or a total computer room peak cooling load less than 600,000 Btu/h (175 kW), the CxA must perform the following activities:

- conduct at least one commissioning verification review of the owner's project requirements, basis of design, and design documents before mid-construction documents development;
- back-check the review comments in all subsequent design submissions; and
- conduct an additional full verification review at 95% completion of the design documents and basis of design.

For projects with peak cooling loads 2,000,000 Btu/h (600 kW) or more, or a total computer room peak cooling load 600,000 Btu/h (175 kW) or more, the CxA must conduct at least three verification reviews of the basis of design:

- one verification review of design documents before the start of design development;
- one verification review of design documents before midconstruction documents; and
- one final verification review of 100% complete design documents, verifying achievement of the owner's project requirements and adjudication of previous review comments.

# EA CREDIT: OPTIMIZE ENERGY PERFORMANCE

BD+C
**1–20 points**
This credit applies to

- New Construction (1–18 points)
- Core & Shell (1–18 points)
- Schools (1–16 points)
- Retail (1–18 points)
- Data Centers (1–18 points)
- Warehouses & Distribution Centers (1–18 points)
- Hospitality (1–18 points)
- Healthcare (1–20 points)

**Intent**

To achieve increasing levels of energy performance beyond the prerequisite standard to reduce environmental and economic harms associated with excessive energy use.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Analyze efficiency measures during the design process and account for the results in design decision making. Use energy simulation of efficiency opportunities, past energy simulation analyses for similar buildings, or published data (e.g., Advanced Energy Design Guides) from analyses for similar buildings.

Analyze efficiency measures, focusing on load reduction and HVAC-related strategies (passive measures are acceptable) appropriate for the facility. Project potential energy savings and holistic project cost implications related to all affected systems.

Choose one of the options below.

**Option 1. Energy Performance Compliance (1–18 points except Schools and Healthcare, 1–16 points Schools, 1–20 points Healthcare)**

Demonstrate a Performance Cost Index (PCI)[1] below the Performance Cost Index Target ($PCI_t$) calculated in accordance with Section 4.2.1.1 of ANSI/ASHRAE/IESNA Standard 90.1-2016, Appendix G, Table 4.2.1.1. For mixed use buildings, the required PCI shall be calculated by using an area weighted average of the building types.

Calculate the PCI, $PCI_t$, and percentage improvement using metrics of cost and greenhouse gas (GHG) emissions. For each energy source serving the building, the GHG emission factors must be identical for the Baseline and Proposed building models.

LEED points are calculated based on the project percent improvement PCI below the $PCI_t$ using metrics of cost and GHG emissions. Total points have been divided equally between the metrics of energy cost and greenhouse gas emissions. Points are awarded according to Table 1 and Table 2.

For project percent improvement for the cost metric, on-site renewable energy may be subtracted from proposed energy cost prior to calculating proposed building performance per ASHRAE Standard 90.1-2016 Section G 2.4.1.

**Table 1. Points for percentage improvement in energy performance – % Cost PCI below $PCI_t$ (1-9 points NC and CS, 1-8 points Schools, 1-10 points Healthcare)**

| New Construction | Healthcare, Major Renovation, CS | Points BD+C (except Schools, Healthcare)* | Points Healthcare | Points Schools |
|---|---|---|---|---|
| 5% | 2% | 1 | 1 | 1 |
| 10% | 5% | 2 | 2 | 2 |
| 15% | 10% | 3 | 3 | 3 |
| 20% | 15% | 4 | 4 | 4 |
| 25% | 20% | 5 | 5 | 5 |
| 30% | 25% | 6 | 6 | 6 |
| 35% | 30% | 7 | 7 | 7 |
| 40% | 35% | 8 | 8 | |
| 45% | 40% | 9 | 9 | 8 |
| 50% | 45% | EP | 10 | EP |
| | 50% | | EP | |

*BD+C projects except Data Centers with unregulated energy cost exceeding 50% of the total proposed building energy cost, and BD+C: Data Centers projects with at least 40% gross colocation data center area may use the "Healthcare, Major Renovation, CS" column in lieu of the "New Construction" column.

On-site renewable energy may be subtracted from proposed greenhouse gas emissions prior to calculating proposed building performance per ASHRAE Standard 90.1-2016 Section G 2.4.1. New off-site renewable energy as defined in EA credit Renewable Energy may be subtracted from proposed greenhouse gas emissions prior to calculating proposed building performance.

**Table 2. Points for percentage improvement in energy performance – % Greenhouse Gas Emissions PCI below $PCI_t$ (1-9 points NC, 1-8 points Schools, 1-10 points Healthcare)**

| New Construction | Healthcare, Major Renovation, CS, | Points BD+C (except Schools, Healthcare)* | Points Healthcare | Points Schools |
|---|---|---|---|---|
| 5% | 2% | 1 | 1 | 1 |
| 10% | 5% | 2 | 2 | 2 |
| 16% | 10% | 3 | 3 | 3 |
| 24% | 16% | 4 | 4 | 4 |

| | | | | |
|---|---|---|---|---|
| 32% | 24% | 5 | 5 | 5 |
| 40% | 32% | 6 | 6 | 6 |
| 50% | 40% | 7 | 7 | 7 |
| 65% | 50% | 8 | 8 | |
| 80% | 65% | 9 | 9 | 8 |
| 100% | 80% | EP | 10 | EP |
| | 100% | | EP | |

BD+C projects except Data Centers with unregulated energy cost exceeding 50% of the total proposed building energy cost, and BD+C: Data Centers projects with at least 40% gross colocation data center area may use the "Healthcare, Major Renovation, CS" column in lieu of the "New Construction" column.

Retail only

For all process loads, define a clear baseline for comparison with the proposed improvements. The baselines in Appendix 3, Tables 1–4, represent industry standards and may be used without additional documentation. Calculate the baseline and design as follows:

- *Appliances and equipment.* For appliances and equipment not covered in Tables 1–4, indicate hourly energy use for proposed and budget equipment, along with estimated daily use hours. Use the total estimated appliance/equipment energy use in the energy simulation model as a plug load. Reduced use time (schedule change) is not a category of energy improvement in this credit. ENERGY STAR ratings and evaluations are a valid basis for performing this calculation.
- *Display lighting.* For display lighting, use the space-by-space method of determining allowed lighting power under ANSI/ASHRAE/IESNA Standard 90.1–2016, with errata (or a USGBC-approved equivalent standard for projects outside the U.S.), to determine the appropriate baseline for both the general building space and the display lighting.
- *Refrigeration.* For hard-wired refrigeration loads, model the effect of energy performance improvements with a simulation program designed to account for refrigeration equipment.

**DATA CENTERS ONLY**

In addition to the requirements above, analyze efficiency measures focused on IT load reduction and HVAC-related strategies. Collocated data centers may use the CS percentage improvement thresholds in lieu of NC thresholds to earn points in Table 1 and Table 2.

OR

**Option 2. Prescriptive Compliance: ASHRAE Advanced Energy Design Guide (1–6 points, 1-4 points CS)**

To be eligible for Option 2, projects must meet the Scope requirements of the applicable AEDGs (or combination of AEDGs for mixed use), and projects must use the ASHRAE 90.1-2016 Prescriptive compliance path in EA Prerequisite Minimum Energy Performance.

Implement and document compliance with the applicable recommendations and standards in Chapter 4, Design Strategies and Recommendations by Climate Zone, for the appropriate ASHRAE 50% Advanced Energy Design Guide and climate zone.

**ASHRAE 50% Advanced Energy Design Guide for Small to Medium Office Buildings**

- *Building envelope:* roofs, walls, floors, slabs, doors, continuous air barriers, and vertical fenestration (1 point, 2 points CS)
- *Interior and exterior lighting,* including daylighting and interior finishes (1 point). CS: Minimum 20% of building area must have a complete lighting design.
- *Plug loads,* including equipment and controls (2 points, 0 points CS)
- *HVAC Systems and Controls (*2 points, 1 point CS). CS scope of work must include at least 20% of the building's projected cooling capacity, heating capacity, and fan volume.

**ASHRAE 50% Advanced Energy Design Guide for Medium to Large Box Retail Buildings**

- *Building envelope:* roofs, walls, floors, slabs, doors, vestibules and fenestration – all orientations (1 point, 2 points CS)
- *Interior and exterior lighting,* excluding lighting power density for sales floor (1 point). CS: Minimum 20% of building area must have a complete lighting design
- *Additional interior lighting* for sales floor (1 point, 0 points CS)
- *Plug loads,* including equipment choices and controls (1 point, 0 points CS)
- *HVAC efficiency and control requirements* (2 points, 1 point CS). CS scope of work must include at least 20% of the building's projected cooling capacity, heating capacity, and fan volume)

**ASHRAE 50% Advanced Energy Design Guide for K–12 School Buildings**

(Not applicable for CS)

- *Building envelope:* roofs, walls, floors, slabs, doors and vertical fenestration (1 point)
- *Interior and exterior lighting,* including daylighting and interior finishes (1 point)
- *Plug loads,* including equipment choices, controls, and kitchen equipment (2 points)
- *HVAC efficiency and control requirements* (2 points)

**ASHRAE 50% Advanced Energy Design Guide for Large Hospitals**

- *Building envelope:* roofs, walls, floors, slabs, doors, vestibules, continuous air barriers, and vertical fenestration (1 point, 2 points CS)
- *Interior and exterior lighting,* including daylighting (form or nonform driven) and interior finishes (1 point). CS: Minimum 20% of building area must have a complete lighting design
- *Plug loads,* including equipment choices, controls, and kitchen equipment (1 point, 0 points CS)
- *HVAC and Service Water Heating Systems and Equipment (2 points) (*1 point CS. CS scope of work must include at least 20% of the building's projected cooling capacity, heating capacity, and fan volume)

**ASHRAE 50% Advanced Energy Design Guide for Grocery Stores**

- *Building envelope:* roofs, walls, floors, slabs, doors, vestibules, continuous air barriers, and vertical fenestration (1 point) (2 points – CS)
- *Interior and exterior lighting, including sales floor (1* points)
- *Refrigeration, Plug, and Process loads,* including equipment choices and controls (2 points)(0 points CS)
- *HVAC efficiency and control requirements (1 point) (*1 point CS. CS scope of work must include at least 20% of the building's projected cooling capacity, heating capacity, and fan volume)

**Option 3. Systems Optimization (1-4 points)**

To be eligible for Option 3, projects must use the ASHRAE 90.1-2016 Prescriptive compliance path in EA Prerequisite Minimum Energy Performance, and must not have more than 2,000 square feet of data center space, laboratory space, or manufacturing space.

Demonstrate an improvement beyond ASHRAE/ASHRAE/IESNA Standard 90.1–2016, with errata, for the following systems: Interior and Exterior Lighting; Daylight controls; Building envelope; HVAC and service water heating equipment efficiency; and Equipment and appliances. Each three strategies selected earns 1 point (*For Core & Shell lighting and HVAC efficiency measures, the project scope of work must include a complete design for at least 20% of the building area to qualify).*

- *Interior and Exterior Lighting:*
  - 15% lighting power reduction
  - 30% lighting power reduction
  - 45% lighting power reduction

- *Daylight controls:*
  Install daylight-responsive controls for a given percentage of connected lighting load (lighting in non-regularly occupied space with occupant sensor controls may be excluded from connected lighting load).
  - 35%
  - 70%

- *Building envelope:*
  - Climate Zones 1 – 2:
    - Thermal Mass Enclosure: More than 70% of opaque above-grade wall area meets ASHRAE 90.1-2016 definition for “mass wall”; and more than 70% of floor area meets ASHRAE 90.1-2016 definition for “mass floor”
    - 25% Envelope UA reduction
    - 50% SHGC reduction (including window shade factors)
  - Climate Zones 3 – 8:
    - 25% Envelope UA reduction
    - 50% Envelope UA reduction
    - 25% reduction in air infiltration measured during commissioning

- *HVAC and Service Water Heating Equipment Efficiency*:
  - Reduction in total fan power allowance of:
    - 15%
    - 30%
    - 45%
  - Improvement in efficiency for at least 75% of the combined cooling, heating, and service water heating capacity
    - 10%
    - 20%
    - 30%

    *Electric resistance heating except heat pump auxiliary heat must be included in total capacity.*

- *Equipment and Appliances*:
  Install a percentage (by rated power) of eligible equipment and appliances meeting the following requirements:
  - *ENERGY STAR* equipment including appliances, office equipment, electronics, and commercial food service equipment (lighting and building envelope products are excluded from this credit). (Electronic Product Environmental Assessment Tool (EPEAT) equipment may be used in lieu of Energy Star equipment where applicable).
  - Prescriptive commercial kitchen and refrigeration equipment requirements listed in Appendix 3, Table 1.

  The project scope of work must include at least 0.25 Watts per square foot of eligible equipment to apply this strategy.

  Percent of Eligible Equipment Installed by Rated Power:

- 20%
- 40%
- 60%
- 80%
- 100%

OR

**Option 4. Data Centers only– System Optimization (1-3 points)**
Calculate an Overall *Systems* Design Value as the sum of the maximum *design Mechanical Load Component (MLC)* and maximum *design Electrical Load Component (ELC)* in accordance with ASHRAE 90.4-2016 Section 6.2, Section 8.2, and Section 11. Document that the Overall Systems Design value is less than the Maximum Overall Systems Value by:

- 10% (1 point)
- 20% (2 points)
- 30% (3 points)

**If the electrical system design is incomplete, the design values shall be assumed to match the values in Table 8.2.1.1 and 8.2.1.2.*

*Collocated data centers:* document that the Overall Systems Design value is less than the Maximum Overall Systems Value by:

- 6% (1 point)
- 12% (2 points)
- 18% (3 points)

# EA CREDIT: ADVANCED ENERGY METERING

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1 point)

**Intent**

To support energy management and identify opportunities for additional energy savings by tracking building-level and system-level energy use.

**Requirements**

**NC, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Install *advanced energy metering* for the following:

- all whole-building energy sources used by the building; and
- any individual energy end uses that represent 10% or more of the total annual consumption of the building.

The advanced energy metering must have the following characteristics.

- Meters must be permanently installed, record at intervals of one hour or less, and transmit data to a remote location.
- Electricity meters must record both consumption and demand. Whole-building electricity meters should record the power factor, if appropriate.
- The data collection system must use a local area network, building automation system, wireless network, or comparable communication infrastructure.
- The system must be capable of storing all meter data for at least 36 months.
- The data must be remotely accessible.
- All meters in the system must be capable of reporting hourly, daily, monthly, and annual energy use.

**CS**

Install meters for future tenant spaces so that tenants will be capable of independently metering energy consumption (electricity, chilled water, etc.) for all systems dedicated to their space. Provide a sufficient number of meters to capture total tenant energy use with a minimum of one meter per energy source per floor.

Install *advanced energy metering* for all base building energy sources used by the building.
The advanced energy metering must have the following characteristics.

- Meters must be permanently installed, record at intervals of one hour or less, and transmit data to a remote location.

- Electricity meters must record both consumption and demand. Whole-building electricity meters should record the power factor, if appropriate.
- The data collection system must use a local area network, building automation system, wireless network, or comparable communication infrastructure.
- The system must be capable of storing all meter data for at least 36 months.
- The data must be remotely accessible.
- All meters in the system must be capable of reporting hourly, daily, monthly, and annual energy use.

# EA CREDIT: GRID HARMONIZATION

BD+C
**1–2 points**
This credit applies to

- New Construction (1–2 points)
- Core & Shell (1–2 points)
- Schools (1–2 points)
- Retail (1–2 points)
- Data Centers (1–2 points)
- Warehouses & Distribution Centers (1–2 points)
- Hospitality (1–2 points)
- Healthcare (1–2 points)

**Intent**

To increase participation in demand response technologies and programs that make energy generation and distribution systems more efficient, increase grid reliability, and reduce greenhouse gas emissions.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Design building and equipment for participation in demand response programs through load shedding or shifting. On-site electricity generation does not meet the intent of this credit.

**Case 1. Demand Response Program Available and Participation (2 points)**

- Participate in an existing demand response (DR) program and complete the following activities. Design a system with the capability for real-time, fully-automated DR based on external initiation by a DR Program Provider. Semi-automated DR may be utilized in practice.
- Enroll in a minimum one-year DR participation amount contractual commitment with a qualified DR program provider, with the intention of multiyear renewal, for at least 10% of the annual on-peak electricity demand. On-peak demand is determined under EA Prerequisite Minimum Energy Performance. The on-peak demand may vary based on the utility climate and pricing structures.
- Develop a comprehensive plan for meeting the contractual commitment during a Demand Response event.
- Include the DR processes in the scope of work for the commissioning authority, including participation in at least one full test of the DR plan.
- Include the DR program and any installed technologies in the building systems manual or include in the current facilities requirements and operations and maintenance plan if the project is not pursuing EA credit Enhanced Commissioning.
- Initiate at least one full test of the DR plan.

OR

**Case 2. Demand Response Capable Building (1 point)**

Have infrastructure in place to take advantage of future demand response programs or dynamic, real-time pricing programs and complete the following activities:

- Install interval recording meters with communications and ability for the building automation system to accept an external price or control signal.

- Develop a comprehensive plan for shedding at least 10% of the annual on-peak electricity demand. On-peak demand is determined under EA Prerequisite Minimum Energy Performance.
- Include the DR processes in the scope of work for the commissioning authority, including participation in at least one full test of the DR plan.
- Include the DR program and any installed technologies in the building systems manual or include in the current facilities requirements and operations and maintenance plan if the project is not pursuing EA credit Enhanced Commissioning.
- Contact local utility representatives to discuss participation in future DR programs.

AND / OR

**Case 3. Load Flexibility and Management Strategies (1-2 points)**

Analyze the building's annual load shape and peak load based as calculated for EA prerequisite Minimum Energy Performance. Review the regional grid load profile using the metric of peak load or peak carbon emissions. The U.S. Environmental Protection Agency's (EPA) AVoided Emissions and geneRation Tool (AVERT) provides regional grid emissions data; local utilities may also provide this data.

Coordinate review of building load shape and peak load with review of the regional grid profile to identify the best value load management strategies that the building can provide.

Implement one or more of the load flexibility and management strategies described below for a maximum of up to two points. All projects must install interval recording meters with communications and the ability for the building automation system to accept an external price signal.

Load Flexibility and Management Strategies:

- Peak Load Optimization: demonstrate that strategy reduces on-peak load by at least 10% as compared to peak electrical demand (1 point)
- Flexible Operating Scenarios: demonstrate that strategy moves at least 10% of peak load by a time period of 2 hours (1 point)
- On-site thermal and/or electricity storage: demonstrate that strategy reduces on-peak load by at least 10% as compared to peak electrical demand (1 point)
- Grid resilience technologies: project served by utilities with resilience programs in place, which leverage strategies such as islanding and part-load operation, automatically achieve this credit (1 point)

Include installed technology in the scope of work for the commissioning authority. Include load flexibility and management strategies and installed technologies in the building systems manual, or include in the current facilities requirements and operations maintenance plan if the project is not pursuing EA credit Enhanced Commissioning

Contact local utility representatives to discuss participation in future DR programs and to inform utility of building load flexibility and management strategies.

# EA CREDIT: RENEWABLE ENERGY

BD+C
**1–5 points**
This credit applies to

- New Construction (1–5 points)
- Core & Shell (1–5 points)
- Schools (1–5 points)
- Retail (1–5 points)
- Data Centers (1–5 points)
- Warehouses & Distribution Centers (1–5 points)
- Hospitality (1–5 points)
- Healthcare (1–5 points)

**Intent**

To reduce the environmental and economic harms associated with fossil fuel energy and reduce greenhouse gas emissions by increasing the supply of renewable energy and carbon mitigation projects.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Use on-site renewable energy systems, procure renewable energy from offsite sources, or offset the greenhouse gas emissions from all or a portion of the building's annual energy use.

Choose one or more strategies for renewable energy procurement from the categories below. Points achieved in each category may be added for a total of 5 points.

- Tier 1: On-site renewable energy generation
- Tier 2: Off-site renewable energy produced by a generation asset(s) built within the last 5 years, or contracted to be operational within one year of building occupancy, and generated by either:
    - an asset(s) in the project's grid subregion
      or
    - an asset(s) in a grid subregion with higher greenhouse gas emissions rates**
- Tier 3: Off-site renewable energy that is produced by a generation asset(s) built within the last 5 years or contracted to be operational within one year of building occupancy
- Tier 4: Off-site renewable energy that is Green-e Energy certified or equivalent
- Tier 5: Off-site renewable energy that is produced by a generation asset(s) that meet Green-e's certification criteria (or equivalent) for eligible renewables, has a mechanism to prevent double counting in place, and is third-party certified to an ecolabel standard.

***Grid subregions with average emissions rates that are higher than the national average emissions rate OR subregions with average marginal emissions rates that are higher than the national average marginal emissions rate.*

For all tiers, renewable energy must be contracted, owned or leased for a period of time between 1 and 15 years. Contract lengths less than 15 years are pro-rated linearly based on 1 and 15 year values in Table 1. Shorter contract lengths may require procurement of renewable energy in quantities that exceed the annual energy use of the building. For all tiers, the attributes (age, grid subregion, etc.) of the generation asset(s) are assessed at the beginning of the contract, and the generation asset(s) retain these attributes for the duration of the initial contract or lease term.

All eligible renewable electricity generation used to comply with this credit shall be substantiated through Energy Attribute Certificates (EACs); the project owner must contract for renewable energy and demonstrate that the EACs are retained on behalf of the LEED project.

EACs must be purchased from renewable energy projects located in the same country or region where the LEED project is located.

Tier 1, Tier 2, and Tier 3 renewable energy may contribute to GHG reductions in EA credit Optimize Energy Performance. When claiming credit in EA credit Optimize Energy Performance, projects are required to calculate GHG emissions and avoided emissions using hourly electricity GHG emissions factors for the project's grid subregion.

Points are awarded according to Table 1, based on the percentage of total site energy use.

Table 1. Points for Renewable Energy Procurement

| **Points** | **Tier 1** | **Tier 2** | | **Tier 3** | | **Tier 4** | | **Tier 5** | |
|---|---|---|---|---|---|---|---|---|---|
| | | 15-Year | 1-Year | 15-Year | 1-Year | 15-Year | 1-Year | 15-Year | 1-Year |
| 1 | 2% | 20% | 150% | 30% | 225% | 40% | 300% | 50% | 375% |
| 2 | 5% | 30% | 225% | 40% | 300% | 60% | 450% | 75% | 562.5% |
| 3 | 10% | 40% | 300% | 50% | 375% | 80% | 600% | | |
| 4 | 20% | 50% | 375% | 60% | 450% | | | | |
| 5 | 40% | 60% | 450% | 70% | 525% | | | | |
| EP | 60% | 70% | 525% | 80% | 600% | 100% | 750% | 100% | 750% |

**AND/OR**

*Carbon Offsets*

Procure carbon offsets to offset annual building greenhouse gas emissions from energy use. Carbon offsets must be contracted for at least 15 years.

Points are awarded according to Table 2, based on the percentage of total annual energy emissions offset by the purchase of carbon offsets.

Table 2. Points for Carbon Offsets Procurement

| Points | Green-e Climate certified or equivalent Carbon Offsets |
|---|---|
| 1 | 100% |
| 2 | 200% |

# EA CREDIT: ENHANCED REFRIGERANT MANAGEMENT

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1 point)

**Intent**

To reduce ozone depletion and support early compliance with the Montreal Protocol while minimizing direct contributions to climate change.

**Requirements**

**NC, CS, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

**Option 1. No Refrigerants or Low-Impact Refrigerants (1 point)**
Do not use refrigerants, or use only refrigerants (naturally occurring or synthetic) that have an ozone depletion potential (ODP) of zero and a global warming potential (GWP) of less than 50.

OR

**Option 2. Calculation of Refrigerant Impact (1 point)**
Select refrigerants that are used in heating, ventilating, air-conditioning, and refrigeration (HVAC&R) equipment to minimize or eliminate the emission of compounds that contribute to ozone depletion and climate change. The combination of all new and existing base building and tenant HVAC&R equipment that serve the project must comply with the following formula:

| **IP units** | **SI units** |
|---|---|
| $\frac{LCGWP}{} + \frac{LCODP}{} \times \frac{10}{5} \leq 100$ | $\frac{LCGWP}{} + \frac{LCODP}{} \times \frac{10}{5} \leq 13$ |
| **Calculation definitions for LCGWP + LCODP x $10^5$ ≤ 100 (IP units)** | **Calculation definitions for LCGWP + LCODP x $10^5$ ≤ 13 (SI units)** |
| LCODP = [ODPr x (Lr x Life +Mr) x Rc]/Life | LCODP = [ODPr x (Lr x Life +Mr) x Rc]/Life |
| LCGWP = [GWPr x (Lr x Life +Mr) x Rc]/Life | LCGWP = [GWPr x (Lr x Life +Mr) x Rc]/Life |
| LCODP: Lifecycle Ozone Depletion Potential (lb CFC 11/Ton-Year) | LCODP: Lifecycle Ozone Depletion Potential (kg CFC 11/(kW/year)) |
| LCGWP: Lifecycle Direct Global Warming Potential (lb $CO_2$/Ton-Year) | LCGWP: Lifecycle Direct Global Warming Potential (kg $CO_2$/kW-year) |

| | |
|---|---|
| GWPr: Global Warming Potential of Refrigerant (0 to 12,000 lb $CO_2$/lbr) | GWPr: Global Warming Potential of Refrigerant (0 to 12,000 kg $CO_2$/kg r) |
| ODPr: Ozone Depletion Potential of Refrigerant (0 to 0.2 lb CFC 11/lbr) | ODPr: Ozone Depletion Potential of Refrigerant (0 to 0.2 kg CFC 11/kg r) |
| Lr: Refrigerant Leakage Rate (2.0%) | Lr: Refrigerant Leakage Rate (2.0%) |
| Mr: End-of-life Refrigerant Loss (10%) | Mr: End-of-life Refrigerant Loss (10%) |
| Rc: Refrigerant Charge (0.5 to 5.0 lbs of refrigerant per ton of gross AHRI rated cooling capacity) | Rc: Refrigerant Charge (0.065 to 0.65 kg of refrigerant per kW of AHRI rated or Eurovent Certified cooling capacity) |
| Life: Equipment Life (10 years; default based on equipment type, unless otherwise demonstrated) | Life: Equipment Life (10 years; default based on equipment type, unless otherwise demonstrated) |

For multiple types of equipment, calculate a weighted average of all base building HVAC&R equipment, using the following formula:

| **IP units** | **SI units** |
|---|---|
| $\frac{[\sum (LCGWP + LCODP \times 10^5) \times Qunit]}{Qtotal} \leq 100$ | $\frac{[\sum (LCGWP + LCODP \times 10^5) \times Qunit]}{Qtotal} \leq 13$ |

| **Calculation definitions for $[\sum (LCGWP + LCODP \times 10^5) \times Qunit] / Qtotal \leq 100$ (IP units)** | **Calculation definitions for $[\sum (LCGWP + LCODP \times 10^5) \times Qunit] / Qtotal \leq 13$ (SI units)** |
|---|---|
| Qunit = Gross AHRI rated cooling capacity of an individual HVAC or refrigeration unit (Tons) | Qunit = Eurovent Certified cooling capacity of an individual HVAC or refrigeration unit (kW) |
| Qtotal = Total gross AHRI rated cooling capacity of all HVAC or refrigeration | Qtotal = Total Eurovent Certified cooling capacity of all HVAC or refrigeration (kW) |

**RETAIL NC**

Meet Option 1 or 2 for all HVAC systems.

Stores with commercial refrigeration systems must comply with the following.

- Use only non-ozone-depleting refrigerants.
- Select equipment with an average HFC refrigerant charge of no more than 1.75 pounds of refrigerant per 1,000 Btu/h (2.72 kg of refrigerant per kW) total evaporator cooling load.

- Demonstrate a predicted store-wide annual refrigerant emissions rate of no more than 15%. Conduct leak testing using the procedures in GreenChill's best practices guideline for leak tightness at installation.

Alternatively, stores with commercial refrigeration systems may provide proof of attainment of EPA GreenChill's silver-level store certification for newly constructed stores.

**BD+C**

# MATERIALS AND RESOURCES (MR)

## MR PREREQUISITE: STORAGE AND COLLECTION OF RECYCLABLES Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**

To reduce the waste that is generated by building occupants and hauled to and disposed of in landfills.

**Requirements**

**NC, CS, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE**

Provide dedicated areas accessible to waste haulers and building occupants for the collection and storage of recyclable materials for the entire building. Collection and storage areas may be separate locations. Recyclable materials must include mixed paper, corrugated cardboard, glass, plastics, and metals. Take appropriate measures for the safe collection, storage, and disposal of two of the following: batteries, mercury-containing lamps, and electronic waste.

**RETAIL NC**

Conduct a waste stream study to identify the retail project's top five recyclable waste streams, by either weight or volume, using consistent metrics. Based on the waste stream study, list the top four waste streams for which collection and storage space will be provided. If no information is available on waste streams for the project, use data from similar operations to make projections. Retailers with existing stores of similar size and function can use historical information from their other locations.

Provide dedicated areas accessible to waste haulers and building occupants for the separation, collection, and storage of recyclable materials for at least the top four recyclable waste streams identified by the waste study. Locate the collection and storage bins close the source of recyclable waste. If any of the top four waste streams are batteries, mercury-containing lamps, or electronic waste, take appropriate measures for safe collection, storage, and disposal.

## MR PREREQUISITE: CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT PLANNING
## Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**

To reduce construction and demolition waste disposed of in landfills and incineration facilities by recovering, reusing, and recycling materials.

**Requirements**

**NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE**

Develop and implement a construction and demolition waste management plan:

- Establish waste diversion goals for the project by identifying at least five materials (both structural and nonstructural) targeted for diversion.
- Specify whether materials will be separated or comingled and describe the diversion strategies planned for the project. Describe where the material will be taken and how the recycling facility will process the material including expected diversion rates for each material stream.

Provide a final report detailing all major waste streams generated, including disposal and diversion rates.

Alternative daily cover (ADC) does not qualify as material diverted from disposal. Include materials destined for ADC in the calculations as waste. Land-clearing debris is not considered construction, demolition, or renovation waste that can contribute to waste diversion.

# MR PREREQUISITE: PBT SOURCE REDUCTION—MERCURY
# Required

BD+C
This prerequisite applies to
- Healthcare

**Intent**

To reduce mercury-containing products and devices and mercury release through product substitution, capture, and recycling.

**Requirements**

**HEALTHCARE**

As part of the project's recycling collection system, identify the following:
- types of mercury-containing products and devices to be collected;
- criteria governing how they are to be handled by a recycling program; and
- disposal methods for captured mercury.

Applicable mercury-containing products and devices include, but are not limited to, lamps (such as linear and circular fluorescents, integrally ballasted and nonintegrally ballasted compact fluorescents and HIDs) and dental wastes (such as scrap amalgam, chair side traps, and separator wastes).

In facilities delivering dental care, specify and install amalgam separation devices that meet or exceed the ISO-11143 standard.

Comply with the mercury elimination requirements outlined below, from the 2010 FGI Guidelines for Design and Construction of Health Care Facilities, Section A1.3- 4b, Mercury Elimination.
- 4.2.1.1. New construction: healthcare facilities may not use mercury-containing equipment, including thermostats, switching devices, and other building system sources. Lamps are excluded.
- 4.2.1.2. Renovation: healthcare facilities must develop a plan to phase out mercury-containing products and upgrade current mercury-containing lamps to high-efficiency, low-mercury, or mercury-free lamp technology.

Do not specify or install preheat, T-9, T-10, or T-12 fluorescents or mercury vapor high-intensity discharge (HID) lamps in the project. Do not specify probe-start metal halide HID lamps in any interior spaces.

Specify and install illuminated exit signs that do not contain mercury and use less than 5 watts of electricity.

Fluorescent and high-pressure sodium lamps must meet the criteria in Table 1.

**Table 1. Maximum mercury content of lamps**

| *Lamp* | *Maximum content* |
|---|---|
| T-8 fluorescent, eight-foot | 10 mg mercury |
| T-8 fluorescent, four-foot | 3.5 mg mercury |
| T-8 fluorescent, U-bent | 6 mg mercury |
| T-5 fluorescent, linear | 2.5 mg mercury |
| T-5 fluorescent, circular | 9 mg mercury |
| Compact fluorescent, nonintegral ballast | 3.5 mg mercury |

| | |
|---|---|
| Compact fluorescent, integral ballast | 3.5 mg mercury, ENERGY STAR qualified |
| High-pressure sodium, up to 400 watts | 10 mg mercury |
| High-pressure sodium, above 400 watts | 32 mg mercury |

mg = milligram

# MR CREDIT: BUILDING LIFE-CYCLE IMPACT REDUCTION

BD+C
**2–6 points**
This credit applies to

- New Construction (2–5 points)
- Core & Shell (2–6 points)
- Schools (2–5 points)
- Retail (2–5 points)
- Data Centers (2–5 points)
- Warehouses & Distribution Centers (2–5 points)
- Hospitality (2–5 points)
- Healthcare (2–5 points)

**Intent**

To encourage adaptive reuse and optimize the environmental performance of products and materials.

**Requirements**

**NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE**

Demonstrate reduced environmental effects during initial project decision-making by reusing existing building resources or demonstrating a reduction in materials use through life-cycle assessment. Achieve one of the following options.

**Option 1. Historic Building Reuse (5 points BD+C, 6 points Core and Shell)**

Maintain the existing building structure, envelope, and interior nonstructural elements of a historic building or contributing building in a historic district. To qualify, the building or historic district must be listed or eligible for listing in the local, state, or national register of historic places. Do not demolish any part of a historic building or contributing building in a historic district unless it is deemed structurally unsound or hazardous. For buildings listed locally, approval of any demolition must be granted by the local historic preservation review board. For buildings listed in a state register or the U.S. National Register of Historic Places (or local equivalent for projects outside the U.S.), approval must appear in a programmatic agreement with the state historic preservation office or National Park Service (or local equivalent for projects outside the U.S.).

Any alteration (preservation, restoration, or rehabilitation) of a historic building or a contributing building in a historic district on the project site must be done in accordance with local or national standards for rehabilitation, whichever are applicable. If building is not subject to historic review, include on the project team a preservation professional who meets U.S. federal qualifications for historic architects (or local equivalent for projects outside the U.S.); the preservation professional must confirm conformance to the Secretary of Interior's Standards for the Treatment of Historic Properties (or local equivalent for projects outside the U.S.).

OR

**Option 2. Renovation of Abandoned or Blighted Building (5 points BD+C, 6 points Core and Shell)**

Maintain at least 50%, by surface area, of the existing building structure, enclosure, and interior structural elements for buildings that meet local criteria of abandoned or are considered blight. The

building must be renovated to a state of productive occupancy. Up to 25% of the building surface area may be excluded from credit calculation because of deterioration or damage.

OR

**Option 3. Building and Material Reuse (1–4 points BD+C, 2-5 points Core and Shell)**

Materials contributing toward this credit may not contribute toward MR Credit Material Disclosure and Optimization – Sourcing of Raw Materials. Path 1 or Path 2 (a/b) may be attempted but combining Path 1 and Path 2 to achieve points is not allowed.

Path 1: Maintain A Combination Of Structural And Non-Structural Elements (2-4 points)

Reuse or salvage building materials from off site or on site as a percentage of the surface area, as listed in Table 1. Include structural elements (e.g., floors, roof decking), enclosure materials (e.g., skin, framing), and permanently installed interior elements (e.g., walls, doors, floor coverings, ceiling systems). Exclude from the calculation window assemblies and any hazardous materials that are remediated as a part of the project.

Table 1. Points for reuse of building materials

| *Percentage of completed project surface area reused* | *Points BD+C* | *Points BD+C (Core and Shell)* |
|---|---|---|
| 25% | 2 | 2 |
| 50% | 3 | 3 |
| 75% | 4 | 5 |

OR

Path 2: Maintain Existing Walls, Floors and Roofs (1-3 points):

Maintain the existing building structure (including floor and roof decking) and envelope (the exterior skin and framing, excluding window assemblies and nonstructural roofing materials).

| *Percent of existing walls, floors and roof reuse* | *Points* |
|---|---|
| 25% | 1 |
| 50% | 2 |
| 75% | 3 |

AND/OR

Path: Maintain Interior Nonstructural Elements (1 point)

Use existing interior nonstructural elements (e.g. interior walls, doors, floor coverings and ceiling systems) in at least 33% (by area) of the completed building, including additions.

OR

**Option 4. Whole-Building Life-Cycle Assessment (1-4 points)**

For new construction (buildings or portions of buildings), conduct a life-cycle assessment of the project's structure and enclosure and select one or more of the following paths below to earn up to 4 points:

Path 1: Conduct a life cycle assessment of the project's structure and enclosure (1 point).

Path 2: Conduct a life cycle assessment of the project's structure and enclosure that demonstrates a minimum of 5% reduction, compared with a baseline building in at least three of the six impact categories listed below, one of which must be global warming potential (2 points).

Path 3: Conduct a life cycle assessment of the project's structure and enclosure that demonstrates a minimum of 10% reduction, compared with a baseline building, in at least three of the six impact categories listed below, one of which must be global warming potential (3 points).

Path 4: Meet requirements of Path 3 and incorporate building reuse and/or salvage materials into the project's structure and enclosure for the proposed design. Demonstrate reductions compared with a baseline building of at least 20% reduction for global warming potential and demonstrate at least 10% reduction in two additional impact categories listed below (4 points).

For Paths 2, 3 and 4 listed above, no impact category assessed as part of the life-cycle assessment may increase by more than 5% compared with the baseline building. Include a narrative of how the life cycle assessment was conducted and if applicable for paths 2, 3 and 4 what changes were made to proposed buildings in order to achieve the related impact reductions.

The baseline and proposed buildings must be of comparable size, function, orientation, and operating energy performance as defined in EA Prerequisite Minimum Energy Performance. The service life of the baseline and proposed buildings must be the same and at least 60 years to fully account for maintenance and replacement. Use the same life-cycle assessment software tools and data sets to evaluate both the baseline building and the proposed building, and report all listed impact categories. Data sets must be compliant with ISO 14044.

Select at least three of the following impact categories for reduction:

- global warming potential (greenhouse gases), in kg $CO_2e$;
- depletion of the stratospheric ozone layer, in kg CFC-11e;
- acidification of land and water sources, in moles H+ or kg $SO_2e$;
- eutrophication, in kg nitrogen eq or kg phosphate eq;
- formation of tropospheric ozone, in kg NOx, kg O3 eq, or kg ethene; and
- depletion of nonrenewable energy resources, in MJ using CML / depletion of fossil fuels in TRACI.

Healthcare only

For all options in this credit, building materials demolished to create courtyards to increase daylighting may be counted as retained in calculations, provided the new courtyards meet the requirements of EQ Credits Daylight and Quality Views.

# MR CREDIT: BUILDING PRODUCT DISCLOSURE AND OPTIMIZATION—ENVIRONMENTAL PRODUCT DECLARATIONS

BD+C
**1–2 points**
This credit applies to

- New Construction (1–2 points)
- Core & Shell (1–2 points)
- Schools (1–2 points)
- Retail (1–2 points)
- Data Centers (1–2 points)
- Warehouses & Distribution Centers (1–2 points)
- Hospitality (1–2 points)
- Healthcare (1–2 points)

**Intent**

To encourage the use of products and materials for which life-cycle information is available and that have environmentally, economically, and socially preferable life-cycle impacts. To reward project teams for selecting products from manufacturers who have verified improved environmental life-cycle impacts.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Achieve one or more of the options below, for a maximum of 2 points.

**Option 1. Environmental Product Declaration (EPD) (1 point)**
Use at least 20 different permanently installed products sourced from at least five different manufacturers that meet one of the disclosure criteria below. (10 different permanently installed products from three different manufacturers for CS and Warehouses & Distribution Centers).

- Life-cycle assessment and environmental product declarations.
    - Products with a publicly available, critically reviewed life-cycle assessment conforming to ISO 14044 that have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.

    - Product-specific Type III EPD -- Internally Reviewed. Products with an internally critically reviewed LCA in accordance with ISO 14071. Products with product-specific internal EPDs which conform to ISO 14025, and EN 15804 or ISO 21930 and have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.

    - Industry-wide Type III EPD -- Products with third-party certification (Type III), including external verification, in which the manufacturer is explicitly recognized as a participant by the program operator. Products with industry-wide EPDs, which conform to ISO 14025, and EN 15804 or ISO 21930 and have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.

- Environmental Product Declarations which conform to ISO 14025 and EN 15804 or ISO 21930 and have at least a cradle to gate scope.

- Product-specific Type III EPD -- Products with third-party certification (Type III), including external verification and external critical review in which the manufacturer is explicitly recognized as the participant by the program operator are valued as 1.5 products for the purposes of credit achievement calculation.

- USGBC approved program – Products that comply with other USGBC approved environmental product declaration frameworks.

**Option 2. Multi-Attribute Optimization (1 point)**
Use products that comply with one of the criteria below for 10%, by cost, of the total value of permanently installed products in the project, or use at least 10 permanently installed products sourced from at least three different manufacturers. Products will be valued as below.

Life Cycle Impact Reduction Action Plan (value at 50% by cost or ½ product)

The manufacturer has produced a product specific LCA using EN 15804 or ISO 21930 for the product and has provided a publicly available action plan to mitigate or reduce life cycle impacts. The action plan must be product-specific using the specified PCR functional unit, be critically reviewed, and must include the following information:

- Description of the LCA conducted including the dataset, software or platform used by manufacturer to complete the analysis.
- Identification of the largest life cycle impact areas identified in the analysis and a narrative description of the impact areas targeted for reduction in the action plan.
- Description of specific steps anticipated in implementation of the action plan. Include proposed changes in formulation or manufacturing processes that are planned as part of impact reduction strategy.
- Specific dates and a full timeline for completion of all the steps described in the action plan.

Life Cycle Impact Reductions in Embodied Carbon.

Products that have demonstrated environmental impact reductions for the specified functional unit based on a current third-party EPD or verified LCA that conforms to the comparability requirements of ISO 14025 and ISO 21930.

- The comparative analysis must show impact reduction in the global warming potential (GWP) impact category and must include a narrative describing how reductions in impacts were achieved. The published comparisons must be third-party verified (value at 100% by cost or 1 product).
- The comparative analysis must show impact reduction(s) of at least 10% in the global warming potential (GWP) impact category and must include a narrative describing how the impact reductions were achieved. The published comparisons must be third-party verified (value at 150% by cost or 1.5 products).
- The comparative analysis must show impact reduction(s) of at least 20% in the global warming potential (GWP) impact category, and demonstrate at least 5% reduction in two additional impact categories. A narrative describing how the impact reductions were achieved is required. The published comparisons must be third-party verified (value at 200% by cost or 2 products).

Impact categories:

- global warming potential (greenhouse gases), in $CO_2e$;
- depletion of the stratospheric ozone layer, in kg CFC-11e;

- acidification of land and water sources, in moles H+ or kg $SO_2$e;
- eutrophication, in kg nitrogen equivalent or kg phosphate equivalent;
- formation of tropospheric ozone, in kg NOx, kg O3 eq, or kg ethene; and
- depletion of nonrenewable energy resources, in MJ using CML / depletion of fossil fuels in TRACI.

USGBC approved program -- Products that comply with other USGBC approved multi-attribute frameworks.

For credit achievement calculation, products sourced (extracted, manufactured, purchased) within 100 miles (160 km) of the project site are valued at twice their base contributing cost (or number of products), up to a maximum of 200% of cost or 2 products.

# MR CREDIT: BUILDING PRODUCT DISCLOSURE AND OPTIMIZATION – SOURCING OF RAW MATERIALS

BD+C
**1–2 points**
This credit applies to

- New Construction (1–2 points)
- Core & Shell (1–2 points)
- Schools (1–2 points)
- Retail (1–2 points)
- Data Centers (1–2 points)
- Warehouses & Distribution Centers (1–2 points)
- Hospitality (1–2 points)
- Healthcare (1–2 points)

**Intent**

To encourage the use of products and materials for which life cycle information is available and that have environmentally, economically, and socially preferable life cycle impacts. To reward project teams for selecting products verified to have been extracted or sourced in a responsible manner.

**Requirements**

**NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE**

**Responsible Sourcing of Raw Materials (1-2points)**
Use products sourced from at least three different manufacturers that meet at least one of the responsible sourcing and extraction criteria below for at least 20%, by cost, of the total value of permanently installed building products in the project (1 point).

Use products sourced from at least five different manufacturers that meet at least one of the responsible sourcing and extraction criteria below for at least 40%, by cost, of the total value of permanently installed building products in the project (2 points).

- *Extended producer responsibility.* Products purchased from a manufacturer (producer) that participates in an extended producer responsibility program or is directly responsible for extended producer responsibility. Products meeting extended producer responsibility criteria are valued at 50% of their cost for the purposes of credit achievement calculation.

- *Bio-based materials.* Bio-based raw materials other than wood must be tested using ASTM Test Method D6866 and be legally harvested, as defined by the exporting and receiving country. Exclude hide products, such as leather and other animal skin material.

    - Bio-based products that meet the criteria above: value at 50% of cost multiplied by the biobased content of the product for the purposes of credit achievement calculation.
    - Bio-based products that meet the Sustainable Agriculture Network's Sustainable Agriculture Standard: value at 100% of cost multiplied by the biobased content of the product for the purposes of credit achievement calculation.

- *Wood products.* Wood products must be certified by the Forest Stewardship Council or USGBC-approved equivalent. Products meeting wood products criteria are valued at 100% of their cost for the purposes of credit achievement calculation.

- *Materials reuse.* Reuse includes salvaged, refurbished, or reused products. Products meeting materials reuse criteria are valued at 200% of their cost for the purposes of credit achievement calculation.

- *Recycled content.* Products meeting recycled content criteria are valued at 100% of their cost for the purposes of credit achievement calculation.

    - Recycled content is the sum of postconsumer recycled content plus one-half the preconsumer recycled content, based on weight.
    - The recycled fraction of the assembly is then multiplied by the cost of assembly to determine the recycled content value.

- *USGBC approved program.* Other USGBC approved programs meeting responsible sourcing and extraction criteria.

For credit achievement calculation, products sourced (extracted, manufactured and purchased) within 100 miles (160 km) of the project site are valued at twice their base contributing cost, upt to a maximum of 200% of cost or 2 products.

## MR CREDIT: BUILDING PRODUCT DISCLOSURE AND OPTIMIZATION – MATERIAL INGREDIENTS

BD+C
**1-2 points**
This credit applies to

- New Construction (1-2 points)
- Core & Shell (1-2 points)
- Schools (1-2 points)
- Retail (1-2 points)
- Data Centers (1-2 points)
- Warehouses & Distribution Centers (1-2 points)
- Hospitality (1-2 points)
- Healthcare (1-2 points)

**Intent**

To encourage the use of products and materials for which life-cycle information is available and that have environmentally, economically, and socially preferable life-cycle impacts. To reward project teams for selecting products for which the chemical ingredients in the product are inventoried using an accepted methodology and for selecting products verified to minimize the use and generation of harmful substances. To reward raw material manufacturers who produce products verified to have improved life-cycle impacts.

**Requirements**

**NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE**

**Option 1. Material Ingredient Reporting (1 point)**
Use at least 20 different permanently installed products from at least five different manufacturers that use any of the following programs to demonstrate the chemical inventory of the product to at least 0.1% (1000

ppm). (10 different permanently installed products from at least three different manufacturers for CS and Warehouses & Distribution Centers)

- *Manufacturer Inventory*. The manufacturer has published complete content inventory for the product following these guidelines:
  - A publicly available inventory of all ingredients identified by name and Chemical Abstract Service Registration Number (CASRN) and/or European Community Number (EC Number).
  - Materials defined as trade secret or intellectual property may withhold the name and/or CASRN/EC Number but must disclose ingredient/chemical role, amount and hazard score/class using either:
    - Greenscreen List Translator (LT) score and/or Full GreenScreen Benchmark (BM)
    - The Globally Harmonized System of Classification and Labeling of Chemicals rev.6 (2015) (GHS)
      - The hazard screen must be applied to each trade secret ingredient and the inventory lists the hazard category for each of the health hazards included in Part 3 of GHS (e.g. "GHS Category 2 Carcinogen").
      - .

- *Health Product Declaration.* The end use product has a published and complete Health Product Declaration with full disclosure of known hazards in compliance with the Health Product Declaration open Standard.
- *Cradle to Cradle.* Product has Material Health Certificate or is Cradle to Cradle Certified™ under standard version 3 or later with a Material Health achievement level at the Bronze level or higher.
- Declare. The Declare product label must meet the following requirements:
  - Declare labels designated as Red List Free or Declared.
  - Declare labels designated as LBC Compliant that demonstrate content inventory to 0.1% (1000 ppm).
- Living Product Challenge. The included Declare product label must demonstrate content inventory to 0.1% (1000 ppm).
- ANSI/BIFMA e3 Furniture Sustainability Standard. The documentation from the assessor or scorecard from BIFMA must demonstrate the product earned at least 3 points under 7.5.1.3 Advanced Level in e3-2014 or 3 points under 7.4.1.3 Advanced Level in e3-2012.
- Product Lens Certification
- Facts – NSF/ANSI 336: Sustainability Assessment for Commercial Furnishings Fabric at any certification level.
- Global Green TAG PHD labels issued after January 1, 2020

Any compliant reports above with third-party verification that includes the verification of content inventory are worth 1.5 products for credit achievement calculations.

AND/OR

**Option 2: Material Ingredient Optimization (1 point)**
Use permanently installed products from at least three different manufacturers that document their material ingredient optimization using the paths below. Choose either 10 compliant products, or select products that constitute at least 10%, by cost, of the total value of permanently installed products in the project.

Material Ingredient Screening and Optimization Action Plan (value at 50% by cost or ½ product)

- The manufacturer has screened the product to at least 1,000 ppm and has provided a publicly available inventory meeting the requirements of Option 1 and completed a detailed action plan to mitigate or reduce known hazards using the principles of green chemistry. The action plan must be product-specific (not company, manufacturer or brand), and must include the following information:
    - Description of the screening or assessment platform used by manufacturer to complete the material ingredient screening and analysis.
    - Identification of the specific green chemistry principles targeted for implementation in the action plan.
    - Description of specific steps anticipated in implementation of the action plan. Include proposed changes in formulation or manufacturing processes that are planned as part of green chemistry optimization strategy.
    - Specific dates and a full timeline for completion of all the steps described in the action plan.

Advanced Inventory & Assessment (value at 100% by cost or 1 product):

- The end use product meets the requirements of any of the following:
    - Manufacturer Inventory or Health Product Declaration: The product has demonstrated a chemical inventory to at least 0.01% by weight (100 ppm) with no GreenScreen LT-1 hazards or GHS Category 1 hazards. The HPD or Manufacturer Inventory must be third party verified.
    - Manufacturer Inventory or HPD: The product has demonstrated a chemical inventory to at least 0.01% by weight (100ppm) and at least 75% by weight of product is assessed using GreenScreen Benchmark assessment. The remaining 25% by weight of product has been inventoried. The GreenScreen assessment must be publicly available. The HPD or Manufacturer Inventory must be third-party verified.
    - Declare labels designated as Red List Free that are third-party verified, or Living Product Challenge certified products that include a Red List Free Declare label.
    - Cradle to Cradle. Product has Material Health Certificate or is Cradel to Cradle Certified™ under standard version 3 or later with a Material Health achievement level at the Bronze level or higher.

Material Ingredient Optimization (value at 150% by cost or 1.5 products)

- The end use product has demonstrated a product inventory and assessment of ingredients using any of the following programs:
    - Manufacturer Inventory or HPD: The product has demonstrated a chemical inventory to at least 0.01% by weight (100ppm) and at least 95% by weight of product is assessed using GreenScreen Benchmark assessment. No Benchmark 1 hazards (BM-1) are present in the end use product. The remaining 5% by weight of product not assessed has been inventoried and screened using GreenScreen List Translator and no GreenScreen LT-1 hazards are present in the end use product. The documents must be third party verified.
    - Cradle to Cradle v3 certified product with Material Health category score of Silver or higher, or a Cradle to Cradle certified Material Health Certificate at Silver level or higher.
    - Living Product Challenge. Products certified to the Living Product Challenge which includes achievement of Imperative 09: Transparent Material Health.

*International Alternative Compliance Path – REACH Optimization (value at 100% of cost or 1 product)*.

- End use products and materials have fully inventoried chemical ingredients to 100 ppm and assess each substance against the Authorization List – Annex XIV, the Restriction list – Annex XVII and the SVHC candidate list, (the version in effect June 2013,) proving that no such substance is included in the product. If the product contains no ingredients listed on the REACH Authorization, Restriction, and Candidate list.

- Global Green Tag International: product has a certified Product Health Declaration (PhD) report. Value at 100% or 1 product.

*USGBC approved program.*

- Products that comply with USGBC approved building product optimization criteria for material ingredient optimization and/or advanced inventory & assessment pathways.

For credit achievement calculation, products sourced (extracted, manufactured, purchased) within 100 miles (160 km) of the project site are valued at twice their base contributing cost (or number of products), up to a maximum of 200% of cost or 2 products.

# MR CREDIT: PBT SOURCE REDUCTION—MERCURY

BD+C
**1 point**
This credit applies to

- Healthcare

**Intent**

To reduce the release of persistent, bioaccumulative, and toxic (PBTs) chemicals associated with the life cycle of building materials.

**Requirements**

**HEALTHCARE**

Specify and install fluorescent lamps with both low mercury content (MR Prerequisite PBT Source Reduction—Mercury) and long lamp life, as listed in Table 1.

**Table 1. Criteria for rated life of low-mercury lamps**

| *Lamp* | Maximum content | *Lamp life (hrs)* |
|---|---|---|
| T-8 fluorescent, eight-foot | 10 mg mercury | Standard output - 24,000 rated hours on instant start ballasts (3-hour starts)<br>High output – 18,000 rated hours on instant start ballasts or program start ballasts (3-hour starts) |
| T-8 fluorescent, four-foot | 3.5 mg mercury | Both standard and high output - 30,000 rated hours on instant start ballasts, or 36,000 rated hours on program start ballasts (3 hour starts) |
| T-8 fluorescent, two-foot and three-foot | 3.5 mg mercury | 24,000 rated hours on instant start ballasts or program start ballasts (3-hour starts) |
| T-8 fluorescent, U-bent | 6 mg mercury | 18,000 rated hours on instant start ballasts, or 24,000 rated hours on program start ballasts (3-hour starts) |
| T-5 fluorescent, linear | 2.5 mg mercury | Both standard and high-output - 25,000 rated hours on program start ballasts |
| T-5 fluorescent, circular | 9 mg mercury | Both standard and high-output – 25,000 rated hours on program start ballasts |
| Compact fluorescent, nonintegral ballast | 3.5 mg mercury | 12,000 rated hours |
| Compact florescent, integral ballast, bare bulb | 3.5 mg mercury, ENERGY STAR qualified | Bare bulb - 10,000 rated hours<br>Covered models such as globes, reflectors, A-19s – 8,000 hours |
| | | |

| | | |
|---|---|---|
| High-pressure sodium, up to 400 watts | 10 mg mercury | Use noncycling type or replace with LED lamps or induction lamps |
| High-pressure sodium, above 400 watts | 32 mg mercury | Use noncycling type or replace with LED lamps or induction lamps |

Do not specify or install circular fluorescent lamps or probe start metal halide lamps.

# MR CREDIT: PBT SOURCE REDUCTION—LEAD, CADMIUM, AND COPPER

BD+C
**2 points**
This credit applies to
- Healthcare

**Intent**

To reduce the release of persistent, bioaccumulative, and toxic (PBT) chemicals associated with the life cycle of building materials.

**Requirements**

**HEALTHCARE**

Specify substitutes for materials manufactured with lead and cadmium, as follows.

**Lead**
- For water intended for human consumption, specify and use solder and flux to connect plumbing pipe on site that meets the California AB1953 standard, which specifies that solder not contain more than 0.2% lead, and flux not more than a weighted average of 0.25% lead for wetted surfaces. The “lead free” label as defined by the Safe Drinking Water Act (SDWA)) does not provide adequate screening for the purposes of this credit because the SDWA defines “lead free” as solders and flux containing 0.2% lead or less.
- For water intended for human consumption, specify and use pipes, pipe fittings, plumbing fittings, and faucets that meet the California law AB1953 of a weighted average lead content of the wetted surface area of not more than 0.25% lead.
- Specify and use lead-free roofing and flashing.
- Specify and use electrical wire and cable with lead content less than 300 parts per million.
- Specify no use of interior or exterior paints containing lead.
- For renovation projects, ensure the removal and appropriate disposal of disconnected wires with lead stabilizers, consistent with the 2002 National Electric Code requirements.

Lead used for radiation shielding and copper used for MRI shielding are exempt.

**Cadmium**
- Specify no use of interior or exterior paints containing intentionally added cadmium.

**Copper**
- For copper pipe applications, reduce or eliminate joint-related sources of copper corrosion:
  - use mechanically crimped copper joint systems; or
  - specify that all solder joints comply with ASTM B828 2002, and specify and use ASTM B813 2010 for flux.

# MR CREDIT: FURNITURE AND MEDICAL FURNISHINGS

BD+C
**1–2 points**
This credit applies to

- Healthcare

**Intent**

To enhance the environmental and human health performance attributes associated with freestanding furniture and medical furnishings.

**Requirements**

**HEALTHCARE**

Use at least 30% (1 point) or 40% (2 points), by cost, of all freestanding furniture and medical furnishings (e.g., mattresses, foams, panel fabrics, cubicle curtains, window coverings, other textiles) that meet the criteria in one of the following three options.

Include built-in casework and built-in millwork in the base building calculations, even if manufactured off site. The dollar value of any individual product may be included in the total qualifying value if the product meets the criteria.

**Option 1. Minimal Chemical Content**

All components that constitute at least 5%, by weight, of a furniture or medical furnishing assembly, including textiles, finishes, and dyes, must contain less than 100 parts per million (ppm) of at least four of the five following chemical groups:

- urea formaldehyde;
- heavy metals, including mercury, cadmium, lead, and antimony;
- hexavalent chromium in plated finishes consistent with the European Union Directive on the Restriction of the Use of Certain Hazardous Substances (EU RoHS);
- stain and nonstick treatments derived from perfluorinated compounds (PFCs), including perfluorooctanoic acid (PFOA); and
- added antimicrobial treatments.

AND/OR

**Option 2. Testing and Modeling of Chemical Content**

All components of a furniture or medical furnishing assembly, including textiles, finishes, and dyes, must contain less than 100 parts per million (ppm) of at least two of the five chemicals or materials listed in Option 1.

New furniture or medical furnishing assemblies must be in accordance with ANSI/BIFMA Standard Method M7.1–2011. Comply with ANSI/BIFMA e3-2010 Furniture Sustainability Standard, Sections 7.6.1 and 7.6.2, using either the concentration modeling approach or the emissions factor approach. Model the test results using the open plan, private office, or seating scenario in ANSI/BIFMA M7.1, as appropriate. USGBC-approved equivalent testing methodologies and contaminant thresholds are also acceptable. Documentation submitted for furniture must indicate the modeling scenario used to determine compliance.

Salvaged and reused furniture more than one year old at the time of use is considered compliant, provided it meets the requirements for any site-applied paints, coatings, adhesives, and sealants.
AND/OR

**Option 3: Multi-Attribute Assessment of Products**

Use products that meet at least one of the criteria below. Each product can receive credit for each criterion met. The scope of any environmental product declaration (EPD) must be at least cradle to gate.

- Life-cycle assessment and environmental product declarations.
    - Products with a publicly available, critically reviewed life-cycle assessment conforming to ISO 14044 that have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.

    - Product-specific Type III EPD -- Internally Reviewed. Products with an internally critically reviewed LCA in accordance with ISO 14071. Products with product-specific internal EPDs which conform to ISO 14025, and EN 15804 or ISO 21930 and have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.

    - Industry-wide Type III EPD -- Products with third-party certification (Type III), including external verification, in which the manufacturer is explicitly recognized as a participant by the program operator. Products with industry-wide EPDs, which conform to ISO 14025, and EN 15804 or ISO 21930 and have at least a cradle to gate scope are valued as one whole product for the purposes of credit achievement calculation.

- Environmental Product Declarations which conform to ISO 14025, 14040, 14044, and EN 15804 or ISO 21930 and have at least a cradle to gate scope.
    - Product-specific Type III EPD -- Products with third-party certification (Type III), including external verification and external critical review in which the manufacturer is explicitly recognized as the participant by the program operator are valued as 1.5 products for the purposes of credit achievement calculation.

- USGBC approved program – Products that comply with other USGBC approved environmental product declaration frameworks.

- *Extended producer responsibility.* Products purchased from a manufacturer (producer) that participates in an extended producer responsibility program or is directly responsible for extended producer responsibility. Products meeting extended producer responsibility criteria are valued at 50% of their cost for the purposes of credit achievement calculation.

- *Bio-based materials.* Bio-based raw materials <u>other than wood</u> must be tested using ASTM Test Method D6866 and be legally harvested, as defined by the exporting and receiving country. Exclude hide products, such as leather and other animal skin material.

    - Bio-based products <u>that meet the criteria above: value at 50% of cost multiplied by the biobased content of the product for the purposes of credit achievement calculation.</u>
    - <u>Bio-based products that meet the Sustainable Agriculture Network's Sustainable Agriculture Standard: value at 100% of cost multiplied by the biobased content of the product for the purposes of credit achievement calculation.</u>

- *Wood products.* Wood products must be certified by the Forest Stewardship Council or USGBC-approved equivalent. Products meeting wood products criteria are valued at 100% of their cost for the purposes of credit achievement calculation.

- *Materials reuse.* Reuse includes salvaged, refurbished, or reused products. Products meeting materials reuse criteria are valued at 200% of their cost for the purposes of credit achievement calculation.

- *Recycled content.* Products meeting recycled content criteria are valued at 100% of their cost for the purposes of credit achievement calculation.

    - Recycled content is the sum of postconsumer recycled content plus one-half the preconsumer recycled content, based on weight.
    - The recycled fraction of the assembly is then multiplied by the cost of assembly to determine the recycled content value.

Products that meet the above criteria are valued according to source location (extraction, manufacture, and purchase point must be within the distances noted below):

For credit achievement calculation, products sourced (extracted, manufactured, purchased) within 100 miles (160 km) of the project site are valued at 200% of their base contributing cost.

# MR CREDIT: DESIGN FOR FLEXIBILITY

BD+C
**1 point**
This credit applies to

- Healthcare

**Intent**

Conserve resources associated with the construction and management of buildings by designing for flexibility and ease of future adaptation and for the service life of components and assemblies.

**Requirements**

**HEALTHCARE**

Increase building flexibility and ease of adaptive use over the life of the structure by employing at least three of the following strategies.

- Use *interstitial space*. Design distribution zone utility systems and equipment including HVAC, plumbing, electrical, information technology, medical gases, and life safety systems to serve the occupied zones and have the capacity to control multiple zones in clinical spaces.
- Provide programmed soft space, such as administration or storage, equal to at least 5% of *departmental gross area* (DGA). Locate soft space adjacent to clinical departments that anticipate growth. Determine a strategy for future accommodation of displaced soft space.
- Provide shell space equal to at least 5% of DGA. Locate it such that it can be occupied without displacing occupied space.
- Identify horizontal expansion capacity for diagnostic and treatment or other clinical space equal to at least 30% of existing floor area (excluding inpatient units) without demolition of occupied space (other than at the connection point). Reconfiguration of additional existing occupied space that has been constructed with demountable partition systems is permitted.
- Design for future vertical expansion on at least 75% of the roof, ensuring that existing operations and service systems can continue at or near capacity during the expansion.
- Designate space for future above-grade parking structures equal to 50% of existing on-grade parking capacity, with direct access to the main hospital lobby or circulation. Vertical transportation pathways that lead directly to the main hospital lobby or circulation are acceptable.
- Use demountable partitions for 50% of applicable areas.
- Use movable or modular casework for at least 50% of casework and custom millwork. Base the calculation on the combined value of casework and millwork, as determined by the cost estimator or contractor.

# MR CREDIT: CONSTRUCTION AND DEMOLITION WASTE MANAGEMENT

BD+C
**1–2 points**
This credit applies to

- New Construction (1–2 points)
- Core & Shell (1–2 points)
- Schools (1–2 points)
- Retail (1–2 points)
- Data Centers (1–2 points)
- Warehouses & Distribution Centers (1–2 points)
- Hospitality (1–2 points)
- Healthcare (1–2 points)

**Intent**

To reduce construction and demolition waste disposed of in landfills and incineration facilities by recovering, reusing, and recycling materials.

**Requirements**

**NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY NC, HEALTHCARE**

Recycle and/or salvage nonhazardous construction and demolition materials. Calculations can be by weight or volume but must be consistent throughout.

Exclude excavated soil, land-clearing debris from calculations. Include materials destined for alternative daily cover (ADC) in the calculations as waste (not diversion). Include wood waste converted to fuel (biofuel) in the calculations; other types of waste-to-energy are not considered diversion for this credit.

However, for international projects that cannot meet credit requirements using reuse and recycling methods, waste-to-energy systems may be considered waste diversion if the European Commission Waste Framework Directive 2008/98/EC and Waste Incineration Directive 2000/76/EC are followed and Waste to Energy facilities meet applicable European Committee for Standardization (CEN) EN 303 standards.

**Option 1. Diversion (1–2 points)**

**Path 1. Divert 50% and Two Material Streams (1 point)**
Divert at least 50% of the total construction and demolition material; diverted materials must include at least two material streams.

OR

**Path 2. Divert 50% using Certified Commingled Recycling Facility (1 Point)**
Divert at least 50% of the total construction and demolition material. All commingled recycling must be sent to offsite sorting facility(ies) certified by the Recycling Certification Institute or approved equivalent.

OR

**Path 3. Divert 75% and Three Material Streams (2 points)**

Divert at least 75% of the total construction and demolition material; diverted materials must include at least three material streams.

OR

**Path 4. Divert 75% using Certified Commingled Recycling Facility and One More Material Stream (2 points)**
Divert at least 75% of the total construction and demolition material; diverted materials must include at least two material streams. All commingled recycling is required to be one of the streams and must be sent to offsite sorting facility(ies) certified by the Recycling Certification Institute or approved equivalent.

OR

**Option 2. Reduction of Total (Construction and Demolition) Waste Material (2 points)**

Salvage or recycle renovation and demolition debris and utilize onsite waste minimizing design strategies for new construction activities. Achieve the waste generation thresholds in Table 1 and create a narrative describing how a project is addressing waste prevention and/or achieving waste generation thresholds via design strategies and onsite waste minimization practices.

Do not generate more than 7.5 pounds of construction waste per square foot (36.6 kilograms of waste per square meter) of the building's floor area for all BD+C projects except Warehouses and Distribution Centers.

**Table 1: Thresholds of compliance for reduction of total waste material**

| Type of Waste | Building Type | Total Waste Reduction Requirements | Points Available |
|---|---|---|---|
| Waste from renovation and demolition Activities | All | Salvage or recycle at least 75%, not including ADC | Required |
| Waste from new construction Activities | All except Warehouses & Distribution Centers | Reduce waste generation to less than 7.5 lbs/sf | 2 |
| | Warehouse & Distribution Center projects | Reduce waste generation to less than 2.5 lbs/sf | |

# INDOOR ENVIRONMENTAL QUALITY (EQ)

## EQ PREREQUISITE: MINIMUM INDOOR AIR QUALITY PERFORMANCE Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**

To contribute to the comfort and well-being of building occupants by establishing minimum standards for indoor air quality (IAQ).

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY**
**Mechanically Ventilated Spaces**
For mechanically ventilated spaces (and for mixed-mode systems when the mechanical ventilation is activated), meet the requirements for both ventilation (option 1 or option 2) and monitoring.

**Mechanical Ventilation**
**Option 1. ASHRAE Standard 62.1–2016**
Meet the requirements of ASHRAE Standard 62.1–2016, Sections 4, 5, 6.2, 6.5, and 7, or a local equivalent, whichever is more stringent.

OR

**Option 2. ISO 17772-1:2017 and EN 16798-3: 2017 and**
Projects outside the U.S. may instead meet the requirements of ISO Standard 17772-1:2017, Section 6.3, using Method 1—Perceived air quality with Category I or II and local standard for ventilation system design such as EN Standard 16798-3: 2017, Sections 7-10.

**Monitoring for mechanical ventilation systems**
Provide outdoor air monitors for all mechanical ventilation systems with outdoor air intake flow greater than 1000 cfm (472 L/s). The monitoring device must be capable of measuring the minimum outdoor air intake flow and be capable of measuring the design minimum outdoor air intake flow with an accuracy of +/–10%. An alarm must indicate when the outdoor airflow value varies by 15% or more from the setpoint.

Alternatively, for constant-volume systems that do not employ demand control ventilation, provide an indicator capable of confirming the intake damper is open to the position needed to maintain the design minimum outdoor airflow as determined during the system startup and balancing.

**Naturally Ventilated Spaces**

For naturally ventilated spaces (and for mixed-mode systems when the mechanical ventilation is inactivated), confirm that natural ventilation is an effective strategy for the project by following the flow diagram in the Chartered Institution of Building Services Engineers (CIBSE) Applications Manual AM10, March 2005, Natural Ventilation in Nondomestic Buildings, Figure 2.8 and meet the requirements for both ventilation (option 1, option 2, or option 3) and monitoring.

**Natural Ventilation**

- **Option 1. ASHRAE Standard 62.1-2016**
  Meet the requirements of ASHRAE 62.1-2016, Sections 4, 6.4, and 6.5.

  OR

- **Option 2. Engineered natural ventilation system**
  Meet the requirements of ASHRAE 62.1-2016, Sections 4 and 6.5, and have an engineered natural ventilation system approved by the authority having jurisdiction (per exception 1 of ASHRAE 62.1-2016 section 6.4).

  OR

- **Option 3. Historic building**
  This option is available to projects located in a building registered as a local or national historic building.

  Meet the requirements of ASHRAE 62.1-2016, Sections 4, 6.4.1, 6.4.2, 6.4.3, and 6.5.

**Monitoring for natural ventilation systems**
Comply with at least one of the following strategies.

- Provide a direct exhaust airflow measurement device capable of measuring the exhaust airflow. This device must measure the exhaust airflow with an accuracy of +/–10% of the design minimum exhaust airflow rate. An alarm must indicate when airflow values vary by 15% or more from the exhaust airflow setpoint. *This strategy is not allowed for projects using Natural Ventilation Option 3. Historic building*.
- Provide automatic indication devices on all natural ventilation openings intended to meet the minimum opening requirements. An alarm must indicate when any one of the openings is closed during occupied hours.
- Monitor carbon dioxide ($CO_2$) concentrations within each thermal zone. $CO_2$ monitors must be between 3 and 6 feet (900 and 1 800 millimeters) above the floor and within the thermal zone. $CO_2$ monitors must have an audible or visual indicator or alert the building automation system if the sensed $CO_2$ concentration exceeds the setpoint by more than 10%. Calculate appropriate $CO_2$ setpoints using the methods in ASHRAE 62.1–2016, Appendix D.

**All Spaces**
The indoor air quality procedure defined in ASHRAE Standard 62.1–2016, Section 6.3 may not be used to comply with this prerequisite.

<u>CS only</u>
Mechanical ventilation systems installed during core and shell construction must be capable of meeting projected ventilation levels and monitoring based on the requirements of anticipated future tenants.

<u>Residential only</u>
In addition to the requirements above, if the project building contains residential units, each dwelling unit must meet the requirements of LEED v4.1 Multifamily EQ Prerequisite: Minimum Indoor Air Quality Performance, EQ Prerequisite Combustion venting and EQ Prerequisite Radon-resistant construction.

**HEALTHCARE**

Meet the requirements of ASHRAE Standard 170-2017, Sections 6-10, and meet the requirements above for monitoring for mechanical ventilation systems.

# EQ PREREQUISITE: ENVIRONMENTAL TOBACCO SMOKE CONTROL
# Required

BD+C
This prerequisite applies to

- New Construction
- Core & Shell
- Schools
- Retail
- Data Centers
- Warehouses & Distribution Centers
- Hospitality
- Healthcare

**Intent**

To prevent or minimize exposure of building occupants, indoor surfaces, and ventilation air distribution systems to environmental tobacco smoke.

**Requirements**

**NC, CS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

For this prerequisite smoking includes tobacco smoke, as well as smoke produced from the combustion of cannabis and controlled substances and the emissions produced by electronic smoking devices.

Prohibit smoking inside the building.

Prohibit smoking outside the building except in designated smoking areas located at least 25 feet (7.5 meters) (or the maximum extent allowable by local codes) from all entries, outdoor air intakes, and operable windows. This smoking requirement also applies to any spaces outside the property line that are used for business purposes.

Communicate the no-smoking policy to occupants. Have in place provisions for enforcement or no-smoking signage.
Residential only

**Option 1. No Smoking**
Meet the requirements above.

OR

**Option 2. Compartmentalization of Smoking Areas**
Meet the requirements above for all areas inside and outside the building except dwelling units and private balconies.

Each dwelling unit where smoking is permitted must be compartmentalized to prevent excessive leakage between units:

- Weather-strip all exterior doors and operable windows in the residential units to minimize leakage from outdoors.
- Weather-strip all doors leading from residential units into common hallways.

- Minimize uncontrolled pathways for the transfer of smoke and other indoor air pollutants between residential units by sealing penetrations in the walls, ceilings, and floors and by sealing vertical chases (including utility chases, garbage chutes, mail drops, and elevator shafts) adjacent to the units.
- Demonstrate a maximum leakage of 0.30 cubic feet per minute per square foot (1.53 liters per second per square meter) at 50 Pa of enclosure (i.e., all surfaces enclosing the apartment, including exterior and party walls, floors, and ceilings). Renovation projects that retain their existing envelope must meet an allowable maximum leakage of 0.50 cfm50 per square foot (2.54 liters per second per square meter) of enclosure area.

**SCHOOLS**

Prohibit smoking on site.

Communicate the no-smoking policy to occupants. Have in place provisions for enforcement or no-smoking signage.

# EQ PREREQUISITE: MINIMUM ACOUSTIC PERFORMANCE
# Required

BD+C
This prerequisite applies to

- Schools

**Intent**

To provide classrooms that facilitate teacher-to-student and student-to-student communication through effective acoustic design.

**Requirements**

**SCHOOLS**

**HVAC Background Noise**
Achieve a maximum background noise level of 40 dBA from heating, ventilating, and air-conditioning (HVAC) systems in classrooms and other core learning spaces. Follow the recommended methodologies and best practices for mechanical system noise control in ANSI Standard S12.60–2010, Part 1, Annex A.1; the 2015 ASHRAE Handbook-- HVAC Applications, Chapter 48, Noise and Vibration Control (with errata); AHRI Standard 885–2008; or a local equivalent for projects outside the U.S.

**Exterior Noise**
For high-noise sites (peak-hour Leq above 60 dBA during school hours), implement acoustic treatment and other measures to minimize noise intrusion from exterior sources and control sound transmission between classrooms and other core learning spaces. Projects at least one-half mile (800 meters) from any significant noise source (e.g., aircraft overflights, highways, trains, industry) are exempt.

**Reverberation Time**
Adhere to the following reverberation time requirements.

**Classrooms and Core Learning Spaces < 20,000 Cubic Feet (566 Cubic Meters)**
Design classrooms and other core learning spaces to include sufficient sound-absorptive finishes for compliance with the reverberation time requirements specified in ANSI Standard S12.60–2010, Part 1, Acoustical Performance Criteria, Design Requirements and Guidelines for Schools, or a local equivalent for projects outside the U.S.

**Option 1**
For each room, confirm that the total surface area of acoustic wall panels, ceiling finishes, and other sound-absorbent finishes equals or exceeds the total ceiling area of the room (excluding lights, diffusers, and grilles). Materials must have an NRC of 0.70 or higher to be included in the calculation.

OR

**Option 2**
Confirm through calculations described in ANSI Standard S12.60-2010 that rooms are designed to meet reverberation time requirements as specified in that standard.

**Classrooms and Core Learning Spaces ≥ 20,000 Cubic Feet (566 Cubic Meters)**
Meet the recommended reverberation times for classrooms and core learning spaces described in the NRC-CNRC Construction Technology Update No. 51, Acoustical Design of Rooms for Speech (2002), or a local equivalent for projects outside the U.S.

**Exceptions**

Exceptions to the requirements because of a limited scope of work or to observe historic preservation requirements will be considered.

# EQ CREDIT: ENHANCED INDOOR AIR QUALITY STRATEGIES

BD+C
**1–2 points**
This credit applies to

- New Construction (1–2 points)
- Core & Shell (1–2 points)
- Schools (1–2 points)
- Retail (1–2 points)
- Data Centers (1–2 points)
- Warehouses & Distribution Centers (1–2 points)
- Hospitality (1–2 points)
- Healthcare (1–2 points)

**Intent**

To promote occupants' comfort, well-being, and productivity by improving indoor air quality.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

**Option 1. Enhanced IAQ Strategies (1 point)**

Comply with all of the following requirements, as applicable.

Mechanically ventilated spaces:

A. entryway systems;
B. interior cross-contamination prevention; and
C. filtration.

Naturally ventilated spaces:

A. entryway systems; and
D. natural ventilation design calculations.

Mixed-mode systems:

A. entryway systems;
B. interior cross-contamination prevention;
C. filtration;
D. natural ventilation design calculations; and
E. mixed-mode design calculations.

***A. Entryway Systems***

Install permanent entryway systems at least 10 feet (3 meters) long in the primary direction of travel to capture dirt and particulates entering the building at regularly used exterior entrances. Acceptable entryway systems include permanently installed grates, grilles, slotted systems that allow for cleaning underneath, rollout mats, and any other materials manufactured as entryway systems with equivalent or better performance. Maintain all on a weekly basis.

Warehouses & Distribution Centers only
Entryway systems are not required at doors leading from the exterior to the loading dock or garage but must be installed between these spaces and adjacent office areas.

Healthcare only
In addition to the entryway system, provide pressurized entryway vestibules at high-volume building entrances.

***B. Interior Cross-Contamination Prevention***

Sufficiently exhaust each space where hazardous gases or chemicals may be present or used (e.g., garages, housekeeping and laundry areas, copying and printing rooms), using the exhaust rates determined in EQ Prerequisite Minimum Indoor Air Quality Performance or a minimum of 0.50 cfm per square foot (2.54 l/s per square meter), to create negative pressure with respect to adjacent spaces when the doors to the room are closed. For each of these spaces, provide self-closing doors and deck-to-deck partitions or a hard-lid ceiling.

***C. Filtration***

Each ventilation system that supplies outdoor air to occupied spaces must have particle filters or air-cleaning devices that meet one of the following filtration media requirements:

- minimum efficiency reporting value (MERV) of 13 or higher, in accordance with ASHRAE Standard 52.2–2017; or
- Equivalent filtration media class of $ePM_1$ 50% or higher, as defined by ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance.

Replace all air filtration media after completion of construction and before occupancy.

Data Centers only
The above filtration media requirements are required only for ventilation systems serving regularly occupied spaces.

***D. Natural Ventilation Design Calculations***

Demonstrate that the system design for occupied spaces employs the appropriate strategies in Chartered Institution of Building Services Engineers (CIBSE) Applications Manual AM10, March 2005, Natural Ventilation in Non-Domestic Buildings, Section 2.4.

***E. Mixed-Mode Design Calculations***

Demonstrate that the system design for occupied spaces complies with CIBSE Applications Manual 13–2000, Mixed Mode Ventilation.

**Option 2. Additional Enhanced IAQ Strategies (1 point)**

Comply with one of the following requirements

Mechanically ventilated spaces (select one):

A. exterior contamination prevention;
B. increased ventilation;
C. carbon dioxide monitoring; or
D. additional source control and monitoring.

Naturally ventilated spaces (select one):

A. exterior contamination prevention;
D. additional source control and monitoring; or
E. natural ventilation room by room calculations.

Mixed-mode systems (select one):

A. exterior contamination prevention;
B. increased ventilation;
D. additional source control and monitoring; or
E. natural ventilation room-by-room calculations.

***A. Exterior Contamination Prevention***

Design the project to minimize and control the entry of pollutants into the building. Ensure through the results of computational fluid dynamics modeling, Gaussian dispersion analyses, wind tunnel modeling, or tracer gas modeling that outdoor air contaminant concentrations at outdoor air intakes are below the thresholds listed in Table 1 (or local equivalent for projects outside the U.S., whichever is more stringent).

Table 1. Maximum concentrations of pollutants at outdoor air intakes

| Pollutants | Maximum concentration | Standard |
|---|---|---|
| Those regulated by National Ambient Air Quality Standards (NAAQS) | Allowable annual average<br>OR<br>8-hour or 24-hour average where an annual standard does not exist<br>OR<br>Rolling 3-month average | National Ambient Air Quality Standards (NAAQS) |

***B. Increased Ventilation***

Increase breathing zone outdoor air ventilation rates to all occupied spaces by at least 30% above the minimum rates as determined in EQ Prerequisite Minimum Indoor Air Quality Performance.

***C. Carbon Dioxide Monitoring***

Monitor $CO_2$ concentrations within all densely occupied spaces. $CO_2$ monitors must be between 3 and 6 feet (900 and 1 800 millimeters) above the floor. $CO_2$ monitors must have an audible or visual indicator or alert the building automation system if the sensed $CO_2$ concentration exceeds the setpoint by more than 10%. Calculate appropriate $CO_2$ setpoints using methods in ASHRAE 62.1–2016, Appendix D.

***D. Additional Source Control and Monitoring***

For spaces where air contaminants are likely, evaluate potential sources of additional air contaminants besides $CO_2$. Develop and implement a materials-handling plan to reduce the likelihood of contaminant release. Install monitoring systems with sensors designed to detect the specific contaminants. An alarm must indicate any unusual or unsafe conditions.

***E. Natural Ventilation Room-by-Room Calculations***

Follow CIBSE AM10, Section 4, Design Calculations, to predict that room-by-room airflows will provide effective natural ventilation.

# EQ CREDIT: LOW-EMITTING MATERIALS

BD+C
**1–3 points**
This credit applies to

- New Construction (1–3 points)
- Core & Shell (1–3 points)
- Schools (1–3 points)
- Retail (1–3 points)
- Data Centers (1–3 points)
- Warehouses & Distribution Centers (1–3 points)
- Hospitality (1–3 points)
- Healthcare (1–3 points)

**Intent**

To reduce concentrations of chemical contaminants that can damage air quality, human health, productivity, and the environment.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Use materials on the building interior (everything within the waterproofing membrane) that meet the low-emitting criteria below. Points are awarded according to Table 1:

| Table 1. Points for low-emitting materials | |
|---|---|
| 2 product categories | 1 point |
| 3 product categories | 2 points |
| 4 product categories | 3 points |
| 5 product categories | 3 points + exemplary performance |
| Reach 90% threshold in at least three product categories | Exemplary performance *or 1 additional point if only 1 or 2 points achieved above.* |

**Paints and Coatings**

At least 75% of all paints and coatings, **by volume or surface area**, meet the *VOC emissions evaluation* AND 100% meet the *VOC content evaluation*.

The paints and coatings product category includes all interior paints and coatings applied on site.

**Adhesives and Sealants**

At least 75% of all adhesives and sealants, **by volume or surface area**, meet the *VOC emissions evaluation* AND 100% meet the *VOC content evaluation*.

The adhesives and sealants product category includes all interior adhesives and sealants applied on site.

**Flooring**

At least 90% of all flooring, **by cost or surface area**, meets the *VOC emissions evaluation OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria*.

The flooring product category includes all types of hard and soft surface flooring (carpet, ceramic, vinyl, rubber, engineered, solid wood, laminates), wall base, underlayments, and other floor coverings.

Subflooring is excluded.

**Wall panels**

At least 75% of all wall panels, **by cost or surface area**, meet the *VOC emissions evaluation, OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria.*

The wall panels product category includes all finish wall treatments (wall coverings, wall paneling, wall tile), surface wall structures such as gypsum or plaster, cubicle/curtain/partition walls, trim, doors, frames, windows, and window treatments.

Removable/interchangeable fabric panels, built-in cabinetry, and vertical structural elements are excluded.

**Ceilings**

At least 90% of all ceilings, **by cost or surface area**, meet the *VOC emissions evaluation, OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria.*

The ceilings product category includes all ceiling panels, ceiling tile, surface ceiling structures such as gypsum or plaster, suspended systems (including canopies and clouds), and glazed skylights.

Overhead structural elements (exposed, finished, and unfinished) are excluded.

**Insulation**

At least 75% of all insulation, by cost or surface area, meets the *VOC emissions evaluation.*

The insulation material category includes all thermal and acoustic boards, batts, rolls, blankets, sound attenuation fire blankets, foamed-in place, loose-fill, blown, and sprayed insulation.

Insulation for HVAC ducts and plumbing piping are excluded.

**Furniture**

At least 75% of all furniture in the project scope of work, **by cost**, meets the *furniture emissions evaluation, OR inherently nonemitting sources criteria, OR salvaged and reused materials criteria.*

The furniture product category includes all stand-alone furniture items purchased for the project.

**Composite Wood**

At least 75% of all composite wood**, by cost or surface area**, meets the *Formaldehyde emissions evaluation OR salvaged and reused materials criteria.*

The composite wood product category includes all particleboard, medium density fiberboard, hardwood veneer plywood, and structural composite wood not included in the flooring, ceiling, wall panels, or furniture material categories.

**Low-emitting criteria**

***Inherently nonemitting sources***
Product is an inherently nonemitting source of VOCs (stone, ceramic, powder-coated metals, plated or anodized metal, glass, concrete, clay brick, and unfinished or untreated solid wood) and has no integral organic-based surface coatings, binders, or sealants.

***Salvaged and reused materials***
Product is more than one year old at the time of use. If finishes are applied to the product on-site, the finishes must meet the *VOC emissions evaluation* AND *VOC content evaluation* requirements.

***VOC emissions evaluation***
Option 1. Product has been tested according to California Department of Public Health (CDPH) Standard Method v1.2–2017 and complies with the VOC limits in Table 4-1 of the method. Additionally, the range of total VOCs after 14 days (336 hours) was measured as specified in the CDPH Standard Method v1.2 and is reported (TVOC ranges: 0.5 mg/m$^3$ or less, between 0.5 and 5 mg/m$^3$, or 5 mg/m$^3$ or more).

Laboratories that conduct the tests must be accredited under ISO/IEC 17025 for the test methods they use. Products used in school classrooms must be evaluated using the classroom scenario, products used in other spaces must be evaluated using the default private office scenario .

The statement of product compliance must include the exposure scenario(s) used, the amount of wet-applied product applied in mass per surface area (if applicable), the range of total VOCs, and follow guidelines in CDPH Standard Method v1.2-2017, Section 8. Organizations that certify manufacturers' claims must be accredited under ISO Guide 17065.

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Option 2. Product has been tested according to EN 16516:2017 and complies with the LCI values from Table 1 of the German AgBB Testing and Evaluation Scheme (2015) and a formaldehyde limit of 10 micrograms per cubic meter. Additionally, the range of total VOCs after 28 days was measured as specified in EN 16516 and reported (TVOC ranges: 0.5 mg/m$^3$ or less, between 0.5 and 5 mg/m$^3$, or 5 mg/m$^3$ or more). Laboratories that conduct the tests must be accredited under ISO/IEC 17025 for the test methods they use.

The statement of product compliance must include the amount of wet-applied product applied in mass per surface area (if applicable) and the range of total VOCs. Organizations that certify manufacturers' claims must be accredited under ISO Guide 17065.

***VOC content evaluation***

Product meets the VOC content limits outlined in one of the applicable standards and for projects in North America, methylene chloride and perchloroethylene may not be intentionally added.

Statement of product compliance must be made by the manufacturer. Any testing must follow the test method specified in the applicable regulation. If the applicable regulation requires subtraction of exempt compounds, any content of intentionally added exempt compounds larger than 1% weight by mass (total exempt compounds) must be disclosed.

- Paints and coatings:

    - California Air Resource Board (CARB) 2007 Suggested Control Measure (SCM) for Architectural Coatings
    - South Coast Air Quality Management District (SCAQMD) Rule 1113, effective February 5, 2016
    - European Decopaint Directive (2004/42/EC)
    - Hong Kong Air pollution control (VOC) Regulation for regulated architectural paints (January 2010)
- Adhesives and sealants:
    - SCAQMD Rule 1168, October 6, 2017
    - Canadian VOC Concentration Limits for Architectural Coatings (SOR/2009-264)
    - Hong Kong Air Pollution Control (VOC) Regulation for regulated adhesives and regulated sealants (April 2012)
    - Free of solvents, as defined in TRGS 610 (January 2011)

***Formaldehyde Emissions Evaluation***

Product meets one of the following:

- EPA TSCA Title VI or California Air Resources Board (CARB) ATCM for formaldehyde requirements for ultra-low-emitting formaldehyde (ULEF) resins or
- EPA TSCA Title VI or CARB ATCM formaldehyde requirements for no added formaldehyde resins (NAF).
- Tested per EN 717-1:2014 for formaldehyde emissions and complies with emissions class E1.Structural composite wood product made with moisture resistant adhesives meeting ASTM 2559, no surface treatments with added urea-formaldehyde resins or coatings, and certified according to one of the following industry standards:
    - Plywood: compliant in accordance with *Voluntary Product Standard - Structural Plywood (PS 1-09), Voluntary Product Standard – Performance Standard for Wood-Based Structural-Use Panels (PS 2-10),* or one of the standards considered by CARB to be equivalent to PS 1 or PS 2: (AS/NZS 2269, EN 636 3S (including CE label), Canadian Standards Association CSA O121 for Douglas fir plywood, CSA O151 for Canadian softwood plywood, for CSA O153 Poplar plywood, or CSAO325 for Construction sheathing)
    - Oriented strand board: specified with the Exposure 1 or Exterior bond classification in accordance with *Voluntary Product Standard – Performance Standard for Wood-Based Structural-Use Panels (PS 2-10)*
    - Structural composite lumber: compliant in accordance with *Standard Specification for Evaluation of Structural Composite Lumber Products (ASTM D 5456-13)*
    - Glued laminated timber: compliant in accordance with *Structural Glued Laminated Timber (ANSI A190.1-2012)*
    - I-joists compliant in accordance with *Standard Specification for Establishing and Monitoring Structural Capacities of Prefabricated Wood I-Joists (ASTM D 5055-13)*
    - Cross-laminated timber: compliant in accordance with *Standard for Performance-Rated Cross-Laminated Timber (PRG 320-15)*
    - Finger-jointed lumber labeled “Heat Resistant Adhesive (HRA)” in accordance with the *American Softwood Lumber Standard (DOC PS-20 2015)*

***Furniture Emissions Evaluation***

Product has been tested in accordance with ANSI/BIFMA Standard Method M7.1–2011 (R2016) and complies with ANSI/BIFMA e3-2014e Furniture Sustainability Standard, Sections 7.6.1 (for half credit, by cost) OR 7.6.2 (for full credit, by cost). If 75% of the furniture also complies with Section 7.6.3 in addition to 7.6.2, the category counts for exemplary level (90%). Laboratories that conduct the tests must be accredited under ISO/IEC 17025 for the test methods they use.

Seating products must be evaluated using the seating scenario. Classroom furniture must be evaluated using the standard school classroom scenario. Other products should be evaluated using the open plan or private office scenario, as appropriate. The open plan scenario is more stringent.

Statements of product compliance must include the exposure scenario(s). Organizations that certify manufacturers' claims must be accredited under ISO Guide 17065.

# EQ CREDIT: CONSTRUCTION INDOOR AIR QUALITY MANAGEMENT PLAN

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1 point)

**Intent**

To promote the well-being of construction workers and building occupants by minimizing indoor air quality problems associated with construction and renovation.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY**

Develop and implement an indoor air quality (IAQ) management plan for the construction and preoccupancy phases of the building. The plan must address all of the following.

During construction, meet or exceed all applicable recommended control measures of the Sheet Metal and Air Conditioning National Contractors Association (SMACNA) IAQ Guidelines for Occupied Buildings under Construction, 2nd edition, 2007, ANSI/SMACNA 008–2008, Chapter 3.

Protect absorptive materials stored on-site and installed from moisture damage.

Do not operate permanently installed air-handling equipment during construction unless filtration media with a minimum efficiency reporting value (MERV) of 8, as determined by ASHRAE 52.2–2017, with errata (or media with $ISO_{coarse}$ 90% or higher, as defined by ISO 16890-2016, Particulate Air Filters for General Ventilation, Determination of the Filtration Performance ), are installed at each return air grille and return or transfer duct inlet opening such that there is no bypass around the filtration media. Immediately before occupancy, replace all filtration media with the final design filtration media, installed in accordance with the manufacturer's recommendations.

Prohibit the use of smoking inside the building and within 25 feet (7.5 meters) of the building openings during construction. Smoking includes tobacco smoke, as well as smoke produced from the combustion of cannabis and controlled substances and the emissions produced by electronic smoking devices.

**HEALTHCARE**
*Moisture.* Develop and implement a moisture control plan to protect stored on-site and installed absorptive materials from moisture damage. Immediately remove from site and properly dispose of any materials susceptible to microbial growth and replace with new, undamaged materials. Also include strategies for protecting the building from moisture intrusion and preventing occupants' exposure to mold spores.

*Particulates.* Do not operate permanently installed air-handling equipment during construction unless filtration media with a minimum efficiency reporting value (MERV) of 8, as determined by ASHRAE 52.2–2017, with errata (or media with $ISO_{coarse}$ 90% or higher, as defined by ISO 16890-2016, Particulate Air

Filters for General Ventilation, Determination of the Filtration Performance ), are installed at each return air grille and return or transfer duct inlet opening such that there is no bypass around the filtration media. Immediately before occupancy, replace all filtration media with the final design filtration media, installed in accordance with the manufacturer's recommendations.

*VOCs.* Schedule construction procedures to minimize exposure of absorbent materials to VOC emissions. Complete painting and sealing before storing or installing "dry" materials, which may accumulate pollutants and release them over time. Store fuels, solvents, and other sources of VOCs separately from absorbent materials.

*Outdoor emissions.* For renovation projects involving waterproofing, repairing asphalt roofing, sealing parking lots, or other outdoor activities that generate high VOC emissions, develop a plan to manage fumes and avoid infiltration to occupied spaces. Comply with the procedures established by NIOSH, Asphalt Fume Exposures during the Application of Hot Asphalt to Roofs (Publication 2003–112).

*Tobacco.* Prohibit the use of tobacco products inside the building and within 25 feet (7.5 meters) of the building entrance during construction.

*Noise and vibration.* Develop a plan based on the British Standard (BS 5228) to reduce noise emissions and vibrations from construction equipment and other nonroad engines by specifying low-noise emission design or the lowest decibel level available that meets performance requirements in the British Standard. Construction crews must wear ear protection in areas where sound levels exceed 85 dB for extended periods.

*Infection control.* For renovations and additions adjacent to occupied facilities or phased occupancy in new construction, follow the FGI 2018 Guidelines for Design and Construction of Hospitals, Guidelines for Design and Construction of Outpatient Facilities, Guidelines for Design and Construction of Residential Health, Care, and Support Facilities and The Joint Commission Standards to establish an integrative infection control team comprising the owner, designer, and contractor to evaluate infection control risk and document the required precautions in a project-specific plan. Use the infection control risk assessment standard published by the American Society of Healthcare Engineering and the U.S. Centers for Disease Control and Prevention (CDC) as a guideline to assess risk and to select mitigation procedures for construction activities.

# EQ CREDIT: INDOOR AIR QUALITY ASSESSMENT

BD+C
**1–2 points**
This credit applies to

- New Construction (1–2 points)
- Schools (1–2 points)
- Retail (1–2 points)
- Data Centers (1–2 points)
- Warehouses & Distribution Centers (1–2 points)
- Hospitality (1–2 points)
- Healthcare (1–2 points)

**Intent**

To establish better quality indoor air in the building after construction and during occupancy.

**Requirements**

**NC, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**
Select one of the following two options, to be implemented after construction ends and the building has been completely cleaned. All interior finishes, such as millwork, doors, paint, carpet, acoustic tiles, and movable furnishings (e.g., workstations, partitions), must be installed, and major VOC punch list items must be finished. The options cannot be combined.

**Option 1. Flush-Out (1 point)**
**Path 1. Before Occupancy**
Install new filtration media and perform a building flush-out by supplying a total air volume of 14,000 cubic feet of outdoor air per square foot (4 267 140 liters of outdoor air per square meter) of gross floor area while maintaining an internal temperature of at least 60°F (15°C) and no higher than 80°F (27°C) and relative humidity no higher than 60%.

OR

**Path 2. During Occupancy**
If occupancy is desired before the flush-out is completed, the space may be occupied only after delivery of a minimum of 3,500 cubic feet of outdoor air per square foot (1 066 260 liters of outdoor air per square meter) of gross floor area while maintaining an internal temperature of at least 60°F (15°C) and no higher than 80°F (27°C) and relative humidity no higher than 60%.

Once the space is occupied, it must be ventilated at a minimum rate of 0.30 cubic foot per minute (cfm) per square foot of outdoor air (1.5 liters per second per square meter of outside air) or the design minimum outdoor air rate determined in EQ Prerequisite Minimum Indoor Air Quality Performance, whichever is greater. During each day of the flush-out period, ventilation must begin at least three hours before occupancy and continue during occupancy. These conditions must be maintained until a total of 14,000 cubic feet per square foot of outdoor air (4 270 liters of outdoor air per square meter) has been delivered to the space.

OR

**Option 2. Air Testing (1-2 points)**
After construction ends and before occupancy, but under ventilation conditions typical for occupancy, conduct baseline IAQ testing in occupied spaces for the contaminants listed in Path 1. Particulate matter

and inorganic gases (for 1 point) and/or Path 2. Volatile organic compounds (for 1 point). Retail projects may conduct the testing within 14 days of occupancy.

**Path 1. Particulate Matter and Inorganic Gases (1 point)**

Test for the particulate matter (PM) and inorganic gases listed in Table 1, using an allowed test method, and demonstrate the contaminants do not exceed the concentration limits listed in the table.

Table 1. Particulate Matter and inorganic gases

| **Contaminant (CAS#)** | **Concentration Limit** **(µg/m³)** | **Allowed Test Methods** |
|---|---|---|
| Carbon monoxide (CO) | 9 ppm; no more than 2 ppm above outdoor levels | ISO 4224<br>EPA Compendium Method IP-3<br>GB/T 18883-2002 for projects in China<br><br>Direct calibrated electrochemical instrument with accuracy of (+/- 2% ppm <50 ppm minimum accuracy). |
| PM 10 | ISO 14644-1:2015, cleanroom class of 8 or lower<br><br>50 µg/m³<br>Healthcare only: 20 µg/m³ | Particulate monitoring device with accuracy greater of 5 micrograms/m3 or 20% of reading and resolution (5 min average data) +/- 5 µg/m³ |
| PM 2.5 | 12 µg/m³ or 35 µg/m³** | |
| Ozone | 0.07 ppm | Monitoring device with accuracy greater of 5 ppb or 20% of reading and resolution (5 min average data) +/- 5 ppb<br><br>ISO 13964<br>ASTM D5149 -– 02<br>EPA designated methods for Ozone |

**Projects in areas with high ambient levels of PM2.5 (known EPA nonattainment areas for PM2.5, or local equivalent) must meet the 35 ug/m³ limit, all other projects should meet the 12 ug/m³ limit.

AND/OR

**Path 2. Volatile Organic Compounds (1 point)**

Perform a screening test for Total Volatile Organic Compounds (TVOC). Use ISO 16000-6, EPA TO-17, or EPA TO-15 to collect and analyze the air sample. Calculate the TVOC value per EN 16516:2017, CDPH Standard Method v1.2 2017 section 3.9.4, or alternative calculation method as long as full method description is included in test report. If the TVOC levels exceed 500 µg/m³, investigate for potential issues by comparing the individual VOC levels from the GC/MS results to associated cognizant authority health-based limits. Correct any identified issues and re-test if necessary.

Additionally, test for the individual volatile organic compounds listed in Table 2 using an allowed test method and demonstrate the contaminants do not exceed the concentration limits listed in the table. Laboratories that conduct the tests must be accredited under ISO/IEC 17025 for the test methods they use.

Exemplary performance is available for projects that test for the additional target volatile organic compounds specified in CDPH Standard Method v1.2-2017, Table 4-1 and do not exceed the full CREL levels for these compounds adopted by Cal/EPA OEHHA in effect on June 2016.

Table 2. Volatile organic compounds

| **Contaminant (CAS#)** | **Concentration Limit ($\mu g/m^3$)** | **Allowed Test Methods** |
|---|---|---|
| Formaldehyde 50-00-0 | 20 $\mu g/m^3$ (16 ppb) | ISO 16000-3, 4;<br>EPA TO-11a,<br>EPA comp. IP-6A<br>ASTM D5197-16 |
| Acetaldehyde 75-07-0 | 140 $\mu g/m^3$ | |
| Benzene 71-43-2 | 3 $\mu g/m^3$ | ISO 16000-6<br>EPA IP-1,<br>EPA TO-17,<br>EPA TO-15<br>ISO 16017-1, 2;<br>ASTM D6196-15 |
| Hexane (n-) 110-54-3 | 7000 $\mu g/m^3$ | |
| Naphthalene 91-20-3 | 9 $\mu g/m^3$ | |
| Phenol 108-95-2 | 200 $\mu g/m^3$ | |
| Styrene 100-42-5 | 900 $\mu g/m^3$ | |
| Tetrachloroethylene 127-18-4 | 35 $\mu g/m^3$ | |
| Toluene 108-88-3 | 300 $\mu g/m^3$ | |
| Vinyl acetate 108-05-4 | 200 $\mu g/m^3$ | |
| Dichlorobenzene (1,4-) 106-46-7 | 800 $\mu g/m^3$ | |
| Xylenes-total 108-38-3, 95-47-6, and 106-42-3 | 700 $\mu g/m^3$ | |

# EQ CREDIT: THERMAL COMFORT

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1 point)

**Intent**

To promote occupants' productivity, comfort, and well-being by providing quality thermal comfort.

**Requirements**

Meet the requirements for both thermal comfort design and thermal comfort control.

**Thermal Comfort Design**

**NC, SCHOOLS, RETAIL, DATA CENTERS, HOSPITALITY, HEALTHCARE**

**Option 1. ASHRAE Standard 55-2017**
Design heating, ventilating, and air-conditioning (HVAC) systems and the building envelope to meet the requirements of ASHRAE Standard 55–2017, Thermal Comfort Conditions for Human Occupancy with errata or a local equivalent.

For natatoriums, demonstrate compliance with ASHRAE HVAC Applications Handbook, 2015 edition, Chapter 5, Places of Assembly, Typical Natatorium Design Conditions, with errata.

OR

**Option 2. ISO Standards**
Design HVAC systems and the building envelope to meet the requirements of the applicable standard:

- ISO 7730:2005, Ergonomics of the Thermal Environment, analytical determination and interpretation of thermal comfort, using calculation of the PMV and PPD indices and local thermal comfort criteria; and
- ISO 17772-2017, Energy Performance of Buildings- Indoor environmental quality- Part 1. Indoor environmental input parameters for the design and assessment of energy performance of buildings Section A2.

Data Centers only
Meet the above requirements for regularly occupied spaces.

**WAREHOUSES & DISTRIBUTION CENTERS**
Meet the above requirements for office portions of the building.

In regularly occupied areas of the building's bulk storage, sorting, and distribution areas, include one or more of the following design alternatives:

- radiant flooring;

- circulating fans;
- passive systems, such as nighttime air, heat venting, or wind flow;
- localized active cooling (refrigerant or evaporative-based systems) or heating systems; and
- localized, hard-wired fans that provide air movement for occupants' comfort.
- other equivalent thermal comfort strategy.

**Thermal Comfort Control**

**NC, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY**

Provide individual thermal comfort controls for at least 50% of individual occupant spaces. Provide group thermal comfort controls for all shared multioccupant spaces.

Thermal comfort controls allow occupants, whether in individual spaces or shared multioccupant spaces, to adjust at least one of the following in their local environment: air temperature, radiant temperature, air speed, and humidity.

Hospitality only
Guest rooms are assumed to provide adequate thermal comfort controls and are therefore not included in the credit calculations.

Retail only
Meet the above requirements for at least 50% of the individual occupant spaces in office and administrative areas.

**HEALTHCARE**

Provide individual thermal comfort controls for every patient room and at least 50% of the remaining individual occupant spaces. Provide group thermal comfort controls for all shared multioccupant spaces.

Thermal comfort controls allow occupants, whether in individual spaces or shared multioccupant spaces, to adjust at least one of the following in their local environment: air temperature, radiant temperature, air speed, and humidity.

# EQ CREDIT: INTERIOR LIGHTING

BD+C
**1–2 points**
This credit applies to

- New Construction (1–2 points)
- Schools (1–2 points)
- Retail (2 points)
- Data Centers (1–2 points)
- Warehouses & Distribution Centers (1–2 points)
- Hospitality (1–2 points)
- Healthcare (1 point)

**Intent**

To promote occupants' productivity, comfort, and well-being by providing high-quality lighting.

**Requirements**

**NC, SCHOOLS, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY**
Select one or both of the following two options.

**Option 1. Lighting Control (1 point)**
For at least 90% of individual occupant spaces, provide individual lighting controls that enable occupants to adjust the lighting to suit their individual tasks and preferences, with at least three lighting levels or scenes (on, off, midlevel). Midlevel is 30% to 70% of the maximum illumination level (not including daylight contributions).

For all shared multioccupant spaces, meet all of the following requirements.

- Have in place multizone control systems that enable occupants to adjust the lighting to meet group needs and preferences, with at least three lighting levels or scenes (on, off, midlevel).
- Lighting for any presentation or projection wall must be separately controlled.
- Switches or manual controls must be located in the same space as the controlled luminaires. A person operating the controls must have a direct line of sight to the controlled luminaires.

Hospitality only
Guest rooms are assumed to provide adequate lighting controls and are therefore not included in the credit calculations.

AND/OR

**Option 2. Lighting Quality (1 point)**
Choose four of the following strategies.

A. For all regularly occupied spaces, use light fixtures with a luminance of less than 2,500 cd/m$^2$ between 45 and 90 degrees from nadir.
   Exceptions include wallwash fixtures properly aimed at walls, as specified by manufacturer's data, indirect uplighting fixtures, provided there is no view down into these uplights from a regularly occupied space above, and any other specific applications (i.e. adjustable fixtures).
B. For the entire project, use light sources with a CRI of 80 or higher. Exceptions include lamps or fixtures specifically designed to provide colored lighting for effect, site lighting, or other special use.
C. For at least 75% of the total connected lighting load, use light sources that have a rated life (or L70 for LED sources) of at least 24,000 hours (at 3-hour per start, if applicable).

D. Use direct-only overhead lighting for 25% or less of the total connected lighting load for all regularly occupied spaces.
E. For at least 90% of the regularly occupied floor area, meet or exceed the following thresholds for area-weighted average surface reflectance: 85% for ceilings, 60% for walls, and 25% for floors.
F. If furniture is included in the scope of work, select furniture finishes to meet or exceed the following thresholds for area-weighted average surface reflectance: 45% for work surfaces, and 50% for movable partitions.
G. For at least 75% of the regularly occupied floor area, meet a ratio of average wall surface illuminance (excluding fenestration) to average work plane (or surface, if defined) illuminance that does not exceed 1:10. Must also meet strategy E, strategy F, or demonstrate area-weighted surface reflectance of at least 60% for walls.
H. For at least 75% of the regularly occupied floor area, meet a ratio of average ceiling illuminance (excluding fenestration) to work surface illuminance that does not exceed 1:10. Must also meet strategy E, strategy F, or demonstrate area-weighted surface reflectance of at least 85% for ceilings.

**RETAIL NC**

For at least 90% of the individual occupant spaces in office and administrative areas, provide individual lighting controls.

In sales areas, provide controls that can reduce the ambient light levels to a midlevel (30% to 70% of the maximum illumination level not including daylight contributions).

**HEALTHCARE**

Provide individual lighting controls for at least 90% of individual occupant spaces in staff areas.

For at least 90% of patient positions, provide lighting controls that are readily accessible from the patient's bed. In multioccupant patient spaces, the controls must be individual lighting controls. In private rooms, also provide exterior window shades, blinds, or curtain controls that are readily accessible from the patient's bed. Exceptions include in-patient critical care, pediatric, and psychiatric patient rooms.

For all shared multioccupant spaces, provide multizone control systems that enable occupants to adjust the lighting to meet group needs and preferences, with at least three lighting levels or scenes (on, off, midlevel). Midlevel is 30% to 70% of the maximum illumination level (not including daylight contributions).

# EQ CREDIT: DAYLIGHT

BD+C
**1–3 points**
This credit applies to

- New Construction (1–3 points)
- Core & Shell (1–3 points)
- Schools (1–3 points)
- Retail (1–3 points)
- Data Centers (1–3 points)
- Warehouses & Distribution Centers (1–3 points)
- Hospitality (1–3 points)
- Healthcare (1–2 points)

**Intent**

To connect building occupants with the outdoors, reinforce circadian rhythms, and reduce the use of electrical lighting by introducing daylight into the space.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Provide manual or automatic (with manual override) glare-control devices for all regularly occupied spaces.

AND

Select one of the following three options.

**Option 1. Simulation: Spatial Daylight Autonomy and Annual Sunlight Exposure (1–3 points, 1-2 points Healthcare)**

Perform annual computer simulations for spatial daylight autonomy$_{300/50\%}$ (sDA$_{300/50\%}$), and annual sunlight exposure$_{1000,250}$ (ASE$_{1000,250}$) as defined in IES LM-83-12 for each regularly occupied space. Healthcare projects must use each regularly occupied space located in the perimeter area determined under EQ Credit Quality Views. Additionally, calculate the average sDA$_{300/50\%}$ value for the total regularly occupied floor area.

For any regularly occupied spaces with ASE$_{1000,250}$ greater than 10%, identify how the space is designed to address glare.

Points are awarded according to Table 1.

**Table 1. Points for Option 1**

| | | |
|---|---|---|
| | *New Construction, Core and Shell, Schools, Retail, Data Centers, Warehouses and Distribution Centers, Hospitality* | Healthcare |

| | | |
|---|---|---|
| The average $sDA_{300/50\%}$ value for the regularly occupied floor area is at least **40%** | 1 point | 1 point |
| The average $sDA_{300/50\%}$ value for the regularly occupied floor area is at least **55%** | 2 points | 2 points |
| The average $sDA_{300/50\%}$ value for the regularly occupied floor area is at least **75%** | 3 points | Exemplary performance |
| Each regularly occupied space achieves $sDA_{300/50\%}$ value of at least **55%** | Exemplary performance *or 1 additional point if only 1 or 2 points achieved above.* | Exemplary performance *or 1 additional point if only 1 point achieved above.* |

The sDA and ASE calculation grids should be no more than 2 feet (600 millimeters) square and laid out across the regularly occupied area at a work plane height of 30 inches (76 millimeters) above finished floor (unless otherwise defined). Use an hourly time-step analysis based on typical meteorological year data, or an equivalent, for the nearest available weather station. Include any permanent interior obstructions. Moveable furniture and partitions may be excluded.

CS only
If the finishes in the space will not be completed, use the following default surface reflectances: 80% for ceilings, 20% for floors, and 50% for walls. Assume that the entire floor plate, except for the core, will be regularly occupied space.

OR

**Option 2. Simulation: Illuminance Calculations (1–3 points, 1-2 points Healthcare)**

Perform computer simulations for illuminance at 9 a.m. and 3 p.m. on a clear-sky day at the equinox for each regularly occupied space. Healthcare projects should use the regularly occupied spaces located in the perimeter area determined under EQ Credit Quality Views.

Demonstrate illuminance levels are between 300 lux and 3,000 lux at both 9 a.m. and 3 p.m. Spaces with view-preserving automatic (with manual override) glare-control devices may demonstrate compliance for only the minimum 300 lux illuminance level.

Points are awarded according to Table 2.

**Table 2. Points for Option 2**

| *New Construction, Core and Shell, Schools, Retail, Data Centers, Warehouses and Distribution Centers, Hospitality* | | *Healthcare* | |
|---|---|---|---|
| *Percentage of regularly occupied floor area* | *Points* | *Percentage of regularly occupied floor area within perimeter area* | *Points* |
| 55% | 1 | 55% | 1 |
| 75% | 2 | 75% | 2 |
| 90% | 3 | 90% | Exemplary performance |

Calculate illuminance intensity for sun (direct component) and sky (diffuse component) for clear-sky conditions as follows:

- Use typical meteorological year data, or an equivalent, for the nearest available weather station.

- Select one day within 15 days of September 21 and one day within 15 days of March 21 that represent the clearest sky condition.
- Use the average of the hourly value for the two selected days.

Exclude blinds or shades from the model. Include any permanent interior obstructions. Moveable furniture and partitions may be excluded.

CS only
Assume the following default surface reflectances if the finishes in the space will not be completed: 80% for ceilings, 20% for floors, and 50% for walls. Assume that the entire floor plate, except for the core, will be regularly occupied space.

OR

**Option 3. Measurement (1-3 points, 1-2 points Healthcare)**

Measure illuminance in each regularly occupied space. Healthcare projects should use the regularly occupied spaces located in the perimeter area determined under EQ Credit Quality Views.

Achieve illuminance levels between 300 lux and 3,000 lux. Spaces with view-preserving automatic (with manual override) glare-control devices may demonstrate compliance for only the minimum 300 lux illuminance level.

Points are awarded according to Table 3.

**Table 3. Points for Option 3**

| *New Construction, Core and Schools, Schools, Retail, Data Centers, Warehouses and Distribution Centers, Hospitality* | | *Healthcare* | |
|---|---|---|---|
| *Percentage of regularly occupied floor area* | *Points* | *Percentage of regularly occupied floor area within perimeter area* | |
| 55% at **one** time in the year | 1 | 55% at **one** time in the year | 1 |
| 75% at **two** times in the year | 2 | 75% at **two** times in the year | 2 |
| 90% at **two** times in the year | 3 | 90% at **two** times in the year | exemplary performance |

With furniture, fixtures, and equipment in place, measure illuminance levels as follows:

- Measure at appropriate work plane height during any hour between 9 a.m. and 3 p.m.
- If pursuing one point, take one measurement in any regularly occupied month. If pursuing two points, take two measurements: one measurement in any regularly occupied month, and take a second as indicated in Table 4.
- For spaces larger than 150 square feet (14 square meters), take measurements on a maximum 10 foot (3 meter) square grid.
- For spaces 150 square feet (14 square meters) or smaller, take measurements on a maximum 3 foot (900 millimeters) square grid.

**Table 4. Timing of measurements for illuminance**

| *If first measurement is taken in …* | *take second measurement in …* |
|---|---|
| January | May-September |
| February | June-October |
| March | June-July, November-December |
| April | August-December |
| May | September-January |

| | |
|---|---|
| June | October-February |
| July | November-March |
| August | December-April |
| September | December-January, May-June |
| October | February-June |
| November | March-July |
| December | April-August |

# EQ CREDIT: QUALITY VIEWS

BD+C
**1–2 points**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1-2 points)

**Intent**

To give building occupants a connection to the natural outdoor environment by providing quality views.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, HOSPITALITY**
Achieve a direct line of sight to the outdoors via vision glazing for 75% of all regularly occupied floor area. View glazing in the contributing area must provide a clear image of the exterior, not obstructed by frits, fibers, patterned glazing, or added tints that distort color balance.

Additionally, 75% of all regularly occupied floor area must have at least two of the following four kinds of views:

- multiple lines of sight to vision glazing in different directions at least 90 degrees apart;
- views that include at least two of the following: (1) flora, fauna, or sky; (2) movement; and (3) objects at least 25 feet (7.5 meters) from the exterior of the glazing;
- unobstructed views located within the distance of three times the head height of the vision glazing; and
- views with a view factor of 3 or greater, as defined in “Windows and Offices; A Study of Office Worker Performance and the Indoor Environment.”

Include in the calculations any permanent interior obstructions. Movable furniture and partitions may be excluded.

Views into interior atria may be used to meet up to 30% of the required area.

**WAREHOUSES & DISTRIBUTION CENTERS**

For the office portion of the building, meet the requirements above.

For the bulk storage, sorting, and distribution portions of the building, meet the requirements above for 25% of the regularly occupied floor area.

**HEALTHCARE**

For inpatient units (IPUs), meet the requirements above (1 point).

For other areas, configure the building floor plates such that the floor area within 15 feet (4.5 meters) of the perimeter exceeds the perimeter area requirement (Table 1), and meet the requirements above for the perimeter area (1 point).

**Table 1. Minimum compliant perimeter area, by floor plate area**

| Floor plate area | | Perimeter area | |
|---|---|---|---|
| **(square feet)** | **(square meters)** | **(square feet)** | **(square meters)** |
| Up to 15,000 | Up to 1 400 | 7,348 | 682 |
| 20,000 | 1 800 | 8,785 | 816 |
| 25,000 | 2 300 | 10,087 | 937 |
| 30,000 | 2 800 | 11,292 | 1 049 |
| 35,000 | 3 300 | 12,425 | 1 154 |
| 40,000 | 3 700 | 13,500 | 1 254 |
| 45,000 | 4 200 | 14,528 | 1 349 |
| 50,000 and larger | 4 600 and larger | 15,516 | 1 441 |

# EQ CREDIT: ACOUSTIC PERFORMANCE

BD+C
**1–2 points**
This credit applies to

- New Construction (1 point)
- Schools (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1–2 points)

**Intent**

To provide workspaces and classrooms that promote occupants' well-being, productivity, and communications through effective acoustic design.

**Requirements**

**NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY**

For all occupied spaces, meet two of the following: HVAC background noise, Sound Transmission, and/or Reverberation time. Meet all three for an exemplary performance point.

Confirm compliance via calculations or measurements in representative rooms, and/or design documentation from a person experienced in the field of acoustics.

**HVAC Background Noise**
Achieve maximum background noise levels from heating, ventilating, and air conditioning (HVAC) systems per 2015 ASHRAE Handbook-- HVAC Applications, Chapter 48, Table 1 ; AHRI Standard 885-2008, Table 15; or a local equivalent.

If confirming compliance via measurements, use a sound level meter that conforms to ANSI S1.4 for type 1 (precision) or type 2 (general purpose) sound measurement instrumentation, the International Electrotechnical Commission (2013) IEC 61672-1:2013 Electroacoustics – Sound Level Meters – Part 1: Specifications, or a local equivalent.

Comply with design criteria for HVAC noise levels resulting from the sound transmission paths listed in 2015 ASHRAE Handbook—HVAC Applications, Chapter 48, Table 6; or a local equivalent.

**Sound Transmission**

Categorize all occupied spaces by use and desired level of acoustic privacy.

Meet the composite sound transmission class ($STC_C$) ratings or noise isolation class (NIC) listed in Table 1. For NIC measurements, use ASTM E336-17a or Annex A.3 of ANSI S12.60-2010.

**Table 1. Minimum composite sound transmission class ratings or noise isolation class for adjacent spaces**

| Adjacency combinations | | **$STC_C$ **** | **NIC**** |
|---|---|---|---|
| Retail | Retail | 50 | 45 |
| Collaborative / multi-use | Hallway, stairway | 25 | 20 |
| Private | Hallway, stairway | 35 | 30 |

| | | | |
|---|---|---|---|
| Confidential | Hallway, stairway | 40 | 35 |
| Collaborative / multi-use | Collaborative / multi-use | 35 | 30 |
| Collaborative / multi-use | Private | 45 | 40 |
| Collaborative / multi-use | Confidential | 50 | 45 |
| Private | Private | 45 | 40 |
| Private | Confidential | 50 | 45 |
| Confidential | Confidential | 50 | 45 |
| Conference room | Conference room | 50 | 45 |
| Mechanical equipment room* | Hallway, stairway | 50 | 45 |
| Mechanical equipment room* | Occupied area | 60 | 55 |

*Minimum STCc or NIC has to be met unless proven that the equipment noise in conjunction with the sound isolation performance of the partitions and doors will not exceed the maximum background noise requirements of the adjacent space.

**If a sound masking system is implemented at a minimum level of 40 dBA, the STCc ratings or NIC values in Table 1 may be lowered by 5 points. This applies to all space types except mechanical equipment rooms. The sound masking system must be designed by an acoustical professional and meet the following criteria:

- The overall level for sound masking must be set by an acoustical professional and must not exceed 48 dBA in open offices, libraries, cafeterias, corridors/hallways, 45 dBA in enclosed offices, and 42 dBA in conference rooms, and wellness rooms. The combined level of masking and HVAC background noise must not exceed these limits.
- The system design and commissioning must provide overall level uniformity of +/-1 dBA and one-third octave band uniformity of +/-2 dB from at least 100 to 5,000 Hz when tested according to ASTM E1573-18
- The sound masking spectrum must conform to the National Research Council of Canada COPE Optimum Masking Spectrum or an alternate spectrum if specified by an acoustical engineer.

**Reverberation Time**

Meet the reverberation time requirements in Table 2 (adapted from Table 9.1 in the Performance Measurement Protocols for Commercial Buildings[2]).

**Table 2. Reverberation time requirements**

| **Room type** | **Application** | **T60 (sec), at 500 Hz, 1000 Hz, and 2000 Hz** |
|---|---|---|
| Hotel/motel | Individual room or suite | < 0.6 |
| | Meeting or banquet room | < 0.8 |
| Office building | Executive or private office | < 0.6 |
| | Conference room | < 0.6 |
| | Teleconference room | < 0.6 |
| | Open-plan office without sound masking | < 0.8 |
| | Open-plan office with sound masking | 0.8 |
| Courtroom | Unamplified speech | < 0.7 |
| | Amplified speech | < 1.0 |
| Performing arts space | Drama theaters, concert and recital halls | Varies by application |
| Laboratories | Testing or research with minimal speech communication | < 1.0 |
| | Extensive phone use and speech communication | < 0.6 |

[2] Adapted from ASHRAE (2007d), ASA (2008), ANSI (2002), and CEN (2007)

| | | |
|---|---|---|
| Church, mosque, synagogue | General assembly with critical music program | Varies by application |
| Library | | < 1.0 |
| Indoor stadium, gymnasium | Gymnasium and natatorium | < 2.0 |
| | Large-capacity space with speech amplification | < 1.5 |

## SCHOOLS

**HVAC Background noise**
Achieve a background noise level of 35 dBA or less from heating, ventilating, and air-conditioning (HVAC) systems in classrooms and other core learning spaces. Follow the recommended methodologies and best practices for mechanical system noise control in ANSI Standard S12.60–2010, Part 1, Annex A.1; the 2015 ASHRAE Handbook—HVAC Applications, Chapter 48, Sound and Vibration Control, with errata; AHRI Standard 885–2008; or a local equivalent.

**Sound Transmission**
Design classrooms and other core learning spaces to meet the sound transmission class (STC) requirements of ANSI S12.60–2010 Part 1, or a local equivalent. Exterior windows must have an STC rating of at least 35, unless outdoor and indoor noise levels can be verified to justify a lower rating.

## HEALTHCARE

Design the facility to meet Option 1 (1 point) and/or Option 2 (1 point).

**Option 1. Speech Privacy, Sound Isolation, and Background Noise (1 point)**

*Speech Privacy and Sound Isolation*
Design sound isolation to achieve speech privacy, acoustical comfort, and minimal annoyance from noise-producing sources. Consider sound levels at both source and receiver locations, the background sound at receiver locations, and the occupants' acoustical privacy and acoustical comfort needs.

Design the facility to meet the criteria outlined in the following sections, as applicable:
- 2018 FGI Guidelines for Design and Construction of Hospitals—Section 1.2-5.1.6 and Section 1.2-5.1.6.2
- 2018 FGI Guidelines for Design and Construction of Outpatient Facilities—Section 1.2-5.1.6 and Section 1.2-5.1.6.2
- 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities--Section 2.5-8.6

*Background Noise*
Consider background noise levels generated by all building mechanical-electrical-plumbing systems, air distribution systems and other facility noise sources under the purview of the project building design-construction team.

Design the facility to meet the criteria outlined in the following sections, as applicable:
- 2018 FGI Guidelines for Design and Construction of Hospitals—Section 1.2-5.1.4 (Table 1.2-5)
- 2018 FGI Guidelines for Design and Construction of Outpatient Facilities—Section 1.2-5.1.4 (Table 1.2-5)
- 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities--Section 2.5-8.4 (Table 2.5-5)

Calculate or measure sound levels in representative rooms and spaces of each type to confirm compliance with criteria in the above-referenced tables using a sound level meter that conforms to ANSI S1.4 for type 1 (precision) or type 2 (general purpose) sound measurement instrumentation. For spaces not listed in Table 1.2-2, refer to ASHRAE 2015 Handbook, Chapter 48, Sound and Vibration Control, Table 1.

**Option 2. Acoustical Finishes and Site Exterior Noise (1 point)**
Meet the requirements for acoustical finishes and site exterior noise.

*Acoustical Finishes*
Specify materials, products systems installation details, and other design features to meet the following:

- 2018 FGI Guidelines for Design and Construction of Hospitals—Section 1.2-5.1.3 (Table 1.2-4)
- 2018 FGI Guidelines for Design and Construction of Outpatient Facilities—Section 1.2-5.1.3 (Table 1.2-4)
- 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities--Section 2.5-8.3 (Table 2.5-4)

Calculate or measure the average sound absorption coefficients for representative unoccupied rooms of each type in the building to confirm conformance with the requirements.

*Site Exterior Noise*
Minimize the effect on building occupants of site exterior noise produced by road traffic, aircraft flyovers, railroads, on-site heliports, emergency power generators during maintenance testing, outdoor facility MEP and building services equipment, etc. Also minimize effects on the surrounding community from all facility MEP equipment and activities as required to meet (1) local applicable codes or (2) background noise requirements above, whichever is more stringent.
Comply with the 2018 FGI Guidelines for the following noise sources:

- heliports, A1.3-3.6.2.2;
- generators, 2.1-8.3.3.1;
- mechanical equipment, 2.1-8.2.1.1; and
- building services, A2.2-5.3

Measure and analyze data to determine the exterior noise classification (A, B, C, or D) of the facility site and design the building envelope to meet the following, as applicable:

- 2018 FGI Guidelines for Design and Construction of Hospitals—Table 1.2-3
- 2018 FGI Guidelines for Design and Construction of Outpatient Facilities— Table 1.2-3
- 2018 FGI Guidelines for Design and Construction of Residential Health, Care, and Support Facilities—Table 2.5-3

.
For exterior site exposure categories B, C, or D, calculate or measure the sound isolation performance of representative elements of the exterior building envelope to determine the composite sound transmission class (STCc) rating for representative façade sections. Measurements should generally conform to ASTM E966-18, Standard Guide for Field Measurements of Airborne Sound Insulation of Building Façades and Façade Elements.

# INNOVATION (IN)

## IN CREDIT: INNOVATION

BD+C
**1–5 points**
This credit applies to

- New Construction (1–5 points)
- Core & Shell (1–5 points)
- Schools (1–5 points)
- Retail (1–5 points)
- Data Centers (1–5 points)
- Warehouses & Distribution Centers (1–5 points)
- Hospitality (1–5 points)
- Healthcare (1–5 points)

**Intent**

To encourage projects to achieve exceptional or innovative performance.

**Requirements**

**NC, CS, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

To achieve all five innovation points, a project team must achieve at least one pilot credit, at least one innovation credit and no more than two exemplary performance credits.

**Option 1. Innovation (1 point)**
Achieve significant, measurable environmental performance using a strategy not addressed in the LEED green building rating system.

Identify the following:

- the intent of the proposed innovation credit;
- proposed requirements for compliance;
- proposed submittals to demonstrate compliance; and
- the design approach or strategies used to meet the requirements.

Examples of innovation may be found in the LEED Innovation Catalog.

AND/OR

**Option 2. Pilot (1 point)**
Achieve one pilot credit from USGBC's LEED Pilot Credit Library.

AND/OR

**Option 3. Additional Strategies**

- **Innovation (1-3 points)**
  Defined in Option 1 above.
- **Pilot (1-3 points)**
  Meet the requirements of Option 2.

- **Exemplary Performance (1–2 points)**
  Achieve exemplary performance in an existing LEED v4 prerequisite or credit that allows exemplary performance, as specified in the LEED Reference Guide, v4 edition. An exemplary performance point is typically earned for achieving double the credit requirements or the next incremental percentage threshold.

## IN CREDIT: LEED ACCREDITED PROFESSIONAL

BD+C
**1 point**
This credit applies to

- New Construction (1 point)
- Core & Shell (1 point)
- Schools (1 point)
- Retail (1 point)
- Data Centers (1 point)
- Warehouses & Distribution Centers (1 point)
- Hospitality (1 point)
- Healthcare (1 point)

### Intent

To encourage the team integration required by a LEED project and to streamline the application and certification process.

### Requirements

#### NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE

At least one principal participant of the project team must be a LEED Accredited Professional (AP) with a specialty appropriate for the project.

# REGIONAL PRIORITY (RP)

## RP CREDIT: REGIONAL PRIORITY

BD+C
**4 points**
This credit applies to

- New Construction (1-4 points)
- Core & Shell (1-4 points)
- Schools (1-4 points)
- Retail (1-4 points)
- Data Centers (1-4 points)
- Warehouses & Distribution Centers (1-4 points)
- Hospitality (1-4 points)
- Healthcare (1-4 points)

### Intent

To provide an incentive for the achievement of credits that address geographically specific environmental, social equity, and public health priorities.

### Requirements

**NC, CS, SCHOOLS, RETAIL NC, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY, HEALTHCARE**

Earn up to four of the six Regional Priority credits. These credits have been identified by the USGBC regional councils and chapters as having additional regional importance for the project's region. A database of Regional Priority credits and their geographic applicability is available on the USGBC website, http://www.usgbc.org.

One point is awarded for each Regional Priority credit achieved, up to a maximum of four.

# APPENDICES

## APPENDIX 1. USE TYPES AND CATEGORIES

**Table 1. Use Types and Categories**

| Category | Use type |
|---|---|
| Food retail | Supermarket |
| | Grocery with produce section |
| Community-serving retail | Convenience store |
| | Farmers market |
| | Hardware store |
| | Pharmacy |
| | Other retail |
| Services | Bank |
| | Family entertainment venue (e.g., theater, sports) |
| | Gym, health club, exercise studio |
| | Hair care |
| | Laundry, dry cleaner |
| | Restaurant, café, diner (excluding those with only drive-thru service) |
| Civic and community facilities | Adult or senior care (licensed) |
| | Child care (licensed) |
| | Community or recreation center |
| | Cultural arts facility (museum, performing arts) |
| | Education facility (e.g., K—12 school, university, adult education center, vocational school, community college) |
| | Government office that serves public on-site |
| | Medical clinic or office that treats patients |
| | Place of worship |
| | Police or fire station |
| | Post office |
| | Public library |
| | Public park |
| | Social services center |
| Community anchor uses (BD+C and ID+C only) | Commercial office (100 or more full-time equivalent jobs) |
| | Housing (100 or more dwelling units) |

Adapted from Criterion Planners, INDEX neighborhood completeness indicator, 2005.

# APPENDIX 2. DEFAULT OCCUPANCY COUNTS

Use Table 1 to calculate default occupancy counts. Only use the occupancy estimates if occupancy is unknown.

For the calculation, use gross floor area, not net or leasable floor area. Gross floor area is defined as the sum of all areas on all floors of a building included within the outside faces of the exterior wall, including common areas, mechanical spaces, circulation areas, and all floor penetrations that connect one floor to another. To determine gross floor area, multiply the building footprint (in square feet or square meters) by the number of floors in the building. Exclude underground or structured parking from the calculation.

**Table 1. Default Occupancy Numbers**

| | Gross square feet per occupant | | Gross square meters per occupant | |
|---|---|---|---|---|
| | Employees | Transients | Employees | Transients |
| General office | 250 | 0 | 23 | 0 |
| Retail, general | 550 | 130 | 51 | 12 |
| Retail or service (e.g., financial, auto) | 600 | 130 | 56 | 12 |
| Restaurant | 435 | 95 | 40 | 9 |
| Grocery store | 550 | 115 | 51 | 11 |
| Medical office | 225 | 330 | 21 | 31 |
| R&D or laboratory | 400 | 0 | 37 | 0 |
| Warehouse, distribution | 2,500 | 0 | 232 | 0 |
| Warehouse, storage | 20,000 | 0 | 1860 | 0 |
| Hotel | 1,500 | 700 | 139 | 65 |
| Educational, daycare | 630 | 105 | 59 | 10 |
| Educational, K–12 | 1,300 | 140 | 121 | 13 |
| Educational, postsecondary | 2,100 | 150 | 195 | 14 |

Sources:
ANSI/ASHRAE/IESNA Standard 90.1–2004 (Atlanta, GA, 2004).
2001 Uniform Plumbing Code (Los Angeles, CA)
California Public Utilities Commission, 2004–2005 Database for Energy Efficiency Resources (DEER) Update Study (2008).
California State University, Capital Planning, Design and Construction Section VI, Standards for Campus Development Programs ( Long Beach, CA, 2002).
City of Boulder Planning Department, Projecting Future Employment—How Much Space per Person (Boulder, 2002).
Metro, 1999 Employment Density Study (Portland, OR 1999).
American Hotel and Lodging Association, Lodging Industry Profile Washington, DC, 2008.
LEED for Core & Shell Core Committee, personal communication (2003 - 2006).
LEED for Retail Core Committee, personal communication (2007)
OWP/P, Medical Office Building Project Averages (Chicago, 2008).
OWP/P, University Master Plan Projects (Chicago, 2008).
U.S. General Services Administration, Childcare Center Design Guide (Washington, DC,2003).

# APPENDIX 3. RETAIL PROCESS LOAD BASELINES

**Table 1a. Commercial kitchen appliance prescriptive measures and baseline for energy cost budget (IP units)**

| | Baseline energy usage for energy modeling path | | | | Levels for prescriptive path | |
|---|---|---|---|---|---|---|
| Appliance type | Fuel | Function | Baseline efficiency | Baseline idle rate | Prescriptive efficiency | Prescriptive idle rate |
| Broiler, underfired | Gas | Cooking | 30% | 16,000 Btu/h/ft$^2$ peak input | 35% | 12,000 Btu/h/ft$^2$ peak input |
| Combination ovens, steam mode (P = pan capacity) | Elec | Cooking | 40% steam mode | 0.37P+4.5 kW | 50% steam mode | 0.133P+0.6400 kW |
| Combination ovens, steam mode | Gas | Cooking | 20% steam mode | 1,210P+35,810 Btu/h | 38% steam mode | 200P+6,511 Btu/h |
| Combination ovens, convection mode | Elec | Cooking | 65% convection mode | 0.1P+1.5 kW | 70% convection mode | 0.080P+0.4989 kW |
| Combination ovens, convection mode | Gas | Cooking | 35% convection mode | 322P+13,563 Btu/h | 44% convection mode | 150P+5,425 Btu/h |
| Convection oven, full-size | Elec | Cooking | 65% | 2.0 kW | 71% | 1.6 kW |
| Convection oven, full-size | Gas | Cooking | 30% | 18,000 Btu/h | 46% | 12,000 Btu/h |
| Convection oven, half-size | Elec | Cooking | 65% | 1.5 kW | 71% | 1.0 kW |
| Conveyor oven, > 25-inch belt | Gas | Cooking | 20% | 70,000 Btu/h | 42% | 57,000 Btu/h |
| Conveyor oven, ≤ 25-inch belt | Gas | Cooking | 20% | 45,000 Btu/h | 42% | 29,000 Btu/h |
| Fryer | Elec | Cooking | 75% | 1.05 kW | 80% | 1.0 kW |
| Fryer | Gas | Cooking | 35% | 14,000 Btu/h | 50% | 9,000 Btu/h |
| Griddle (based on 3 ft model) | Elec | Cooking | 60% | 400 W/ft$^2$ | 70% | 320 W/ft$^2$ |
| Griddle (based on 3 ft model) | Gas | Cooking | 30% | 3,500 Btu/h/ft$^2$ | 38% | 2,650 Btu/h/ft$^2$ |

| | | | | | | |
|---|---|---|---|---|---|---|
| Hot food holding cabinets (excluding drawer warmers and heated display), 0 < V < 13 ft³ (V = volume) | Elec | Cooking | na | 40 W/ft³ | Na | 21.5V Watts |
| Hot food holding cabinets (excluding drawer warmers and heated display), 13 ≤ V < 28 ft³ | Elec | Cooking | na | 40 W/ft³ | Na | 2.0V + 254 Watts |
| Hot food holding cabinets (excluding drawer warmers and heated display), 28 ft³ ≤ V | Elec | Cooking | na | 40 W/ft³ | Na | 3.8V + 203.5 Watts |
| Large vat fryer | Elec | Cooking | 75% | 1.35 kW | 80% | 1.1 kW |
| Large vat fryer | Gas | Cooking | 35% | 20,000 Btu/h | 50% | 12,000 Btu/h |
| Rack oven, double | Gas | Cooking | 30% | 65,000 Btu/h | 50% | 35,000 Btu/h |
| Rack oven, single | Gas | Cooking | 30% | 43,000 Btu/h | 50% | 29,000 Btu/h |
| Range | Elec | Cooking | 70% | | 80% | |
| Range | Gas | Cooking | 35% | na | 40% and no standing pilots | na |
| Steam cooker, batch cooking | Elec | Cooking | 26% | 200 W/pan | 50% | 135 W/pan |
| Steam cooker, batch cooking | Gas | Cooking | 15% | 2,500 Btu/h/pan | 38% | 2,100 Btu/h/pan |
| Steam cooker, high production or cook to order | Elec | Cooking | 26% | 330 W/pan | 50% | 275 W/pan |

| | | | | | | |
|---|---|---|---|---|---|---|
| Steam cooker, high production or cook to order | Gas | Cooking | 15% | 5,000 Btu/h/pan | 38% | 4,300 Btu/h/pan |
| Toaster | Elec | Cooking | — | 1.8 kW average operating energy rate | Na | 1.2 kW average operating energy rate |
| Ice machine, IMH (ice-making head, H = ice harvest), H ≥ 450 lb/day | Elec | Ice | 6.89 - 0.0011H kWh/100 lb ice | na | $37.72*H^{-0.298}$ kWh/100 lb ice | na |
| Ice machine, IMH (ice-making head), H ≤ 450 lb/day | Elec | Ice | 10.26 – 0.0086H kWh/100 lb ice | na | $37.72*H^{-0.298}$ kWh/100 lb ice | na |
| Ice machine, RCU (remote condensing unit, w/o remote compressor, H < 1,000 lb/day | Elec | Ice | 8.85 - 0.0038H kWh/100lb ice | na | $22.95*H^{-0.258}$ + 1.00 kWh/100 lb ice | na |
| Ice machine, RCU (remote condensing unit), 1600 > H ≥ 1000 lb/day | Elec | ice | 5.10 kWh/100 lb ice | Na | $22.95*H^{-0.258}$ + 1.00 kWh/100 lb ice | na |
| Ice machine, RCU (remote condensing unit), H ≥ 1600 lb/day | Elec | Ice | 5.10 kWh/100lb ice | Na | -0.00011*H + 4.60 kWh/100 lb ice | na |
| Ice machine, SCU (self-contained unit), H < 175 lb/day | Elec | Ice | 18.0 - 0.0469H kWh/100lb ice | Na | $48.66*H^{-0.326}$ + 0.08 kWh/100 lb ice | na |

| | | | | | | |
|---|---|---|---|---|---|---|
| Ice machine self-contained unit, H ≥ 175 lb/day | Elec | Ice | 9.80 kWh/100 lb ice | Na | $48.66*H^{-0.326}$ + 0.08 kWh/100 lb ice | na |
| Ice machine, water-cooled ice-making head, H ≥ 1436 lb/day (must be on chilled loop) | Elec | Ice | 4.0 kWh/100 lb ice | Na | 3.68 kWh/100 lb ice | na |
| Ice machine, water-cooled ice-making head, 500 lb/day < H < 1436 (must be on chilled loop) | Elec | Ice | 5.58 – 0.0011H kWh/100 lb ice | Na | 5.13 - 0.001H kWh/100 lb ice | na |
| Ice machine, water-cooled ice-making head, H < 500 lb/day (must be on chilled loop) | Elec | Ice | 7.80 – 0.0055H kWh/100 lb ice | Na | 7.02 - 0.0049H kWh/100 lb ice | na |
| Ice machine, water-cooled once-through (open loop) | Elec | Ice | Banned | Banned | Banned | Banned |
| Ice machine, water-cooled SCU (self-contained unit), H < 200 lb/day (must be on chilled loop) | Elec | Ice | 11.4 – 0.0190H kWh/100 lb ice | Na | 10.6 - 0.177H kWh/100 lb ice | na |
| Ice machine, water-cooled self-contained | Elec | Ice | 7.6 kWh/100 lb ice | Na | 7.07 kWh/100 lb ice | na |

| | | | | | | |
|---|---|---|---|---|---|---|
| unit, H ≥ 200 lb/day (must be on chilled loop) | | | | | | |
| Chest freezer, solid or glass door | Elec | Refrig | 0.45V + 0.943 kWh/day | Na | ≤ 0.270V + 0.130 kWh/day | na |
| Chest refrigerator, solid or glass door | Elec | Refrig | 0.1V + 2.04 kWh/day | Na | ≤ 0.125V + 0.475 kWh/day | na |
| Glass-door reach-in freezer, 0 < V < 15 $ft^3$ | Elec | Refrig | 0.75V + 4.10 kWh/day | Na | ≤ 0.607V + 0.893 kWh/day | na |
| Glass-door reach-in freezer, 15 ≤ V < 30 $ft^3$ | Elec | Refrig | .75V + 4.10 kWh/day | Na | ≤ 0.733V – 1.00 kWh/day | na |
| Glass-door reach-in freezer, 30 ≤ V < 50 $ft^3$ | Elec | Refrig | .75V + 4.10 kWh/day | Na | ≤ 0.250V + 13.50 kWh/day | na |
| Glass-door reach-in freezer, 50 ≤ V $ft^3$ | Elec | Refrig | 0.75V + 4.10 kWh/day | Na | ≤ 0.450V + 3.50 kWh/day | na |
| Glass-door reach-in refrigerator, 0 < V < 15 $ft^3$ | Elec | Refrig | 0.12V + 3.34 kWh/day | Na | ≤ 0.118V + 1.382 kWh/day | na |
| Glass-door reach-in refrigerator, 15 ≤ V < 30 $ft^3$ | Elec | Refrig | 0.12V + 3.34 kWh/day | Na | ≤ 0.140V + 1.050 kWh/day | na |
| Glass-door reach-in refrigerator, 30 ≤ V < 50 $ft^3$ | Elec | Refrig | 0.12V + 3.34 kWh/day | Na | ≤ 0.088V + 2.625 kWh/day | na |
| Glass-door reach-in refrigerator, 50 ≤ V $ft^3$ | Elec | Refrig | 0.12V + 3.34 kWh/day | Na | ≤ 0.110V + 1.500 kWh/day | na |
| Solid-door reach-in freezer, 0 < V < 15 $ft^3$ | Elec | Refrig | 0.4V + 1.38 kWh/day | Na | ≤ 0.250V + 1.25 kWh/day | na |

| | | | | | | |
|---|---|---|---|---|---|---|
| Solid-door reach-in freezer, 15 ≤ V < 30 ft³ | Elec | Refrig | 0.4V + 1.38 kWh/day | Na | ≤ 0.400V – 1.000 kWh/day | na |
| Solid-door reach-in freezer, 30 ≤ V < 50 ft³ | Elec | Refrig | 0.4V + 1.38 kWh/day | Na | ≤ 0.163V + 6.125 kWh/day | na |
| Solid-door reach-in freezer, 50 ≤ V ft³ | Elec | Refrig | 0.4V + 1.38 kWh/day | Na | ≤ 0.158V + 6.333 kWh/day | na |
| Solid-door reach-in refrigerator, 0 < V < 15 ft³ | Elec | Refrig | 0.1V + 2.04 kWh/day | Na | ≤ 0.089V + 1.411 kWh/day | na |
| Solid-door reach-in refrigerator, 15 ≤ V < 30 ft³ | Elec | Refrig | 0.1V + 2.04 kWh/day | Na | ≤ 0.037V + 2.200 kWh/day | na |
| Solid-door reach-in refrigerator, 30 ≤ V < 50 ft³ | Elec | Refrig | 0.1V + 2.04 kWh/day | Na | ≤ 0.056V + 1.635 kWh/day | na |
| Solid-door reach-in refrigerator, 50 ≤ V ft³ | Elec | Refrig | 0.1V + 2.04 kWh/day | Na | ≤ 0.060V + 1.416 kWh/day | na |
| Clothes washer | Gas | Sanitation | 1.72 MEF | Na | 2.00 MEF | na |
| Door-type dish machine, high temp | Elec | Sanitation | na | 1.0 kW | Na | 0.70 kW |
| Door-type dish machine, low temp | Elec | Sanitation | na | 0.6 kW | Na | 0.6 kW |
| Multitank rack conveyor dish machine, high temp | Elec | Sanitation | na | 2.6 kW | Na | 2.25 kW |
| Multitank rack conveyor dish machine, low temp | Elec | Sanitation | na | 2.0 kW | Na | 2.0 kW |

| | | | | | | |
|---|---|---|---|---|---|---|
| Single-tank rack conveyor dish machine, high temp | Elec | Sanitation | na | 2.0 kW | Na | 1.5 kW |
| Single-tank rack conveyor dish machine, low temp | Elec | Sanitation | na | 1.6 kW | Na | 1.5 kW |
| Undercounter dish machine, high temp | Elec | Sanitation | na | 0.9 kW | Na | 0.5 kW |
| Undercounter dish machine, low temp | Elec | Sanitation | na | 0.5 kW | Na | 0.5 kW |

The energy efficiency, idle energy rates, and water use requirements, where applicable, are based on the following test methods:
ASTM F1275 Standard Test Method for Performance of Griddles
ASTM F1361 Standard Test Method for Performance of Open Deep Fat Fryers
ASTM F1484 Standard Test Methods for Performance of Steam Cookers
ASTM F1496 Standard Test Method for Performance of Convection Ovens
ASTM F1521 Standard Test Methods for Performance of Range Tops
ASTM F1605 Standard Test Method for Performance of Double-Sided Griddles
ASTM F1639 Standard Test Method for Performance of Combination Ovens
ASTM F1695 Standard Test Method for Performance of Underfired Broilers
ASTM F1696 Standard Test Method for Energy Performance of Single-Rack Hot Water Sanitizing, ASTM Door-Type Commercial Dishwashing Machines
ASTM F1704 Standard Test Method for Capture and Containment Performance of Commercial Kitchen Exhaust Ventilation Systems
ASTM F1817 Standard Test Method for Performance of Conveyor Ovens
ASTM F1920 Standard Test Method for Energy Performance of Rack Conveyor, Hot Water Sanitizing, Commercial Dishwashing Machines
ASTM F2093 Standard Test Method for Performance of Rack Ovens
ASTM F2140 Standard Test Method for Performance of Hot Food Holding Cabinets
ASTM F2144 Standard Test Method for Performance of Large Open Vat Fryers
ASTM F2324 Standard Test Method for Prerinse Spray Valves
ASTM F2380 Standard Test Method for Performance of Conveyor Toasters
ARI 810-2007: Performance Rating of Automatic Commercial Ice Makers
ANSI/ASHRAE Standard 72–2005: Method of Testing Commercial Refrigerators and Freezers with temperature setpoints at 38°F for medium-temp refrigerators, 0°F for low-temp freezers, and -15°F for ice cream freezers

**Table 1b. Commercial Kitchen Appliance Prescriptive Measures and Baseline for Energy Cost Budget (SI units)**

| | Baseline energy usage for energy modeling path | | | | Levels for prescriptive path | |
|---|---|---|---|---|---|---|
| Appliance type | Fuel | Function | Baseline efficiency | Baseline idle rate | Prescriptive efficiency | Prescriptive idle rate |
| Broiler, underfired | Gas | Cooking | 30% | 50.5 kW/$m^2$ | 35% | 37.9 kW/$m^2$ |
| Combination oven, steam mode (P = pan capacity) | Elec | Cooking | 40% steam mode | 0.37P+4.5 kW | 50% steam mode | 0.133P+0.6400 kW |
| Combination oven, steam mode | Gas | Cooking | 20% steam mode | (1 210P+ 35 810)/3 412 kW | 38% steam mode | (200P+6 511)/ 3 412 kW |
| Combination oven, convection mode | Elec | Cooking | 65% convection mode | 0.1P+1.5 kW | 70% convection mode | 0.080P+0.4989 kW |
| Combination oven, convection mode | Gas | Cooking | 35% convection mode | (322P+ 13 563)/ 3412 kW | 44% convection mode | (150P+5 425)/ 3412 kW |
| Convection oven, full-size | Elec | Cooking | 65% | 2.0 kW | 71% | 1.6 kW |
| Convection oven, full-size | Gas | Cooking | 30% | 5.3 kW | 46% | 3.5 kW |
| Convection oven, half-size | Elec | Cooking | 65% | 1.5 kW | 71% | 1.0 kW |
| Conveyor oven, > 63.5 cm belt | Gas | Cooking | 20% | 20.5 kW | 42% | 16.7 kW |
| Conveyor oven, < 63.5 cm belt | Gas | Cooking | 20% | 13.2 kW | 42% | 8.5 kW |
| Fryer | Elec | Cooking | 75% | 1,05 kW | 80% | 1.0 kW |
| Fryer | Gas | Cooking | 35% | 4.1 kW | 50% | 2.64 kW |
| Griddle (based on 90-cm model) | Elec | Cooking | 60% | 4.3 kW/$m^2$ | 70% | 3 .45 kW/$m^2$ |
| Griddle (based on 90-cm model) | Gas | Cooking | 30% | 11 kW/$m^2$ | 33% | 8.35 kW/$m^2$ |
| Hot food holding cabinets (excluding drawer warmers and heated display) 0 < V < 0.368 $m^3$ (V = volume) | Elec | Cooking | na | 1.4 kW/$m^3$ | Na | (21.5*V)/0.0283 kW/$m^3$ |

| | | | | | | |
|---|---|---|---|---|---|---|
| Hot food holding cabinets (excluding drawer warmers and heated display), 0.368 ≤ V < 0.793 $m^3$ | Elec | Cooking | na | 1.4 kW/$m^3$ | Na | (2.0*V + 254)/0.0283 kW/$m^3$ |
| Hot food holding cabinets (excluding drawer warmers and heated display), 0.793 $m^3$ ≤ V | Elec | Cooking | na | 1.4 kW/$m^3$ | Na | (3.8*V + 203.5)/0.0283 kW/$m^3$ |
| Large vat fryer | Elec | Cooking | 75% | 1.35 kW | 80% | 1.1 kW |
| Large vat fryer | Gas | Cooking | 35% | 5.86 kW | 50% | 3.5 kW |
| Rack oven, double | Gas | Cooking | 30% | 19 kW | 50% | 10.25 kW |
| Rack oven, single | Gas | Cooking | 30% | 12.6 kW | 50% | 8.5 kW |
| Range | Elec | Cooking | 70% | na | 80% | na |
| Range | Gas | Cooking | 35% | na | 40% and no standing pilots | **na** |
| Steam cooker, batch cooking | Elec | Cooking | 26% | 200 W/pan | 50% | 135 W/pan |
| Steam cooker, batch cooking | Gas | Cooking | 15% | 733 W/pan | 38% | 615 W/pan |
| Steam cooker, high production or cook to order | Elec | Cooking | 26% | 330 W/pan | 50% | 275 W/pan |
| Steam cooker, high production or cook to order | Gas | Cooking | 15% | 1.47 kW/pan | 38% | 1.26 kW/pan |
| Toaster | Elec | Cooking | **na** | 1.8 kW average operating energy rate | **Na** | 1.2 kW average operating energy rate |
| Ice machine, IMH (ice making head, H = ice harvest) H ≧ 204 kg/day | Elec | Ice | 0.0015 - 5.3464E$^{-07}$ kWh/kg ice | na | ≤ 13.52*H$^{-0.298}$ kWh/100 kg ice | na |

| | | | | | | |
|---|---|---|---|---|---|---|
| Ice machine, IMH (ice making head), H < 204 kg/day | Elec | Ice | 0.2262 - 4.18E$^{-04}$ kWh/kg ice | na | ≤ 13.52*H$^{-0.298}$ kWh/100 kg ice | na |
| Ice machine, RCU (remote condensing unit, w/o remote compressor) H < 454 kg/day | Elec | Ice | 0.1951 - 1.85E$^{-04}$ kWh/kg ice | na | ≤ 111.5835H$^{-0.258}$) + 2.205 kWh/100 kg ice | na |
| Ice machine, RCU (remote condensing unit) 726 > H ≥ 454 kg/day | Elec | Ice | 0.1124 kWh/kg ice | na | ≤ 111.5835H$^{-0.258}$) + 2.205 kWh/100 kg ice | na |
| Ice machine, RCU (remote condensing unit), H ≥ 726kg/day | Elec | Ice | 0.1124 kWh/kg ice | na | ≤ -0.00024H + 4.60 kWh/100 kg ice | na |
| Ice machine, SCU (self contained unit), H < 79 kg/day | Elec | Ice | 0.3968 - 2.28E$^{-03}$ kWh/kg ice | na | 236.59H$^{-0.326}$ +0.176 kWh/100 kg ice | na |
| Ice machine, SCU (self contained unit), H ≥ 79 kg/day | Elec | Ice | 0.2161 kWh/kg ice | na | 236.59H$^{-0.326}$ +0.176 kWh/100 kg ice | na |
| Ice machine, water-cooled ice-making head, H ≥ 651 kg/day(must be on a chilled loop) | Elec | Ice | 0.0882 kWh/kg ice | na | ≤ 8.11 kWh/100 kg ice | na |
| Ice machine, water-cooled ice-making head, 227 ≤ H < 651 kg/day (must be on a chilled loop) | Elec | Ice | 0.1230 - 5.35E$^{-05}$ kWh/kg ice | na | ≤ 11.31 - 0.065H kWh/100 kg ice | na |
| Ice machine, water-cooled ice-making head, H < | Elec | Ice | 0.1720 - 2.67E$^{-04}$ kWh/kg ice | na | ≤ 15.48 - 0.0238H kWh/100 kg ice | na |

| | | | | | | |
|---|---|---|---|---|---|---|
| 227 kg/day( must be on a chilled loop) | | | | | | |
| Ice machine, water-cooled once-through (open loop) | Elec | Ice | Banned | Banned | Banned | Banned |
| Ice machine, water cooled SCU (self-contained unit) H < 91 kg/day (must be on a chilled loop) | Elec | Ice | 0.2513 - $9.23E^{-04}$ kWh/kg ice | na | ≤ 23.37-0.086H kWh/100 kg ice | na |
| Ice machine, water cooled SCU (self-contained unit) H ≥ 91 kg/day (must be on a chilled loop) | Elec | Ice | 0.1676 kWh/kg ice | na | 15.57 kWh/100 kg ice | na |
| Chest freezer, solid or glass door | Elec | Refrig | 15.90V + 0.943 kWh/day | na | 9.541V + 0.130 kWh/day | na |
| Chest refrigerator, solid or glass door | Elec | Refrig | 3.53V + 2.04 kWh/day | na | ≤ 4.417 V + 0.475 kWh/day | na |
| Glass-door reach-in freezer, 0 < V < 0.42 $m^3$ | Elec | Refrig | 26.50V + 4.1 kWh/day | na | ≤ 21.449V + 0.893 kWh/day | na |
| Glass-door reach-in freezer, 0.42 ≤ V < 0.85 $m^3$ | Elec | Refrig | 26.50V + 4.1 kWh/day | na | ≤ 25.901V – 1.00 kWh/day | na |
| Glass-door reach-in freezer, 0.85 ≤ V < 1.42 $m^3$ | Elec | Refrig | 26.50V + 4.1 kWh/day | na | ≤ 8.834V + 13.50 kWh/day | na |
| Glass-door reach-in freezer, 1.42 ≤ V $m^3$ | Elec | Refrig | 26.50V + 4.1 kWh/day | na | ≤ 15.90V + 3.50 kWh/day | na |
| Glass-door reach-in refrigerator, 0 < V < 0.42$m^3$ | Elec | Refrig | 4.24V + 3.34 kWh/day | na | ≤ 4.169V + 1.382 kWh/day | na |
| Glass-door reach-in refrigerator, 0.42 ≤ V < 0.85 $m^3$ | Elec | Refrig | 4.24V + 3.34 kWh/day | na | ≤ 4.947V + 1.050 kWh/day | na |

| | | | | | | |
|---|---|---|---|---|---|---|
| Glass-door reach-in refrigerator, 0.85 ≤ V < 1.42 $m^3$ | Elec | Refrig | 4.24V + 3.34 kWh/day | na | ≤ 3.109V + 2.625 kWh/day | na |
| Glass-door reach-in refrigerator, 1.42 ≤ V $m^3$ | Elec | Refrig | 4.24V + 3.34 kWh/day | na | ≤ 3.887V + 1.500 kWh/day | na |
| Solid-door reach-in freezer, 0 < V < 0.42 $m^3$ | Elec | Refrig | 14.13V + 1.38 kWh/day | na | ≤ 8.834V + 1.25 kWh/day | na |
| Solid-door reach-in freezer, 0.42 ≤ V < 0.85 $m^3$ | Elec | Refrig | 14.13V + 1.38 kWh/day | na | ≤ 4.819V – 1.000 kWh/day | na |
| Solid-door reach-in freezer, 0.85 ≤ V < 1.42 $m^3$ | Elec | Refrig | 14.13V + 1.38 kWh/day | na | ≤ 5.760V + 6.125 kWh/day | na |
| Solid-door reach-in freezer, 1.42 ≤ V $m^3$ | Elec | Refrig | 14.13V + 1.38 kWh/day | na | ≤ 5.583V + 6.333 kWh/day | na |
| Solid-door reach-in refrigerator, 0 < V < 0.42$m^3$ | Elec | Refrig | 3.53V + 2.04 kWh/day | na | ≤ 3.145V + 1.411 kWh/day | na |
| Solid-door reach-in refrigerator, 0.42 ≤ V < 0.85 $m^3$ | Elec | Refrig | 3.53V + 2.04 kWh/day | na | ≤ 1.307V + 2.200 kWh/day | na |
| Solid-door reach-in refrigerator, 0.85 ≤ V < 1.42 $m^3$ | Elec | Refrig | 3.53V + 2.04 kWh/day | na | ≤ 1.979V + 1.635 kWh/day | na |
| Solid-door reach-in refrigerator, 1.42 ≤ V $m^3$ | Elec | Refrig | 3.53V + 2.04 kWh/day | na | ≤ 2.120V + 1.416 kWh/day | na |
| Clothes washer | Gas | Sanitation | 1.72 MEF | | 2.00 MEF | |
| Door-type dish machine, high temp | Elec | Sanitation | na | 1.0 kW | Na | 0.70 kW |
| Door-type dish machine, low temp | Elec | Sanitation | na | 0.6 kW | Na | 0.6 kW |

| | | | | | | |
|---|---|---|---|---|---|---|
| Multitank rack conveyor dish machine, high temp | Elec | Sanitation | na | 2.6 kW | Na | 2.25 kW |
| Multitank rack conveyor dish machine, low temp | Elec | Sanitation | na | 2.0 kW | Na | 2.0 kW |
| Single-tank rack conveyor dish machine, high temp | Elec | Sanitation | **na** | 2.0 kW | Na | 1.5 kW |
| Single-tank rack conveyor dish machine, low temp | Elec | Sanitation | na | 1.6 kW | Na | 1.5 kW |
| Undercounter dish machine, high temp | Elec | Sanitation | na | 0.9 kW | Na | 0.5 kW |
| Undercounter dish machine, low temp | Elec | Sanitation | na | 0.5 kW | Na | 0.5 kW |

The energy efficiency, idle energy rates, and water use requirements, where applicable, are based on the following test methods:

ASTM F1275 Standard Test Method for Performance of Griddles
ASTM F1361 Standard Test Method for Performance of Open Deep Fat Fryers
ASTM F1484 Standard Test Methods for Performance of Steam Cookers
ASTM F1496 Standard Test Method for Performance of Convection Ovens
ASTM F1521 Standard Test Methods for Performance of Range Tops
ASTM F1605 Standard Test Method for Performance of Double-Sided Griddles
ASTM F1639 Standard Test Method for Performance of Combination Ovens
ASTM F1695 Standard Test Method for Performance of Underfired Broilers
ASTM F1696 Standard Test Method for Energy Performance of Single-Rack Hot Water Sanitizing, ASTM Door-Type Commercial Dishwashing Machines
ASTM F1704 Standard Test Method for Capture and Containment Performance of Commercial Kitchen Exhaust Ventilation Systems
ASTM F1817 Standard Test Method for Performance of Conveyor Ovens
ASTM F1920 Standard Test Method for Energy Performance of Rack Conveyor, Hot Water Sanitizing, Commercial Dishwashing Machines
ASTM F2093 Standard Test Method for Performance of Rack Ovens
ASTM F2140 Standard Test Method for Performance of Hot Food Holding Cabinets
ASTM F2144 Standard Test Method for Performance of Large Open Vat Fryers
ASTM F2324 Standard Test Method for Prerinse Spray Valves
ASTM F2380 Standard Test Method for Performance of Conveyor Toasters
ARI 810-2007: Performance Rating of Automatic Commercial Ice Makers

ANSI/ASHRAE Standard 72–2005: Method of Testing Commercial Refrigerators and Freezers with temperature setpoints at 38°F (3°C) for mediumtemp refrigerators, -18°C for low-temp freezers, and -26°C for ice cream freezers.

**Table 2. Supermarket refrigeration prescriptive measures and baseline for energy cost budget**

| Item | Attribute | Prescriptive measure | Baseline for energy modeling path |
|---|---|---|---|
| Commercial Refrigerator and Freezers | Energy Use Limits | ASHRAE 90.1-2010 Addendum g. Table 6.8.1L | ASHRAE 90.1-2010 Addendum g. Table 6.8.1L |
| Commercial Refrigeration Equipment | Energy Use Limits | ASHRAE 90.1-2010 Addendum g. Table 6.8.1M | ASHRAE 90.1-2010 Addendum g. Table 6.8.1M |

**Table 3. Walk-in coolers and freezers prescriptive measures and baseline for energy cost budget**

| Item | Attribute | Prescriptive measure | Baseline for energy modeling path |
|---|---|---|---|
| Envelope | Freezer insulation | R-46 | R-36 |
| | Cooler insulation | R-36 | R-20 |
| | Automatic closer doors | Yes | No |
| | High-efficiency low- or no-heat reach-in doors | 40W/ft (130W/m) of door frame (low temperature), 17W/ft (55W/m) of door frame (medium temperature) | 40W/ft (130W/m) of door frame (low temperature), 17W/ft (55W/m) of door frame (medium temperature) |
| Evaporator | Evaporator fan motor and control | Shaded pole and split phase motors prohibited; use PSC or EMC motors | Constant-speed fan |
| | Hot gas defrost | No electric defrosting. | Electric defrosting |
| Condenser | Air-cooled condenser fan motor and control | Shaded pole and split phase motors prohibited; use PSC or EMC motors; add condenser fan controllers | Cycling one-speed fan |
| | Air Cooled condenser design approach | Floating head pressure controls or ambient subcooling | 10°F (-12°C) to 15°F (-9°C) dependent on suction temperature |
| Lighting | Lighting power density (W/sq.ft.) | 0.6 W/sq.ft. (6.5 W/sq. meter) | 0.6 W/sq.ft. (6.5 W/sq. meter) |
| Commercial Refrigerator and Freezers | Energy Use Limits | N/A | Use an Exceptional Calculation Method if attempting to take savings |
| Commercial Refrigerator and Freezers | Energy Use Limits | N/A | Use an Exceptional Calculation Method if attempting to take savings |

**Table 4. Commercial kitchen ventilation prescriptive measures and baseline for energy cost budget**

| Strategies | Prescriptive measure | Baseline |
|---|---|---|
| Kitchen hood control | ASHRAE 90.1-2010 Section 6.5.7.1, except that Section 6.5.7.1.3 and Section 6.5.7.1.4 shall apply if the total kitchen exhaust airflow rate exceeds 2,000 cfm (960 L/s) (as opposed to 5,000 cfm (2,400 L/s) noted in the ASHRAE 90.1-2010 requirements) | ASHRAE 90.1-2010 Section 6.5.7.1 and Section G3.1.1 Exception (d) where applicable |